UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

(Mark One)

☒ **ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the fiscal year ended December 31, 2023
OR

☐ **TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934**

For the transition period from _____ to _____

Commission File Number: 001-34756

Tesla, Inc.

(Exact name of registrant as specified in its charter)

Delaware	**91-2197729**
(State or other jurisdiction of incorporation or organization)	**(I.R.S. Employer Identification No.)**
1 Tesla Road Austin, Texas	**78725**
(Address of principal executive offices)	**(Zip Code)**

(512) 516-8177
(Registrant's telephone number, including area code)

Securities registered pursuant to Section 12(b) of the Act:

Title of each class	Trading Symbol(s)	Name of each exchange on which registered
Common stock	TSLA	The Nasdaq Global Select Market

Securities registered pursuant to Section 12(g) of the Act:
None

Indicate by check mark whether the registrant is a well-known seasoned issuer, as defined in Rule 405 of the Securities Act. Yes ☒ No ☐

Indicate by check mark if the registrant is not required to file reports pursuant to Section 13 or 15(d) of the Act. Yes ☐ No ☒

Indicate by check mark whether the registrant (1) has filed all reports required to be filed by Section 13 or 15(d) of the Securities Exchange Act of 1934 ("Exchange Act") during the preceding 12 months (or for such shorter period that the registrant was required to file such reports), and (2) has been subject to such filing requirements for the past 90 days. Yes ☒ No ☐

Indicate by check mark whether the registrant has submitted electronically every Interactive Data File required to be submitted pursuant to Rule 405 of Regulation S-T (§232.405 of this chapter) during the preceding 12 months (or for such shorter period that the registrant was required to submit such files). Yes ☒ No ☐

Indicate by check mark whether the registrant is a large accelerated filer, an accelerated filer, a non-accelerated filer, a smaller reporting company, or an emerging growth company. See the definitions of "large accelerated filer," "accelerated filer," "smaller reporting company" and "emerging growth company" in Rule 12b-2 of the Exchange Act:

Large accelerated filer	☒	Accelerated filer	☐
Non-accelerated filer	☐	Smaller reporting company	☐
Emerging growth company	☐		

If an emerging growth company, indicate by check mark if the registrant has elected not to use the extended transition period for complying with any new or revised financial accounting standards provided pursuant to Section 13(a) of the Exchange Act. ☐

Indicate by check mark whether the Registrant has filed a report on and attestation to its management's assessment of the effectiveness of its internal control over financial reporting under Section 404(b) of the Sarbanes-Oxley Act (15 U.S.C. 7262(b)) by the registered public accounting firm that prepared or issued its audit report. ☒

If securities are registered pursuant to Section 12(b) of the Act, indicate by check mark whether the financial statements of the registrant included in the filing reflect the correction of an error to previously issued financial statements. ☐

Indicate by check mark whether any of those error corrections are restatements that required a recovery analysis of incentive-based compensation received by any of the registrant's executive officers during the relevant recovery period pursuant to §240.10D-1(b). ☐

Indicate by check mark whether the registrant is a shell company (as defined in Rule 12b-2 of the Exchange Act). Yes ☐ No ☒

The aggregate market value of voting stock held by non-affiliates of the registrant, as of June 30, 2023, the last day of the registrant's most recently completed second fiscal quarter, was $722.52 billion (based on the closing price for shares of the registrant's Common Stock as reported by the NASDAQ Global Select Market on June 30, 2023). Shares of Common Stock held by each executive officer and director have been excluded in that such persons may be deemed to be affiliates. This determination of affiliate status is not necessarily a conclusive determination for other purposes.

As of January 22, 2024, there were 3,184,790,415 shares of the registrant's common stock outstanding.

DOCUMENTS INCORPORATED BY REFERENCE

Portions of the registrant's Proxy Statement for the 2024 Annual Meeting of Stockholders are incorporated herein by reference in Part III of this Annual Report on Form 10-K to the extent stated herein. Such proxy statement will be filed with the Securities and Exchange Commission within 120 days of the registrant's fiscal year ended December 31, 2023.

TESLA, INC.

ANNUAL REPORT ON FORM 10-K FOR THE YEAR ENDED DECEMBER 31, 2023

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Forward-Looking Statements

The discussions in this Annual Report on Form 10-K contain forward-looking statements reflecting our current expectations that involve risks and uncertainties. These forward-looking statements include, but are not limited to, statements concerning supply chain constraints, our strategy, competition, future operations and production capacity, future financial position, future revenues, projected costs, profitability, expected cost reductions, capital adequacy, expectations regarding demand and acceptance for our technologies, growth opportunities and trends in the markets in which we operate, prospects and plans and objectives of management. The words "anticipates," "believes," "could," "estimates," "expects," "intends," "may," "plans," "projects," "will," "would" and similar expressions are intended to identify forward-looking statements, although not all forward-looking statements contain these identifying words. We may not actually achieve the plans, intentions or expectations disclosed in our forward-looking statements and you should not place undue reliance on our forward-looking statements. Actual results or events could differ materially from the plans, intentions and expectations disclosed in the forward-looking statements that we make. These forward-looking statements involve risks and uncertainties that could cause our actual results to differ materially from those in the forward-looking statements, including, without limitation, the risks set forth in Part I, Item 1A, "Risk Factors" of the Annual Report on Form 10-K for the fiscal year ended December 31, 2023 and that are otherwise described or updated from time to time in our other filings with the Securities and Exchange Commission (the "SEC"). The discussion of such risks is not an indication that any such risks have occurred at the time of this filing. We do not assume any obligation to update any forward-looking statements.

PART I

ITEM 1. BUSINESS

Overview

We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. We generally sell our products directly to customers, and continue to grow our customer-facing infrastructure through a global network of vehicle showrooms and service centers, Mobile Service, body shops, Supercharger stations and Destination Chargers to accelerate the widespread adoption of our products. We emphasize performance, attractive styling and the safety of our users and workforce in the design and manufacture of our products and are continuing to develop full self-driving technology for improved safety. We also strive to lower the cost of ownership for our customers through continuous efforts to reduce manufacturing costs and by offering financial and other services tailored to our products.

Our mission is to accelerate the world's transition to sustainable energy. We believe that this mission, along with our engineering expertise, vertically integrated business model and focus on user experience differentiate us from other companies.

Segment Information

We operate as two reportable segments: (i) automotive and (ii) energy generation and storage.

The automotive segment includes the design, development, manufacturing, sales and leasing of high-performance fully electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment also includes services and other, which includes sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives.

Our Products and Services

Automotive

We currently manufacture five different consumer vehicles – the Model 3, Y, S, X and Cybertruck. Model 3 is a four-door mid-size sedan that we designed for manufacturability with a base price for mass-market appeal. Model Y is a compact sport utility vehicle ("SUV") built on the Model 3 platform with seating for up to seven adults. Model S is a four-door full-size sedan and Model X is a mid-size SUV with seating for up to seven adults. Model S and Model X feature the highest performance characteristics and longest ranges that we offer in a sedan and SUV, respectively. In November 2023, we entered the consumer pickup truck market with first deliveries of the Cybertruck, a full-size electric pickup truck with a stainless steel exterior that has the utility and strength of a truck while featuring the speed of a sports car.

In 2022, we also began early production and deliveries of a commercial electric vehicle, the Tesla Semi. We have planned electric vehicles to address additional vehicle markets, and to continue leveraging developments in our proprietary Full Self-Driving ("FSD") Capability features, battery cell and other technologies.

Energy Generation and Storage

Energy Storage Products

Powerwall and Megapack are our lithium-ion battery energy storage products. Powerwall, which we sell directly to customers, as well as through channel partners, is designed to store energy at a home or small commercial facility. Megapack is an energy storage solution for commercial, industrial, utility and energy generation customers, multiple of which may be grouped together to form larger installations of gigawatt hours ("GWh") or greater capacity.

We also continue to develop software capabilities for remotely controlling and dispatching our energy storage systems across a wide range of markets and applications, including through our real-time energy control and optimization platforms.

Solar Energy Offerings

We sell retrofit solar energy systems to customers and channel partners and also make them available through power purchase agreement ("PPA") arrangements. We purchase most of the components for our retrofit solar energy systems from multiple sources to ensure competitive pricing and adequate supply. We also design and manufacture certain components for our solar energy products.

We sell our Solar Roof, which combines premium glass roof tiles with energy generation, directly to customers, as well as through channel customers. We continue to improve our installation capability and efficiency, including through collaboration with real estate developers and builders on new homes.

Technology

Automotive

Battery and Powertrain

Our core vehicle technology competencies include powertrain engineering and manufacturing and our ability to design vehicles that utilize the unique advantages of an electric powertrain. We have designed our proprietary powertrain systems to be adaptable, efficient, reliable and cost-effective while withstanding the rigors of an automotive environment. We offer dual motor powertrain vehicles, which use two electric motors to maximize traction and performance in an all-wheel drive configuration, as well as vehicle powertrain technology featuring three electric motors for further increased performance in certain versions of Model S and Model X, Cybertruck and the Tesla Semi.

We maintain extensive testing and R&D capabilities for battery cells, packs and systems, and have built an expansive body of knowledge on lithium-ion cell chemistry types and performance characteristics. In order to enable a greater supply of cells for our products with higher energy density at lower costs, we have developed a new proprietary lithium-ion battery cell and improved manufacturing processes.

Vehicle Control and Infotainment Software

The performance and safety systems of our vehicles and their battery packs utilize sophisticated control software. Control systems in our vehicles optimize performance, customize vehicle behavior, manage charging and control all infotainment functions. We develop almost all of this software, including most of the user interfaces, internally and update our vehicles' software regularly through over-the-air updates.

Self-Driving Development and Artificial Intelligence

We have expertise in developing technologies, systems and software to enable self-driving vehicles using primarily vision-based technologies. Our FSD Computer runs our neural networks in our vehicles, and we are also developing additional computer hardware to better enable the massive amounts of field data captured by our vehicles to continually train and improve these neural networks for real-world performance.

Currently, we offer in our vehicles certain advanced driver assist systems under our Autopilot and FSD Capability options. Although at present the driver is ultimately responsible for controlling the vehicle, our systems provide safety and convenience functionality that relieves drivers of the most tedious and potentially dangerous aspects of road travel much like the system that airplane pilots use, when conditions permit. As with other vehicle systems, we improve these functions in our vehicles over time through over-the-air updates.

We intend to establish in the future an autonomous Tesla ride-hailing network, which we expect would also allow us to access a new customer base even as modes of transportation evolve.

We are also applying our artificial intelligence learnings from self-driving technology to the field of robotics, such as through Optimus, a robotic humanoid in development, which is controlled by the same AI system.

Energy Generation and Storage

Energy Storage Products

We leverage many of the component-level technologies from our vehicles in our energy storage products. By taking a modular approach to the design of battery systems, we can optimize manufacturing capacity of our energy storage products. Additionally, our expertise in power electronics enables our battery systems to interconnect with electricity grids while providing fast-acting systems for power injection and absorption. We have also developed software to remotely control and dispatch our energy storage systems.

Solar Energy Systems

We have engineered Solar Roof over numerous iterations to combine aesthetic appeal and durability with power generation. The efficiency of our solar energy products is aided by our own solar inverter, which incorporates our power electronics technologies. We designed both products to integrate with Powerwall.

Design and Engineering

Automotive

We have established significant in-house capabilities in the design and test engineering of electric vehicles and their components and systems. Our team has significant experience in computer-aided design as well as durability, strength and crash test simulations, which reduces the product development time of new models. We have also achieved complex engineering feats in stamping, casting and thermal systems, and developed a method to integrate batteries directly with vehicle body structures without separate battery packs to optimize manufacturability, weight, range and cost characteristics.

We are also expanding our manufacturing operations globally while taking action to localize our vehicle designs and production for particular markets, including country-specific market demands and factory optimizations for local workforces. As we increase our capabilities, particularly in the areas of automation, die-making and line-building, we are also making strides in the simulations modeling these capabilities prior to construction.

Energy Generation and Storage

Our expertise in electrical, mechanical, civil and software engineering allows us to design, engineer, manufacture and install energy generating and storage products and components, including at the residential through utility scale. For example, the modular design of our Megapack utility-scale battery line is intended to significantly reduce the amount of assembly required in the field. We also customize solutions including our energy storage products, solar energy systems and/or Solar Roof for customers to meet their specific needs.

Sales and Marketing

Historically, we have been able to achieve sales without traditional advertising and at relatively low marketing costs. We continue to monitor our public narrative and brand, and tailor our marketing efforts accordingly, including through investments in customer education and advertising as necessary.

Automotive

Direct Sales

Our vehicle sales channels currently include our website and an international network of company-owned stores. In some jurisdictions, we also have galleries to educate and inform customers about our products, but such locations do not transact in the sale of vehicles. We believe this infrastructure enables us to better control costs of inventory, manage warranty service and pricing, educate consumers about electric vehicles, make our vehicles more affordable, maintain and strengthen the Tesla brand and obtain rapid customer feedback.

We reevaluate our sales strategy both globally and at a location-by-location level from time to time to optimize our sales channels. However, sales of vehicles in the automobile industry tend to be cyclical in many markets, which may expose us to volatility from time to time.

Used Vehicle Sales

Our used vehicle business supports new vehicle sales by integrating the trade-in of a customer's existing Tesla or non-Tesla vehicle with the sale of a new or used Tesla vehicle. The Tesla and non-Tesla vehicles we acquire as trade-ins are subsequently remarketed, either directly by us or through third parties. We also remarket used Tesla vehicles acquired from other sources including lease returns.

Public Charging

We have a growing global network of Tesla Superchargers, which are our industrial-grade, high-speed vehicle chargers. Where possible, we co-locate Superchargers with our solar and energy storage systems to reduce costs and promote renewable power. Supercharger stations are typically placed along well-traveled routes and in and around dense city centers to allow vehicle owners the ability to enjoy quick, reliable charging along an extensive network with convenient stops. Use of the Supercharger network either requires payment of a fee or is free under certain sales programs. In November 2021, we began to offer Supercharger access to non-Tesla vehicles in certain locations in support of our mission to accelerate the world's transition to sustainable energy, and in November 2022, we opened up our previously proprietary charging connector as the North American Charging Standard (NACS). This enables all electric vehicles and charging stations to interoperate — which makes charging easier and more efficient for everyone and advances our mission to accelerate the world's transition to sustainable energy. Following this, a number of major automotive companies announced their adoption of NACS, with their access to the Supercharger network beginning in phases in 2024 and their production of NACS vehicles beginning no later than 2025. We also engaged SAE International to govern NACS as an industry standard, now named J3400. We continue to monitor and increase our network of Tesla Superchargers in anticipation of future demand.

We also work with a wide variety of hospitality, retail and public destinations, as well as businesses with commuting employees, to offer additional charging options for our customers, as well as single-family homeowners and multi-family residential entities, to deploy home charging solutions.

In-App Upgrades

As our vehicles are capable of being updated remotely over-the-air, our customers may purchase additional paid options and features through the Tesla app or through the in-vehicle user interface. We expect that this functionality will also allow us to offer certain options and features on a subscription basis in the future.

Energy Generation and Storage

We market and sell our solar and energy storage products to residential, commercial and industrial customers and utilities through a variety of channels, including through our website, stores and galleries, as well as through our network of channel partners, and in the case of some commercial customers, through PPA transactions. We emphasize simplicity, standardization and accessibility to make it easy and cost-effective for customers to adopt clean energy, while reducing our customer acquisition costs.

Service and Warranty

Automotive

Service

We provide service for our electric vehicles at our company-owned service locations and through Tesla Mobile Service technicians who perform work remotely at customers' homes or other locations. Servicing the vehicles ourselves allows us to identify problems and implement solutions and improvements faster than traditional automobile manufacturers and their dealer networks. The connectivity of our vehicles also allows us to diagnose and remedy many problems remotely and proactively.

Vehicle Limited Warranties and Extended Service Plans

We provide a manufacturer's limited warranty on all new and used Tesla vehicles we sell directly to consumers, which may include limited warranties on certain components, specific types of damage or battery capacity retention. We also currently offer optional extended service plans that provide coverage beyond the new vehicle limited warranties for certain models in specified regions.

Energy Generation and Storage

We provide service and repairs to our energy product customers, including under warranty where applicable. We generally provide manufacturer's limited warranties with our energy storage products and offer certain extended limited warranties that are available at the time of purchase of the system. If we install a system, we also provide certain limited warranties on our installation workmanship.

For retrofit solar energy systems, we provide separate limited warranties for workmanship and against roof leaks, and for Solar Roof, we also provide limited warranties for defects and weatherization. For components not manufactured by us, we generally pass-through the applicable manufacturers' warranties.

As part of our solar energy system and energy storage contracts, we may provide the customer with performance guarantees that commit that the underlying system will meet or exceed the minimum energy generation or performance requirements specified in the contract.

Financial Services

Automotive

Purchase Financing and Leases

We offer leasing and/or loan financing arrangements for our vehicles in certain jurisdictions in North America, Europe and Asia ourselves and through various financial institutions. Under certain of such programs, we have provided resale value guarantees or buyback guarantees that may obligate us to cover a resale loss up to a certain limit or repurchase the subject vehicles at pre-determined values.

Insurance

In 2021, we launched our insurance product using real-time driving behavior in select states, which offers rates that are often better than other alternatives and promotes safer driving. Our insurance products are currently available in 12 states and we plan to expand the markets in which we offer insurance products, as part of our ongoing effort to decrease the total cost of ownership for our customers.

Energy Generation and Storage

We offer certain financing options to our solar customers, which enable the customer to purchase and own a solar energy system, Solar Roof or integrated solar and Powerwall system. Our solar PPAs, offered primarily to commercial customers, charge a fee per kilowatt-hour based on the amount of electricity produced by our solar energy systems.

Manufacturing

We currently have manufacturing facilities in the U.S. in Northern California, in Buffalo, New York, Gigafactory New York; in Austin, Texas, Gigafactory Texas and near Reno, Nevada, Gigafactory Nevada. At these facilities, we manufacture and assemble, among other things, vehicles, certain vehicle parts and components, such as our battery packs and battery cells, energy storage components and solar products and components.

Internationally, we also have manufacturing facilities in China (Gigafactory Shanghai) and Germany (Gigafactory Berlin-Brandenburg), which allows us to increase the affordability of our vehicles for customers in local markets by reducing transportation and manufacturing costs and eliminating the impact of unfavorable tariffs. In March 2023, we announced the location of our next Gigafactory in Monterrey, Mexico. Generally, we continue to expand production capacity at our existing facilities. We also intend to further increase cost-competitiveness in our significant markets by strategically adding local manufacturing.

Supply Chain

Our products use thousands of parts that are sourced from hundreds of suppliers across the world. We have developed close relationships with vendors of key parts such as battery cells, electronics and complex vehicle assemblies. Certain components purchased from these suppliers are shared or are similar across many product lines, allowing us to take advantage of pricing efficiencies from economies of scale.

As is the case for some automotive companies, some of our procured components and systems are sourced from single suppliers. Where multiple sources are available for certain key components, we work to qualify multiple suppliers for them where it is sensible to do so in order to minimize potential production risks due to disruptions in their supply. We also mitigate risk by maintaining safety stock for key parts and assemblies and die banks for components with lengthy procurement lead times.

Our products use various raw materials including aluminum, steel, cobalt, lithium, nickel and copper. Pricing for these materials is governed by market conditions and may fluctuate due to various factors outside of our control, such as supply and demand and market speculation. We strive to execute long-term supply contracts for such materials at competitive pricing when feasible, and we currently believe that we have adequate access to raw materials supplies to meet the needs of our operations.

Governmental Programs, Incentives and Regulations

Globally, the ownership of our products by our customers is impacted by various government credits, incentives, and policies. Our business and products are also subject to numerous governmental regulations that vary among jurisdictions.

The operation of our business is also impacted by various government programs, incentives, and other arrangements. See Note 2,*Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Programs and Incentives

Inflation Reduction Act

On August 16, 2022, the Inflation Reduction Act of 2022 (“IRA”) was enacted into law and is effective for taxable years beginning after December 31, 2022, and remains subject to future guidance releases. The IRA includes multiple incentives to promote clean energy, electric vehicles, battery and energy storage manufacture or purchase, including through providing tax credits to consumers. For example, qualifying Tesla customers may receive up to $7,500 in federal tax credits for the purchase of qualified electric vehicles in the U.S. through 2032.

Automotive Regulatory Credits

We earn tradable credits in the operation of our business under various regulations related to zero-emission vehicles (“ZEVs”), greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements. Sales of these credits are recognized within automotive regulatory credits revenue in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Energy Storage System Incentives and Policies

While the regulatory regime for energy storage projects is still under development, there are various policies, incentives and financial mechanisms at the federal, state and local levels that support the adoption of energy storage.

For example, energy storage systems that are charged using solar energy may be eligible for the solar energy-related U.S. federal tax credits described below. The Federal Energy Regulatory Commission (“FERC”) has also taken steps to enable the participation of energy storage in wholesale energy markets. In addition, California and a number of other states have adopted procurement targets for energy storage, and behind-the-meter energy storage systems qualify for funding under the California Self Generation Incentive Program. Our customers primarily benefit directly under these programs. In certain instances our customers may transfer such credits to us as contract consideration. In such transactions, they are included as a component of energy generation and storage revenues in our consolidated statements of operations included elsewhere in this Annual Report on Form 10-K.

Pursuant to the IRA, under Sections 48, 48E and 25D of the Internal Revenue Code (”IRC”), standalone energy storage technology is eligible for a tax credit between 6% and 50% of qualified expenditures, regardless of the source of energy, which may be claimed by our customers for storage systems they purchase or by us for arrangements where we own the systems. These tax credits are primarily for the benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Solar Energy System Incentives and Policies

U.S. federal, state and local governments have established various policies, incentives and financial mechanisms to reduce the cost of solar energy and to accelerate the adoption of solar energy. These incentives include tax credits, cash grants, tax abatements and rebates.

In particular, pursuant to the IRA, Sections 48, 48E and 25D of the IRC provides a tax credit between 6% and 70% of qualified commercial or residential expenditures for solar energy systems, which may be claimed by our customers for systems they purchase, or by us for arrangements where we own the systems for properties that meet statutory requirements. These tax credits are primarily for the direct benefit of our customers and are currently scheduled to phase-out starting in 2032 or later.

Regulations

Vehicle Safety and Testing

In the U.S., our vehicles are subject to regulation by the National Highway Traffic Safety Administration ("NHTSA"), including all applicable Federal Motor Vehicle Safety Standards ("FMVSS") and the NHTSA bumper standard. Numerous FMVSS apply to our vehicles, such as crash-worthiness and occupant protection requirements. Our current vehicles fully comply and we expect that our vehicles in the future will fully comply with all applicable FMVSS with limited or no exemptions, however, FMVSS are subject to change from time to time. As a manufacturer, we must self-certify that our vehicles meet all applicable FMVSS and the NHTSA bumper standard, or otherwise are exempt, before the vehicles may be imported or sold in the U.S.

We are also required to comply with other federal laws administered by NHTSA, including the Corporate Average Fuel Economy standards, Theft Prevention Act requirements, labeling requirements and other information provided to customers in writing, Early Warning Reporting requirements regarding warranty claims, field reports, death and injury reports and foreign recalls, a Standing General Order requiring reports regarding crashes involving vehicles equipped with advanced driver assistance systems, and additional requirements for cooperating with compliance and safety investigations and recall reporting. The U.S. Automobile Information and Disclosure Act also requires manufacturers of motor vehicles to disclose certain information regarding the manufacturer's suggested retail price, optional equipment and pricing. In addition, federal law requires inclusion of fuel economy ratings, as determined by the U.S. Department of Transportation and the Environmental Protection Agency (the "EPA"), and New Car Assessment Program ratings as determined by NHTSA, if available.

Our vehicles sold outside of the U.S. are subject to similar foreign compliance, safety, environmental and other regulations. Many of those regulations are different from those applicable in the U.S. and may require redesign and/or retesting. Some of those regulations impact or prevent the rollout of new vehicle features.

Self-Driving Vehicles

Generally, laws pertaining to self-driving vehicles are evolving globally, and in some cases may create restrictions on features or vehicle designs that we develop. While there are currently no federal U.S. regulations pertaining specifically to self-driving vehicles or self-driving equipment, NHTSA has published recommended guidelines on self-driving vehicles, apart from the FMVSS and manufacturer reporting obligations, and retains the authority to investigate and/or take action on the safety or compliance of any vehicle, equipment or features operating on public roads. Certain U.S. states also have legal restrictions on the operation, registration or licensure of self-driving vehicles, and many other states are considering them. This regulatory patchwork increases the legal complexity with respect to self-driving vehicles in the U.S.

In markets that follow the regulations of the United Nations Economic Commission for Europe ("ECE markets"), some requirements restrict the design of advanced driver-assistance or self-driving features, which can compromise or prevent their use entirely. Other applicable laws, both current and proposed, may hinder the path and timeline to introducing self-driving vehicles for sale and use in the markets where they apply.

Other key markets, including China, continue to consider self-driving regulation. Any implemented regulations may differ materially from the U.S. and ECE markets, which may further increase the legal complexity of self-driving vehicles and limit or prevent certain features.

Automobile Manufacturer and Dealer Regulation

In the U.S., state laws regulate the manufacture, distribution, sale and service of automobiles, and generally require motor vehicle manufacturers and dealers to be licensed in order to sell vehicles directly to residents. Certain states have asserted that the laws in such states do not permit automobile manufacturers to be licensed as dealers or to act in the capacity of a dealer, or that they otherwise restrict a manufacturer's ability to deliver or perform warranty repairs on vehicles. To sell vehicles to residents of states where we are not licensed as a dealer, we generally conduct the sale out of the state. In certain such states, we have opened "galleries" that serve an educational purpose and where sales may not occur.

Some automobile dealer trade associations have both challenged the legality of our operations in court and used administrative and legislative processes to attempt to prohibit or limit our ability to operate existing stores or expand to new locations. Certain dealer associations have also actively lobbied state licensing agencies and legislators to interpret existing laws or enact new laws in ways not favorable to our ownership and operation of our own retail and service locations. We expect such challenges to continue, and we intend to actively fight any such efforts.

Battery Safety and Testing

Our battery packs are subject to various U.S. and international regulations that govern transport of "dangerous goods," defined to include lithium-ion batteries, which may present a risk in transportation. We conduct testing to demonstrate our compliance with such regulations.

We use lithium-ion cells in our high voltage battery packs in our vehicles and energy storage products. The use, storage and disposal of our battery packs are regulated under existing laws and are the subject of ongoing regulatory changes that may add additional requirements in the future. We have agreements with third party battery recycling companies to recycle our battery packs, and we are also piloting our own recycling technology.

Solar Energy—General

We are subject to certain state and federal regulations applicable to solar and battery storage providers and sellers of electricity. To operate our systems, we enter into standard interconnection agreements with applicable utilities. Sales of electricity and non-sale equipment leases by third parties, such as our leases and PPAs, have faced regulatory challenges in some states and jurisdictions.

Solar Energy—Net Metering

Most states in the U.S. make net energy metering, or net metering, available to solar customers. Net metering typically allows solar customers to interconnect their solar energy systems to the utility grid and offset their utility electricity purchases by receiving a bill credit for excess energy generated by their solar energy system that is exported to the grid. In certain jurisdictions, regulators or utilities have reduced or eliminated the benefit available under net metering or have proposed to do so.

Competition

Automotive

The worldwide automotive market is highly competitive and we expect it will become even more competitive in the future as a significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter the electric vehicle market.

We believe that our vehicles compete in the market based on both their traditional segment classification as well as their propulsion technology. For example, Cybertruck competes with other pickup trucks, Model S and Model X compete primarily with premium sedans and premium SUVs and Model 3 and Model Y compete with small to medium-sized sedans and compact SUVs, which are extremely competitive markets. Competing products typically include internal combustion vehicles from more established automobile manufacturers; however, many established and new automobile manufacturers have entered or have announced plans to enter the market for electric and other alternative fuel vehicles. Overall, we believe these announcements and vehicle introductions, including the introduction of electric vehicles into rental car company fleets, promote the development of the electric vehicle market by highlighting the attractiveness of electric vehicles relative to the internal combustion vehicle. Many major automobile manufacturers have electric vehicles available today in major markets including the U.S., China and Europe, and other current and prospective automobile manufacturers are also developing electric vehicles. In addition, several manufacturers offer hybrid vehicles, including plug-in versions.

We believe that there is also increasing competition for our vehicle offerings as a platform for delivering self-driving technologies, charging solutions and other features and services, and we expect to compete in this developing market through continued progress on our Autopilot, FSD and neural network capabilities, Supercharger network and our infotainment offerings.

Energy Generation and Storage

Energy Storage Systems

The market for energy storage products is also highly competitive, and both established and emerging companies have introduced products that are similar to our product portfolio or that are alternatives to the elements of our systems. We compete with these companies based on price, energy density and efficiency. We believe that the specifications and features of our products, our strong brand and the modular, scalable nature of our energy storage products give us a competitive advantage in our markets.

Solar Energy Systems

The primary competitors to our solar energy business are the traditional local utility companies that supply energy to our potential customers. We compete with these traditional utility companies primarily based on price and the ease by which customers can switch to electricity generated by our solar energy systems. We also compete with solar energy companies that provide products and services similar to ours. Many solar energy companies only install solar energy systems, while others only provide financing for these installations. We believe we have a significant expansion opportunity with our offerings and that the regulatory environment is increasingly conducive to the adoption of renewable energy systems.

Intellectual Property

We place a strong emphasis on our innovative approach and proprietary designs which bring intrinsic value and uniqueness to our product portfolio. As part of our business, we seek to protect the underlying intellectual property rights of these innovations and designs such as with respect to patents, trademarks, copyrights, trade secrets, confidential information and other measures, including through employee and third-party nondisclosure agreements and other contractual arrangements. For example, we place a high priority on obtaining patents to provide the broadest and strongest possible protection to enable our freedom to operate our innovations and designs across all of our products and technologies as well as to protect and defend our product portfolio. We have also adopted a patent policy in which we irrevocably pledged that we will not initiate a lawsuit against any party for infringing our patents through activity relating to electric vehicles or related equipment for so long as such party is acting in good faith. We made this pledge in order to encourage the advancement of a common, rapidly-evolving platform for electric vehicles, thereby benefiting ourselves, other companies making electric vehicles and the world.

Environmental, Social and Governance (ESG) and Human Capital Resources

ESG

The very purpose of Tesla's existence is to accelerate the world's transition to sustainable energy. We believe the world cannot reduce carbon emissions without addressing both energy generation and consumption, and we are designing and manufacturing a complete energy and transportation ecosystem to achieve this goal. As we expand, we are building each new factory to be more efficient and sustainably designed than the previous one, including with respect to per-unit waste reduction and resource consumption, including water and energy usage. We are focused on further enhancing sustainability of operations outside of our direct control, including reducing the carbon footprint of our supply chain.

We are committed to sourcing only responsibly produced materials, and our suppliers are required to provide evidence of management systems that ensure social, environmental and sustainability best practices in their own operations, as well as to demonstrate a commitment to responsible sourcing into their supply chains. We have a zero-tolerance policy when it comes to child or forced labor and human trafficking by our suppliers and we look to the Organization for Economic Co-operation and Development Due Diligence Guidelines to inform our process and use feedback from our internal and external stakeholders to find ways to continually improve. We are also driving safety in our own factories by focusing on worker engagement. Our incidents per vehicle continue to drop even as our production volumes increase. We also strive to be an employer of choice by offering compelling, impactful jobs with best in-industry benefits.

We believe that sound corporate governance is critical to helping us achieve our goals, including with respect to ESG. We continue to evolve a governance framework that exercises appropriate oversight of responsibilities at all levels throughout the company and manages its affairs consistent with high principles of business ethics. Our ESG Sustainability Council is made up of leaders from across our company, and regularly presents to our Board of Directors, which oversees our ESG impacts, initiatives and priorities.

Human Capital Resources

A competitive edge for Tesla is its ability to attract and retain high quality employees. During the past year, Tesla made substantial investments in its workforce, further strengthening its standing as one of the most desirable and innovative companies to work for. As of December 31, 2023, our employee headcount worldwide was 140,473.

We have created an environment that fosters growth opportunities, and as of this report, nearly two-thirds (65%) of our managers were promoted from an internal, non-manager position, and 43% of our management employees have been with Tesla for more than five years. Tesla's growth of 35% over the past two years has offered internal career development to our employees as well as the ability to make a meaningful contribution to a sustainable future.

We are able to retain our employees, in part, not only because employees can enjoy ownership in Tesla through stock (of which 89% have been given the opportunity to), but because we also provide them with excellent health benefits such as free counseling, paid parental leave, paid time off and zero-premium medical plan options that are made available on the first day of employment.

We recognize the positive impact that leaders can have on their teams and offer fundamental skills training and continuous development to all leaders through various programs globally.

We don't stop there. Tesla has several other programs strategically designed to increase paths for greater career opportunity such as:

- **Technician Trainee (Service)** – The Tesla Technician Trainee Program provides on-the-job automotive maintenance training at Tesla, resulting in an industry certification. Targeted at individuals with limited experience, whether in industry or vocational schools, the program prepares trainees for employment as technicians. In 2023, we hired over 1,900 Technician Trainees across the U.S., Germany and China.

- **START (Manufacturing and Service)** – Tesla START is an intensive training program that complements the Technician Trainee program and equips individuals with the skills needed for a successful technician role at Tesla. We have partnered with colleges and technical academies to launch Tesla START in the U.S., United Kingdom and Germany. In 2023, we hired over 350 trainees for manufacturing and service roles through this program, providing an opportunity to transition into full-time employment.

- **Internships** – Annually, Tesla hires over 6,000 university and college students from around the world. We recruit from diverse student organizations and campuses, seeking top talent passionate about our mission. Our interns engage in meaningful work from day one, and we often offer them full-time positions post-internship.

- **Military Fellowship and Transition Programs –** The Military Fellowship and Transition Programs are designed to offer exiting military service members in the U.S. and Europe with career guidance on transitioning into the civil workforce. We partner with the career transition services of European Defence Ministries across five countries, as well as the U.S. Chamber of Commerce's Hire our Heroes. These programs aim to convert high-performing individuals to full-time roles and create a veteran talent pipeline.

- **Apprenticeships** – Tesla Apprenticeships are offered globally, providing academic and on-the-job training to prepare specialists in skilled trades. Apprentices will complete between one to four years of on-the-job training. Apprentice programs have seen skilled trade hires across the U.S., Australia, Hong Kong, Korea and Germany.

- **Manufacturing Development Program** – Tesla's manufacturing pathway program is designed to provide graduating high school seniors with the financial resources, coursework and experience they need to start a successful manufacturing career at Tesla. We hired 373 graduates through this program in 2023, and our goal in 2024 is grow this program to over 600 students annually across our Fremont Factory, Gigafactory Nevada, Gigafactory Texas and Gigafactory New York.

- **Engineering Development Program** – Launched in January 2024, this program targets recent college and university graduates for specialized engineering fields. In collaboration with Austin Community College, the program educates early-career engineers in controls engineering, enhancing their knowledge of high-demand technologies for U.S. manufacturing.

We will continue to expand the opportunities for our employees to add skills and develop professionally with a new Employee Educational Assistance Program launching in the U.S. in the spring of 2024 to help employees pursue select certificates or degrees. With virtual, self-paced education options available, employees can pursue a new path or expand their knowledge while continuing to grow their career.

At Tesla, our employees show up passionate about making a difference in the world and for each other. We remain unwavering in our demand that our factories, offices, stores and service centers are places where our employees feel respected and appreciated. Our policies are designed to promote fairness and respect for everyone. We hire, evaluate and promote employees based on their skills and performance. Everyone is expected to be trustworthy, demonstrate excellence in their performance and collaborate with others. With this in mind, we will not tolerate certain behaviors. These include harassment, retaliation, violence, intimidation and discrimination of any kind on the basis of race, color, religion, national origin, gender, sexual orientation, gender identity, gender expression, age, disability or veteran status.

Anti-harassment training is conducted on day one of new hire orientation for all employees and reoccurring for leaders. In addition, we run various leadership development programs throughout the year aimed at enhancing leaders' skills, and in particular, helping them to understand how to appropriately respond to and address employee concerns.

Employees are encouraged to speak up both in regard to misconduct and safety concerns and can do so by contacting the integrity line, submitting concerns through our Take Charge process, or notifying their Human Resource Partner or any member of management. Concerns are reviewed in accordance with established protocols by investigators with expertise, who also review for trends and outcomes for remediation and appropriate controls. Responding to questions timely is key so Human Resource Partners for each functional area are visible throughout facilities and are actively involved in driving culture and engagement alongside business leaders.

Available Information

We file or furnish periodic reports and amendments thereto, including our Annual Reports on Form 10-K, our Quarterly Reports on Form 10-Q and Current Reports on Form 8-K, proxy statements and other information with the SEC. In addition, the SEC maintains a website (www.sec.gov) that contains reports, proxy and information statements, and other information regarding issuers that file electronically. Our website is located at www.tesla.com, and our reports, amendments thereto, proxy statements and other information are also made available, free of charge, on our investor relations website at ir.tesla.com as soon as reasonably practicable after we electronically file or furnish such information with the SEC. The information posted on our website is not incorporated by reference into this Annual Report on Form 10-K.

ITEM 1A. RISK FACTORS

You should carefully consider the risks described below together with the other information set forth in this report, which could materially affect our business, financial condition and future results. The risks described below are not the only risks facing our company. Risks and uncertainties not currently known to us or that we currently deem to be immaterial also may materially adversely affect our business, financial condition and operating results.

Risks Related to Our Ability to Grow Our Business

We may experience delays in launching and ramping the production of our products and features, or we may be unable to control our manufacturing costs.

We have previously experienced and may in the future experience launch and production ramp delays for new products and features. For example, we encountered unanticipated supplier issues that led to delays during the initial ramp of our first Model X and experienced challenges with a supplier and with ramping full automation for certain of our initial Model 3 manufacturing processes. In addition, we may introduce in the future new or unique manufacturing processes and design features for our products. As we expand our vehicle offerings and global footprint, there is no guarantee that we will be able to successfully and timely introduce and scale such processes or features.

In particular, our future business depends in large part on increasing the production of mass-market vehicles. In order to be successful, we will need to implement, maintain and ramp efficient and cost-effective manufacturing capabilities, processes and supply chains and achieve the design tolerances, high quality and output rates we have planned at our manufacturing facilities in California, Nevada, Texas, China, Germany and any future sites such as Mexico. We will also need to hire, train and compensate skilled employees to operate these facilities. Bottlenecks and other unexpected challenges such as those we experienced in the past may arise during our production ramps, and we must address them promptly while continuing to improve manufacturing processes and reducing costs. If we are not successful in achieving these goals, we could face delays in establishing and/or sustaining our product ramps or be unable to meet our related cost and profitability targets.

We have experienced, and may also experience similar future delays in launching and/or ramping production of our energy storage products and Solar Roof; new product versions or variants; new vehicles; and future features and services based on artificial intelligence. Likewise, we may encounter delays with the design, construction and regulatory or other approvals necessary to build and bring online future manufacturing facilities and products.

Any delay or other complication in ramping the production of our current products or the development, manufacture, launch and production ramp of our future products, features and services, or in doing so cost-effectively and with high quality, may harm our brand, business, prospects, financial condition and operating results.

Our suppliers may fail to deliver components according to schedules, prices, quality and volumes that are acceptable to us, or we may be unable to manage these components effectively.

Our products contain thousands of parts purchased globally from hundreds of suppliers, including single-source direct suppliers, which exposes us to multiple potential sources of component shortages. Unexpected changes in business conditions, materials pricing, including inflation of raw material costs, labor issues, wars, trade policies, natural disasters, health epidemics such as the global COVID-19 pandemic, trade and shipping disruptions, port congestions, cyberattacks and other factors beyond our or our suppliers' control could also affect these suppliers' ability to deliver components to us or to remain solvent and operational. For example, a global shortage of semiconductors beginning in early 2021 has caused challenges in the manufacturing industry and impacted our supply chain and production. Additionally, if our suppliers do not accurately forecast and effectively allocate production or if they are not willing to allocate sufficient production to us, or face other challenges such as insolvency, it may reduce our access to components and require us to search for new suppliers. The unavailability of any component or supplier could result in production delays, idle manufacturing facilities, product design changes and loss of access to important technology and tools for producing and supporting our products, as well as impact our capacity expansion and our ability to fulfill our obligations under customer contracts. Moreover, significant increases in our production or product design changes by us have required and may in the future require us to procure additional components in a short amount of time. We have faced in the past, and may face suppliers who are unwilling or unable to sustainably meet our timelines or our cost, quality and volume needs, which may increase our costs or require us to replace them with other sources. Finally, as we construct new manufacturing facilities and add production lines to existing facilities, we may experience issues in correspondingly increasing the level of localized procurement at those facilities. While we believe that we will be able to secure additional or alternate sources or develop our own replacements for most of our components, there is no assurance that we will be able to do so quickly or at all. Additionally, we may be unsuccessful in our continuous efforts to negotiate with existing suppliers to obtain cost reductions and avoid unfavorable changes to terms, source less expensive suppliers for certain parts and redesign certain parts to make them less expensive to produce, especially in the case of increases in materials pricing. Any of these occurrences may harm our business, prospects, financial condition and operating results.

As the scale of our vehicle production increases, we will also need to accurately forecast, purchase, warehouse and transport components at high volumes to our manufacturing facilities and servicing locations internationally. If we are unable to accurately match the timing and quantities of component purchases to our actual needs or successfully implement automation, inventory management and other systems to accommodate the increased complexity in our supply chain and parts management, we may incur unexpected production disruption, storage, transportation and write-off costs, which may harm our business and operating results.

We may be unable to meet our projected construction timelines, costs and production ramps at new factories, or we may experience difficulties in generating and maintaining demand for products manufactured there.

Our ability to increase production of our vehicles on a sustained basis, make them affordable globally by accessing local supply chains and workforces and streamline delivery logistics is dependent on the construction and ramp of our current and future factories. The construction of and commencement and ramp of production at these factories are subject

to a number of uncertainties inherent in all new manufacturing operations, including ongoing compliance with regulatory requirements, procurement and maintenance of construction, environmental and operational licenses and approvals for additional expansion, supply chain constraints, hiring, training and retention of qualified employees and the pace of bringing production equipment and processes online with the capability to manufacture high-quality units at scale. Moreover, we will have to establish and ramp production of our proprietary battery cells and packs at our new factories, and we additionally intend to incorporate sequential design and manufacturing changes into vehicles manufactured at each new factory. If we experience any issues or delays in meeting our projected timelines, costs, capital efficiency and production capacity for our new factories, expanding and managing teams to implement iterative design and production changes there, maintaining and complying with the terms of any debt financing that we obtain to fund them or generating and maintaining demand for the vehicles we manufacture there, our business, prospects, operating results and financial condition may be harmed.

We may be unable to grow our global product sales, delivery and installation capabilities and our servicing and vehicle charging networks, or we may be unable to accurately project and effectively manage our growth.

Our success will depend on our ability to continue to expand our sales capabilities. We are targeting a global mass demographic with a broad range of potential customers, in which we have relatively limited experience projecting demand and pricing our products. We currently produce numerous international variants at a limited number of factories, and if our specific demand expectations for these variants prove inaccurate, we may not be able to timely generate deliveries matched to the vehicles that we produce in the same timeframe or that are commensurate with the size of our operations in a given region. Likewise, as we develop and grow our energy products and services worldwide, our success will depend on our ability to correctly forecast demand in various markets.

Because we do not have independent dealer networks, we are responsible for delivering all of our vehicles to our customers. As our production volumes continue to grow, we have faced in the past, and may face challenges with deliveries at increasing volumes, particularly in international markets requiring significant transit times. We have also deployed a number of delivery models, such as deliveries to customers' homes and workplaces and touchless deliveries, but there is no guarantee that such models will be scalable or be accepted globally. Likewise, as we ramp our energy products, we are working to substantially increase our production and installation capabilities. If we experience production delays or inaccurately forecast demand, our business, financial condition and operating results may be harmed.

Moreover, because of our unique expertise with our vehicles, we recommend that our vehicles be serviced by us or by certain authorized professionals. If we experience delays in adding servicing capacity or servicing our vehicles efficiently, or experience unforeseen issues with the reliability of our vehicles, particularly higher-volume additions to our fleet such as Model 3 and Model Y, it could overburden our servicing capabilities and parts inventory. Similarly, the increasing number of Tesla vehicles also requires us to continue to rapidly increase the number of our Supercharger stations and connectors throughout the world.

There is no assurance that we will be able to ramp our business to meet our sales, delivery, installation, servicing and vehicle charging targets globally, that our projections on which such targets are based will prove accurate or that the pace of growth or coverage of our customer infrastructure network will meet customer expectations. These plans require significant cash investments and management resources and there is no guarantee that they will generate additional sales or installations of our products, or that we will be able to avoid cost overruns or be able to hire additional personnel to support them. As we expand, we will also need to ensure our compliance with regulatory requirements in various jurisdictions applicable to the sale, installation and servicing of our products, the sale or dispatch of electricity related to our energy products and the operation of Superchargers. If we fail to manage our growth effectively, it may harm our brand, business, prospects, financial condition and operating results.

We will need to maintain and significantly grow our access to battery cells, including through the development and manufacture of our own cells, and control our related costs.

We are dependent on the continued supply of lithium-ion battery cells for our vehicles and energy storage products, and we will require substantially more cells to grow our business according to our plans. Currently, we rely on suppliers such as Panasonic and Contemporary Amperex Technology Co. Limited (CATL) for these cells. We have to date fully qualified only a very limited number of such suppliers and have limited flexibility in changing suppliers. Any disruption in the supply of battery cells from our suppliers could limit production of our vehicles and energy storage products. In the long term, we intend to supplement cells from our suppliers with cells manufactured by us, which we believe will be more efficient, manufacturable at greater volumes and more cost-effective than currently available cells. However, our efforts to develop and manufacture such battery cells have required, and may continue to require, significant investments, and there can be no assurance that we will be able to achieve these targets in the timeframes that we have planned or at all. If we are

unable to do so, we may have to curtail our planned vehicle and energy storage product production or procure additional cells from suppliers at potentially greater costs, either of which may harm our business and operating results.

In addition, the cost and mass production of battery cells, whether manufactured by our suppliers or by us, depends in part upon the prices and availability of raw materials such as lithium, nickel, cobalt and/or other metals. The prices for these materials fluctuate and their available supply may be unstable, depending on market conditions and global demand for these materials. For example, as a result of increased global production of electric vehicles and energy storage products, suppliers of these raw materials may be unable to meet our volume needs. Additionally, our suppliers may not be willing or able to reliably meet our timelines or our cost and quality needs, which may require us to replace them with other sources. Any reduced availability of these materials may impact our access to cells and our growth, and any increases in their prices may reduce our profitability if we cannot recoup such costs through increased prices. Moreover, our inability to meet demand and any product price increases may harm our brand, growth, prospects and operating results.

Our future growth and success are dependent upon consumers' demand for electric vehicles and specifically our vehicles in an automotive industry that is generally competitive, cyclical and volatile.

Though we continue to see increased interest and adoption of electric vehicles, if the market for electric vehicles in general and Tesla vehicles in particular does not develop as we expect, develops more slowly than we expect, or if demand for our vehicles decreases in our markets or our vehicles compete with each other, our business, prospects, financial condition and operating results may be harmed.

In addition, electric vehicles still constitute a small percentage of overall vehicle sales. As a result, the market for our vehicles could be negatively affected by numerous factors, such as:

- perceptions about electric vehicle features, quality, safety, performance and cost;
- perceptions about the limited range over which electric vehicles may be driven on a single battery charge, and access to charging facilities;
- competition, including from other types of alternative fuel vehicles, plug-in hybrid electric vehicles and high fuel-economy internal combustion engine vehicles;
- volatility in the cost of oil, gasoline and energy;
- government regulations and economic incentives and conditions; and
- concerns about our future viability.

The target demographics for our vehicles are highly competitive. Sales of vehicles in the automotive industry tend to be cyclical in many markets, which may expose us to further volatility. We also cannot predict the duration or direction of current global trends or their sustained impact on consumer demand. Ultimately, we continue to monitor macroeconomic conditions to remain flexible and to optimize and evolve our business as appropriate, and attempt to accurately project demand and infrastructure requirements globally and deploy our production, workforce and other resources accordingly. Rising interest rates may lead to consumers to increasingly pull back spending, including on our products, which may harm our demand, business and operating results. If we experience unfavorable global market conditions, or if we cannot or do not maintain operations at a scope that is commensurate with such conditions or are later required to or choose to suspend such operations again, our business, prospects, financial condition and operating results may be harmed.

We face strong competition for our products and services from a growing list of established and new competitors.

The worldwide automotive market is highly competitive today and we expect it will become even more so in the future. A significant and growing number of established and new automobile manufacturers, as well as other companies, have entered, or are reported to have plans to enter, the market for electric and other alternative fuel vehicles, including hybrid, plug-in hybrid and fully electric vehicles, as well as the market for self-driving technology and other vehicle applications and software platforms. In some cases, our competitors offer or will offer electric vehicles in important markets such as China and Europe, and/or have announced an intention to produce electric vehicles exclusively at some point in the future. In addition, certain government and economic incentives which provide benefits to manufacturers who assemble domestically or have local suppliers, may provide a greater benefit to our competitors, which could negatively impact our profitability. Many of our competitors have significantly more or better-established resources than we do to devote to the design, development, manufacturing, distribution, promotion, sale and support of their products. Increased competition could result in our lower vehicle unit sales, price reductions, revenue shortfalls, loss of customers and loss of market share, which may harm our business, financial condition and operating results.

We also face competition in our energy generation and storage business from other manufacturers, developers, installers and service providers of competing energy technologies, as well as from large utilities. Decreases in the retail or wholesale prices of electricity from utilities or other renewable energy sources could make our products less attractive to customers and lead to an increased rate of customer defaults.

Risks Related to Our Operations

We may experience issues with lithium-ion cells or other components manufactured at our Gigafactories, which may harm the production and profitability of our vehicle and energy storage products.

Our plan to grow the volume and profitability of our vehicles and energy storage products depends on significant lithium-ion battery cell production, including by our partner Panasonic at Gigafactory Nevada. We also produce several vehicle components at our Gigafactories, such as battery modules and packs and drive units, and manufacture energy storage products. If we are unable to or otherwise do not maintain and grow our respective operations, or if we are unable to do so cost-effectively or hire and retain highly-skilled personnel there, our ability to manufacture our products profitably would be limited, which may harm our business and operating results.

Finally, the high volumes of lithium-ion cells and battery modules and packs manufactured by us and by our suppliers are stored and recycled at our various facilities. Any mishandling of these products may cause disruption to the operation of such facilities. While we have implemented safety procedures related to the handling of the cells, there can be no assurance that a safety issue or fire related to the cells would not disrupt our operations. Any such disruptions or issues may harm our brand and business.

We face risks associated with maintaining and expanding our international operations, including unfavorable and uncertain regulatory, political, economic, tax and labor conditions.

We are subject to legal and regulatory requirements, political uncertainty and social, environmental and economic conditions in numerous jurisdictions, including markets in which we generate significant sales, over which we have little control and which are inherently unpredictable. Our operations in such jurisdictions, particularly as a company based in the U.S., create risks relating to conforming our products to regulatory and safety requirements and charging and other electric infrastructures; organizing local operating entities; establishing, staffing and managing foreign business locations; attracting local customers; navigating foreign government taxes, regulations and permit requirements; enforceability of our contractual rights; trade restrictions, customs regulations, tariffs and price or exchange controls; and preferences in foreign nations for domestically manufactured products. For example, we monitor tax legislation changes on a global basis, including changes arising as a result of the Organization for Economic Cooperation and Development's multi-jurisdictional plan of action to address base erosion and profit shifting. Such conditions may increase our costs, impact our ability to sell our products and require significant management attention, and may harm our business if we are unable to manage them effectively.

Our business may suffer if our products or features contain defects, fail to perform as expected or take longer than expected to become fully functional.

If our products contain design or manufacturing defects, whether relating to our software or hardware, that cause them not to perform as designed or intended or that require repair, or certain features of our vehicles such as new Autopilot or FSD Capability features take longer than expected to become enabled, are legally restricted or become subject to onerous regulation, our ability to develop, market and sell our products and services may be harmed, and we may experience delivery delays, product recalls, allegations of product liability, breach of warranty and related consumer protection claims and significant warranty and other expenses. While we are continuously working to develop and improve our products' capability and performance, there is no guarantee that any incremental changes in the specific software or equipment we deploy in our vehicles over time will not result in initial functional disparities from prior iterations or will perform as forecast in the timeframe we anticipate, or at all. Although we attempt to remedy any issues we observe in our products as effectively and rapidly as possible, such efforts may not be timely, may hamper production or may not completely satisfy our customers. We have performed, and continue to perform, extensive internal testing on our products and features, though, like the rest of the industry, we currently have a limited frame of reference by which to evaluate certain aspects of their long-term quality, reliability, durability and performance characteristics, including exposure to or consequence of external attacks. While we attempt to identify and address or remedy defects we identify pre-production and sale, there may be latent defects that we may be unable to detect or control for in our products, and thereby address, prior to their sale to or installation for customers.

We may be required to defend or insure against product liability claims.

The automobile industry generally experiences significant product liability claims, and as such we face the risk of such claims in the event our vehicles do not perform or are claimed to not have performed as expected. As is true for other automakers, our vehicles have been involved and we expect in the future will be involved in accidents resulting in death or personal injury, and such accidents where Autopilot, Enhanced Autopilot or FSD Capability features are engaged are the subject of significant public attention, especially in light of NHTSA's Standing General Order requiring reports regarding crashes involving vehicles with advanced driver assistance systems. We have experienced, and we expect to continue to face, claims and regulatory scrutiny arising from or related to misuse or claimed failures or alleged misrepresentations of such new technologies that we are pioneering. In addition, the battery packs that we produce make use of lithium-ion cells. On rare occasions, lithium-ion cells can rapidly release the energy they contain by venting smoke and flames in a manner that can ignite nearby materials as well as other lithium-ion cells. While we have designed our battery packs to passively contain any single cell's release of energy without spreading to neighboring cells, there can be no assurance that a field or testing failure of our vehicles or other battery packs that we produce will not occur, in particular due to a high-speed crash. Likewise, as our solar energy systems and energy storage products generate and store electricity, they have the potential to fail or cause injury to people or property. Any product liability claim may subject us to lawsuits and substantial monetary damages, product recalls or redesign efforts, and even a meritless claim may require us to defend it, all of which may generate negative publicity and be expensive and time-consuming. In most jurisdictions, we generally self-insure against the risk of product liability claims for vehicle exposure, meaning that any product liability claims will likely have to be paid from company funds and not by insurance.

We will need to maintain public credibility and confidence in our long-term business prospects in order to succeed.

In order to maintain and grow our business, we must maintain credibility and confidence among customers, suppliers, analysts, investors, ratings agencies and other parties in our long-term financial viability and business prospects. Maintaining such confidence may be challenging due to our limited operating history relative to established competitors; customer unfamiliarity with our products; any delays we may experience in scaling manufacturing, delivery and service operations to meet demand; competition and uncertainty regarding the future of electric vehicles or our other products and services; our quarterly production and sales performance compared with market expectations; and other factors including those over which we have no control. In particular, Tesla's products, business, results of operations, and statements and actions of Tesla and its management are subject to significant amounts of commentary by a range of third parties. Such attention can include criticism, which may be exaggerated or unfounded, such as speculation regarding the sufficiency or stability of our management team. Any such negative perceptions, whether caused by us or not, may harm our business and make it more difficult to raise additional funds if needed.

We may be unable to effectively grow, or manage the compliance, residual value, financing and credit risks related to, our various financing programs.

We offer financing arrangements for our vehicles in North America, Europe and Asia primarily ourselves and through various financial institutions. We also currently offer vehicle financing arrangements directly through our local subsidiaries in certain markets. Depending on the country, such arrangements are available for specified models and may include operating leases directly with us under which we typically receive only a very small portion of the total vehicle purchase price at the time of lease, followed by a stream of payments over the term of the lease. We have also offered various arrangements for customers of our solar energy systems whereby they pay us a fixed payment to lease or finance the purchase of such systems or purchase electricity generated by them. If we do not successfully monitor and comply with applicable national, state and/or local financial regulations and consumer protection laws governing these transactions, we may become subject to enforcement actions or penalties.

The profitability of any directly-leased vehicles returned to us at the end of their leases depends on our ability to accurately project our vehicles' residual values at the outset of the leases, and such values may fluctuate prior to the end of their terms depending on various factors such as supply and demand of our used vehicles, economic cycles and the pricing of new vehicles. We have made in the past and may make in the future certain adjustments to our prices from time to time in the ordinary course of business, which may impact the residual values of our vehicles and reduce the profitability of our vehicle leasing program. The funding and growth of this program also rely on our ability to secure adequate financing and/or business partners. If we are unable to adequately fund our leasing program through internal funds, partners or other financing sources, and compelling alternative financing programs are not available for our customers who may expect or need such options, we may be unable to grow our vehicle deliveries. Furthermore, if our vehicle leasing business grows substantially, our business may suffer if we cannot effectively manage the resulting greater levels of residual risk.

Similarly, we have provided resale value guarantees to vehicle customers and partners for certain financing programs, under which such counterparties may sell their vehicles back to us at certain points in time at pre-determined amounts. However, actual resale values are subject to fluctuations over the term of the financing arrangements, such as from the vehicle pricing changes discussed above. If the actual resale values of any vehicles resold or returned to us pursuant to these programs are materially lower than the pre-determined amounts we have offered, our financial condition and operating results may be harmed.

Finally, our vehicle and solar energy system financing programs and our energy storage sales programs also expose us to customer credit risk. In the event of a widespread economic downturn or other catastrophic event, our customers may be unable or unwilling to satisfy their payment obligations to us on a timely basis or at all. If a significant number of our customers default, we may incur substantial credit losses and/or impairment charges with respect to the underlying assets.

We must manage ongoing obligations under our agreement with the Research Foundation for the State University of New York relating to our Gigafactory New York.

We are party to an operating lease and a research and development agreement through the State University of New York (the "SUNY Foundation"). These agreements provide for the construction and use of our Gigafactory New York, which we have primarily used for the development and production of our Solar Roof and other solar products and components, energy storage components and Supercharger components, and for other lessor-approved functions. Under this agreement, we are obligated to, among other things, meet employment targets as well as specified minimum numbers of personnel in the State of New York and in Buffalo, New York and spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York during a period that was initially 10 years beginning April 30, 2018. As of December 31, 2023, we are currently in excess of such targets relating to investments and personnel in the State of New York and Buffalo. While we expect to have and grow significant operations at Gigafactory New York and the surrounding Buffalo area, any failure by us in any year over the course of the term of the agreement to meet all applicable future obligations may result in our obligation to pay a "program payment" of $41 million to the SUNY Foundation for such year, the termination of our lease at Gigafactory New York which may require us to pay additional penalties, and/or the need to adjust certain of our operations. Any of the foregoing events may harm our business, financial condition and operating results.

If we are unable to attract, hire and retain key employees and qualified personnel, our ability to compete may be harmed.

The loss of the services of any of our key employees or any significant portion of our workforce could disrupt our operations or delay the development, introduction and ramp of our products and services. In particular, we are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer. None of our key employees is bound by an employment agreement for any specific term and we may not be able to successfully attract and retain senior leadership necessary to grow our business. Our future success also depends upon our ability to attract, hire and retain a large number of engineering, manufacturing, marketing, sales and delivery, service, installation, technology and support personnel, especially to support our planned high-volume product sales, market and geographical expansion and technological innovations. If we are not successful in managing these risks, our business, financial condition and operating results may be harmed.

Employees may leave Tesla or choose other employers over Tesla due to various factors, such as a very competitive labor market for talented individuals with automotive or technology experience, or any negative publicity related to us. In regions where we have or will have operations, particularly significant engineering and manufacturing centers, there is strong competition for individuals with skillsets needed for our business, including specialized knowledge of electric vehicles, engineering and electrical and building construction expertise. We also compete with both mature and prosperous companies that have far greater financial resources than we do and start-ups and emerging companies that promise short-term growth opportunities.

Finally, our compensation philosophy for all of our personnel reflects our startup origins, with an emphasis on equity-based awards and benefits in order to closely align their incentives with the long-term interests of our stockholders. We periodically seek and obtain approval from our stockholders for future increases to the number of awards available under our equity incentive and employee stock purchase plans. If we are unable to obtain the requisite stockholder approvals for such future increases, we may have to expend additional cash to compensate our employees and our ability to retain and hire qualified personnel may be harmed.

We are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer.

We are highly dependent on the services of Elon Musk, Technoking of Tesla and our Chief Executive Officer. Although Mr. Musk spends significant time with Tesla and is highly active in our management, he does not devote his full time and attention to Tesla. Mr. Musk also currently serves as Chief Executive Officer and Chief Technical Officer of Space Exploration Technologies Corp., a developer and manufacturer of space launch vehicles, Chairman and Chief Technical Officer of X Corp., a social media company, and is involved in other emerging technology ventures.

Our information technology systems or data, or those of our service providers or customers or users could be subject to cyber-attacks or other security incidents, which could result in data breaches, intellectual property theft, claims, litigation, regulatory investigations, significant liability, reputational damage and other adverse consequences.

We continue to expand our information technology systems as our operations grow, such as product data management, procurement, inventory management, production planning and execution, sales, service and logistics, dealer management, financial, tax and regulatory compliance systems. This includes the implementation of new internally developed systems and the deployment of such systems in the U.S. and abroad. While, we maintain information technology measures designed to protect us against intellectual property theft, data breaches, sabotage and other external or internal cyber-attacks or misappropriation, our systems and those of our service providers are potentially vulnerable to malware, ransomware, viruses, denial-of-service attacks, phishing attacks, social engineering, computer hacking, unauthorized access, exploitation of bugs, defects and vulnerabilities, breakdowns, damage, interruptions, system malfunctions, power outages, terrorism, acts of vandalism, security breaches, security incidents, inadvertent or intentional actions by employees or other third parties, and other cyber-attacks.

To the extent any security incident results in unauthorized access or damage to or acquisition, use, corruption, loss, destruction, alteration or dissemination of our data, including intellectual property and personal information, or our products or vehicles, or for it to be believed or reported that any of these occurred, it could disrupt our business, harm our reputation, compel us to comply with applicable data breach notification laws, subject us to time consuming, distracting and expensive litigation, regulatory investigation and oversight, mandatory corrective action, require us to verify the correctness of database contents, or otherwise subject us to liability under laws, regulations and contractual obligations, including those that protect the privacy and security of personal information. This could result in increased costs to us and result in significant legal and financial exposure and/or reputational harm.

We also rely on service providers, and similar incidents relating to their information technology systems could also have a material adverse effect on our business. There have been and may continue to be significant supply chain attacks. Our service providers, including our workforce management software provider, have been subject to ransomware and other security incidents, and we cannot guarantee that our or our service providers' systems have not been breached or that they do not contain exploitable defects, bugs, or vulnerabilities that could result in a security incident, or other disruption to, our or our service providers' systems. Our ability to monitor our service providers' security measures is limited, and, in any event, malicious third parties may be able to circumvent those security measures.

Further, the implementation, maintenance, segregation and improvement of these systems require significant management time, support and cost, and there are inherent risks associated with developing, improving and expanding our core systems as well as implementing new systems and updating current systems, including disruptions to the related areas of business operation. These risks may affect our ability to manage our data and inventory, procure parts or supplies or manufacture, sell, deliver and service products, adequately protect our intellectual property or achieve and maintain compliance with, or realize available benefits under, tax laws and other applicable regulations.

Moreover, if we do not successfully implement, maintain or expand these systems as planned, our operations may be disrupted, our ability to accurately and/or timely report our financial results could be impaired and deficiencies may arise in our internal control over financial reporting, which may impact our ability to certify our financial results. Moreover, our proprietary information, including intellectual property and personal information, could be compromised or misappropriated and our reputation may be adversely affected. If these systems or their functionality do not operate as we expect them to, we may be required to expend significant resources to make corrections or find alternative sources for performing these functions.

Any unauthorized control or manipulation of our products' systems could result in loss of confidence in us and our products.

Our products contain complex information technology systems. For example, our vehicles and energy storage products are designed with built-in data connectivity to accept and install periodic remote updates from us to improve or update their functionality. While we have implemented security measures intended to prevent unauthorized access to our information technology networks, our products and their systems, malicious entities have reportedly attempted, and may attempt in the future, to gain unauthorized access to modify, alter and use such networks, products and systems to gain control of, or to change, our products' functionality, user interface and performance characteristics or to gain access to data stored in or generated by our products. We encourage reporting of potential vulnerabilities in the security of our products through our security vulnerability reporting policy, and we aim to remedy any reported and verified vulnerability. However, there can be no assurance that any vulnerabilities will not be exploited before they can be identified, or that our remediation efforts are or will be successful.

Any unauthorized access to or control of our products or their systems or any loss of data could result in legal claims or government investigations. In addition, regardless of their veracity, reports of unauthorized access to our products, their systems or data, as well as other factors that may result in the perception that our products, their systems or data are capable of being hacked, may harm our brand, prospects and operating results. We have been the subject of such reports in the past.

Our business may be adversely affected by any disruptions caused by union activities.

It is not uncommon for employees of certain trades at companies such as ours to belong to a union, which can result in higher employee costs and increased risk of work stoppages. Moreover, regulations in some jurisdictions outside of the U.S. mandate employee participation in industrial collective bargaining agreements and work councils with certain consultation rights with respect to the relevant companies' operations. Although we work diligently to provide the best possible work environment for our employees, they may still decide to join or seek recognition to form a labor union, or we may be required to become a union signatory. From time to time, labor unions have engaged in campaigns to organize certain of our operations, as part of which such unions have filed unfair labor practice charges against us with the National Labor Relations Board (the "NLRB"), and they may do so in the future. Any unfavorable ultimate outcome for Tesla may have a negative impact on the perception of Tesla's treatment of our employees. Furthermore, we are directly or indirectly dependent upon companies with unionized work forces, such as suppliers and trucking and freight companies. Any work stoppages or strikes organized by such unions could delay the manufacture and sale of our products and may harm our business and operating results.

We may choose to or be compelled to undertake product recalls or take other similar actions.

As a manufacturing company, we must manage the risk of product recalls with respect to our products. Recalls for our vehicles have resulted from various hardware and software-related safety concerns or non-compliance determinations. In addition to recalls initiated by us for various causes, testing of or investigations into our products by government regulators or industry groups may compel us to initiate product recalls or may result in negative public perceptions about the safety of our products, even if we disagree with the defect determination or have data that contradicts it. In the future, we may voluntarily or involuntarily initiate recalls if any of our products are determined by us or a regulator to contain a safety defect or be noncompliant with applicable laws and regulations, such as U.S. Federal Motor Vehicle Safety Standards. Such recalls, whether voluntary or involuntary or caused by systems or components engineered or manufactured by us or our suppliers, could result in significant expense, supply chain complications and service burdens, and may harm our brand, business, prospects, financial condition and operating results.

Our current and future warranty reserves may be insufficient to cover future warranty claims.

We provide a manufacturer's warranty on all new and used Tesla vehicles we sell directly to customers. We also provide certain warranties with respect to the energy generation and storage systems we sell, including on their installation and maintenance. For components not manufactured by us, we generally pass through to our customers the applicable manufacturers' warranties, but may retain some warranty responsibilities for some or all of the life of such components. As part of our energy generation and storage system contracts, we may provide the customer with performance guarantees that guarantee that the underlying system will meet or exceed the minimum energy generation or other energy performance requirements specified in the contract. Under these performance guarantees, we generally bear the risk of electricity production or other performance shortfalls, including in some cases shortfalls caused by failures in components from third party manufacturers. These risks are exacerbated in the event such manufacturers cease operations or fail to honor their warranties.

If our warranty reserves are inadequate to cover future warranty claims on our products, our financial condition and operating results may be harmed. Warranty reserves include our management's best estimates of the projected costs to repair or to replace items under warranty, which are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. Such estimates are inherently uncertain and changes to our historical or projected experience, especially with respect to products that we have introduced relatively recently and/or that we expect to produce at significantly greater volumes than our past products, may cause material changes to our warranty reserves in the future.

Our insurance coverage strategy may not be adequate to protect us from all business risks.

We may be subject, in the ordinary course of business, to losses resulting from products liability, accidents, acts of God and other claims against us, for which we may have no insurance coverage. As a general matter, we do not maintain as much insurance coverage as many other companies do, and in some cases, we do not maintain any at all. Additionally, the policies that we do have may include significant deductibles or self-insured retentions, policy limitations and exclusions, and we cannot be certain that our insurance coverage will be sufficient to cover all future losses or claims against us. A loss that is uninsured or which exceeds policy limits may require us to pay substantial amounts, which may harm our financial condition and operating results.

Our debt agreements contain covenant restrictions that may limit our ability to operate our business.

The terms of certain of our debt facilities contain, and any of our other future debt agreements may contain, covenant restrictions that may limit our ability to operate our business, including restrictions on our and/or our subsidiaries' ability to, among other things, incur additional debt or create liens. In addition, under certain circumstances we are required to maintain a certain amount of liquidity. As a result of these covenants, our ability to respond to changes in business and economic conditions and engage in beneficial transactions, including to obtain additional financing as needed, may be restricted. Furthermore, our failure to comply with our debt covenants could result in a default under our debt agreements, which could permit the holders to accelerate our obligation to repay the debt. If any of our debt is accelerated, we may not have sufficient funds available to repay it.

Additional funds may not be available to us when we need or want them.

Our business and our future plans for expansion are capital-intensive, and the specific timing of cash inflows and outflows may fluctuate substantially from period to period. We may need or want to raise additional funds through the issuance of equity, equity-related or debt securities or through obtaining credit from financial institutions to fund, together with our principal sources of liquidity, the costs of developing and manufacturing our current or future products, to pay any significant unplanned or accelerated expenses or for new significant strategic investments, or to refinance our significant consolidated indebtedness, even if not required to do so by the terms of such indebtedness. We cannot be certain that additional funds will be available to us on favorable terms when required, or at all. If we cannot raise additional funds when we need them, our financial condition, results of operations, business and prospects could be materially and adversely affected.

We may be negatively impacted by any early obsolescence of our manufacturing equipment.

We depreciate the cost of our manufacturing equipment over their expected useful lives. However, product cycles or manufacturing technology may change periodically, and we may decide to update our products or manufacturing processes more quickly than expected. Moreover, improvements in engineering and manufacturing expertise and efficiency may result in our ability to manufacture our products using less of our currently installed equipment. Alternatively, as we ramp and mature the production of our products to higher levels, we may discontinue the use of already installed equipment in favor of different or additional equipment. The useful life of any equipment that would be retired early as a result would be shortened, causing the depreciation on such equipment to be accelerated, and our results of operations may be harmed.

There is no guarantee that we will have sufficient cash flow from our business to pay our indebtedness or that we will not incur additional indebtedness.

As of December 31, 2023, we and our subsidiaries had outstanding $4.68 billion in aggregate principal amount of indebtedness (see Note 11*Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K). Our consolidated indebtedness may increase our vulnerability to any generally adverse economic and industry conditions. We and our subsidiaries may, subject to the limitations in the terms of our existing and future indebtedness, incur additional debt, secure existing or future debt or recapitalize our debt.

Our ability to make scheduled payments of the principal and interest on our indebtedness when due, to make payments upon conversion or repurchase demands with respect to our convertible senior notes or to refinance our indebtedness as we may need or desire, depends on our future performance, which is subject to economic, financial, competitive and other factors beyond our control. Our business may not continue to generate cash flow from operations in the future sufficient to satisfy our obligations under our existing indebtedness and any future indebtedness we may incur, and to make necessary capital expenditures. If we are unable to generate such cash flow, we may be required to adopt one or more alternatives, such as reducing or delaying investments or capital expenditures, selling assets, refinancing or obtaining additional equity capital on terms that may be onerous or highly dilutive. Our ability to refinance existing or future indebtedness will depend on the capital markets and our financial condition at such time. In addition, our ability to make payments may be limited by law, by regulatory authority or by agreements governing our future indebtedness. We may not be able to engage in these activities on desirable terms or at all, which may result in a default on our existing or future indebtedness and harm our financial condition and operating results.

We are exposed to fluctuations in currency exchange rates.

We transact business globally in multiple currencies and have foreign currency risks related to our revenue, costs of revenue, operating expenses and localized subsidiary debt denominated in currencies other than the U.S. dollar. To the extent we have significant revenues denominated in such foreign currencies, any strengthening of the U.S. dollar would tend to reduce our revenues as measured in U.S. dollars, as we have historically experienced, and are currently experiencing. In addition, a portion of our costs and expenses have been, and we anticipate will continue to be, denominated in foreign currencies. If we do not have fully offsetting revenues in these currencies and if the value of the U.S. dollar depreciates significantly against these currencies, our costs as measured in U.S. dollars as a percent of our revenues will correspondingly increase and our margins will suffer. As a result, our operating results may be harmed.

We may not be able to adequately protect or defend ourselves against intellectual property infringement claims, which may be time-consuming and expensive, or affect the freedom to operate our business.

Our competitors or other third parties may hold or obtain patents, copyrights, trademarks or other proprietary rights that could prevent, limit or interfere with our ability to make, use, develop, sell or market our products and services, which could make it more difficult for us to operate our business. From time to time, the holders of such intellectual property rights may assert their rights and urge us to take licenses and/or may bring suits alleging infringement or misappropriation of such rights, which could result in substantial costs, negative publicity and management attention, regardless of merit.

In addition, the effective protection for our brands, technologies, and proprietary information may be limited or unavailable in certain countries, making it difficult to protect our intellectual property from misappropriation or infringement. Although we make reasonable efforts to maintain the confidentiality of our proprietary information, we cannot guarantee that these actions will deter or prevent misappropriation of our intellectual property. The theft or unauthorized use or publication of our trade secrets and confidential information could affect our competitive position.

While we endeavor to obtain and protect the intellectual property rights that we expect will allow us to retain or advance our strategic initiatives in these circumstances, there can be no assurance that we will be able to adequately identify and protect the portions of intellectual property that are strategic to our business, or mitigate the risk of potential suits or other legal demands by third parties. Accordingly, we may consider the entering into licensing agreements with respect to such rights, although no assurance can be given that such licenses can be obtained on acceptable terms or that litigation will not occur, and such licenses and associated litigation could significantly increase our operating expenses. Further, if we are determined to have or believe there is a high likelihood that we have infringed upon a third party's intellectual property rights, we may be required to cease making, selling or incorporating certain components or intellectual property into the goods and services we offer, to pay substantial damages and/or license royalties, to redesign our products and services and/or to establish and maintain alternative branding for our products and services. In the event that we are required to take one or more such actions, our brand, business, financial condition and operating results may be harmed.

Increased scrutiny and changing expectations from stakeholders with respect to the Company's ESG practices may result in additional costs or risks.

Companies across many industries are facing increasing scrutiny related to their environmental, social and governance (ESG) practices. Investor advocacy groups, certain institutional investors, investment funds and other influential investors are also increasingly focused on ESG practices and in recent years have placed increasing importance on the non-financial impacts of their investments. While our mission is to accelerate the world's transition to sustainable energy, if our ESG practices do not meet investor or other industry stakeholder expectations, which continue to evolve, we may incur additional costs and our brand, ability to attract and retain qualified employees and business may be harmed.

Our operations could be adversely affected by events outside of our control, such as natural disasters, wars or health epidemics.

We may be impacted by natural disasters, wars, health epidemics, weather conditions, the long-term effects of climate change, power outages or other events outside of our control. For example, our Fremont Factory and Gigafactory Nevada are located in seismically active regions in Northern California and Nevada, and our Gigafactory Shanghai is located in a flood-prone area. Moreover, the area in which our Gigafactory Texas is located experienced severe winter storms in the first quarter of 2021 that had a widespread impact on utilities and transportation. If major disasters such as earthquakes, floods or other climate-related events occur, or our information system or communication breaks down or operates improperly, our headquarters and production facilities may be seriously damaged, or we may have to stop or delay production and shipment of our products. In addition, the global COVID-19 pandemic has impacted economic markets, manufacturing operations, supply chains, employment and consumer behavior in nearly every geographic region and industry across the world, and we have been, and may in the future be, adversely affected as a result. Also, the broader consequences in the current conflict between Russia and Ukraine, which may include further embargoes, regional instability and geopolitical shifts; airspace bans relating to certain routes, or strategic decisions to alter certain routes; and potential retaliatory action by the Russian government against companies, and the extent of the conflict on our business and operating results cannot be predicted. We may incur expenses or delays relating to such events outside of our control, which could have a material adverse impact on our business, operating results and financial condition.

Risks Related to Government Laws and Regulations

Demand for our products and services may be impacted by the status of government and economic incentives supporting the development and adoption of such products.

Government and economic incentives that support the development and adoption of electric vehicles in the U.S. and abroad, including certain tax exemptions, tax credits and rebates, may be reduced, eliminated, amended or exhausted from time to time. For example, previously available incentives favoring electric vehicles in certain areas have expired or were cancelled or temporarily unavailable, and in some cases were not eventually replaced or reinstituted, which may have negatively impacted sales. In addition, certain government and economic incentives may also be implemented or amended to provide benefits to manufacturers who assemble domestically, have local suppliers or have other characteristics that may not apply to Tesla. Such developments could negatively impact demand for our vehicles, and we and our customers may have to adjust to them, including through pricing modifications.

In addition, certain governmental rebates, tax credits and other financial incentives that are currently available with respect to our solar and energy storage product businesses allow us to lower our costs and encourage customers to buy our products and investors to invest in our solar financing funds. However, these incentives may expire when the allocated funding is exhausted, reduced or terminated as renewable energy adoption rates increase, sometimes without warning. Likewise, in jurisdictions where net metering is currently available, our customers receive bill credits from utilities for energy that their solar energy systems generate and export to the grid in excess of the electric load they use. The benefit available under net metering has been or has been proposed to be reduced, altered or eliminated in several jurisdictions, and has also been contested and may continue to be contested before the Federal Energy Regulatory Commission. Any reductions or terminations of such incentives may harm our business, prospects, financial condition and operating results by making our products less competitive for customers, increasing our cost of capital and adversely impacting our ability to attract investment partners and to form new financing funds for our solar and energy storage assets.

Finally, we and our fund investors claim these U.S. federal tax credits and certain state incentives in amounts based on independently appraised fair market values of our solar and energy storage systems. Some governmental authorities have audited such values and in certain cases have determined that these values should be lower, and they may do so again in the future. Such determinations may result in adverse tax consequences and/or our obligation to make indemnification or other payments to our funds or fund investors.

We are subject to evolving laws and regulations that could impose substantial costs, legal prohibitions or unfavorable changes upon our operations or products.

As we grow our manufacturing operations in additional regions, we are or will be subject to complex environmental, manufacturing, health and safety laws and regulations at numerous jurisdictional levels in the U.S., China, Germany and other locations abroad, including laws relating to the use, handling, storage, recycling, disposal and/or human exposure to hazardous materials, product material inputs and post-consumer products and with respect to constructing, expanding and maintaining our facilities. New, or changes in, environmental and climate change laws, regulations or rules could also lead to increased costs of compliance, including remediations of any discovered issues, and changes to our operations, which may be significant, and any failures to comply could result in significant expenses, delays or fines. In addition, as we have

increased our employee headcount and operations, we are and may continue to be subject to increased scrutiny, including litigation and government investigations, that we will need to defend against. If we are unable to successfully defend ourselves in such litigation or government investigations, it may harm our brand, ability to attract and retain qualified employees, business and financial condition. We are also subject to laws and regulations applicable to the supply, manufacture, import, sale, service and performance of our products both domestically and abroad. For example, in countries outside of the U.S., we are required to meet standards relating to vehicle safety, fuel economy and emissions that are often materially different from equivalent requirements in the U.S., thus resulting in additional investment into the vehicles and systems to ensure regulatory compliance in all countries. This process may include official review and certification of our vehicles by foreign regulatory agencies prior to market entry, as well as compliance with foreign reporting and recall management systems requirements.

In particular, we offer in our vehicles in certain markets Autopilot and FSD Capability features that today assist drivers with certain tedious and potentially dangerous aspects of road travel, but which currently require drivers to remain fully engaged in the driving operation. We are continuing to develop our Autopilot and FSD Capability technology. There are a variety of international, federal and state regulations that may apply to, and may adversely affect, the design and performance, sale, marketing, registration and operation of Autopilot and FSD Capability, and future capability, including full self-driving vehicles that may not be operated by a human driver. This includes many existing vehicle standards that were not originally intended to apply to vehicles that may not be operated by a human driver. Such regulations continue to rapidly change, which increases the likelihood of a patchwork of complex or conflicting regulations, or may delay, restrict or prohibit the availability of certain functionalities and vehicle designs, which could adversely affect our business.

Finally, as a manufacturer, installer and service provider with respect to solar generation and energy storage systems, a supplier of electricity generated and stored by certain of the solar energy and energy storage systems we install for customers, and a provider of grid services through virtual power plant models, we are impacted by federal, state and local regulations and policies concerning the import or export of components, electricity pricing, the interconnection of electricity generation and storage equipment with the electrical grid and the sale of electricity generated by third party-owned systems. If regulations and policies are introduced that adversely impact the import or export of components, or the interconnection, maintenance or use of our solar and energy storage systems, they could deter potential customers from purchasing our solar and energy storage products and services, threaten the economics of our existing contracts and cause us to cease solar and energy storage system sales and services in the relevant jurisdictions, which may harm our business, financial condition and operating results.

Any failure by us to comply with a variety of U.S. and international privacy and consumer protection laws may harm us.

Any failure by us or our vendors or other business partners to comply with our public privacy notice or with federal, state or international privacy, data protection or security laws or regulations relating to the processing, collection, use, retention, security and transfer of personally identifiable information could result in regulatory or litigation-related actions against us, legal liability, fines, damages, ongoing audit requirements and other significant costs. Substantial expenses and operational changes may be required in connection with maintaining compliance with such laws, and even an unsuccessful challenge by customers or regulatory authorities of our activities could result in adverse publicity and could require a costly response from and defense by us. In addition, certain privacy laws are still subject to a high degree of uncertainty as to their interpretation, application and impact, and may require extensive system and operational changes, be difficult to implement, increase our operating costs, adversely impact the cost or attractiveness of the products or services we offer, or result in adverse publicity and harm our reputation. For example, the General Data Protection Regulation applies to the processing of personal information collected from individuals located in the European Union requiring certain data protection measures when handling, with a significant risk of fines for noncompliance. Similarly, our North American operations are subject to complex and changing federal and US state-specific data privacy laws and regulations, such as the California Consumer Privacy Act which imposes certain legal obligations on our use and processing of personal information related to California residents. Finally, additional privacy and cybersecurity laws have come into effect in China.

These laws continue to develop and may be inconsistent from jurisdiction to jurisdiction. Complying with emerging and changing requirements may cause us to incur substantial costs and make enhancements to relevant data practices. Noncompliance could result in significant penalties or legal liability.

In addition to the risks related to general privacy regulation, we may also be subject to specific vehicle manufacturer obligations relating to cybersecurity, data privacy and data localization requirements which place additional risks to our international operations. Risks and penalties could include ongoing audit requirements, data protection

authority investigations, legal proceedings by international governmental entities or others resulting in mandated disclosure of sensitive data or other commercially unfavorable terms. Notwithstanding our efforts to protect the security and integrity of our customers' personal information, we may be required to expend significant resources to comply with data breach requirements if, for example, third parties improperly obtain and use the personal information of our customers or we otherwise experience a data loss with respect to the personal information we process and handle. A major breach of our network security and systems may occur despite defensive measures, and may result in fines, penalties and damages and harm our brand, prospects and operating results.

We could be subject to liability, penalties and other restrictive sanctions and adverse consequences arising out of certain governmental investigations and proceedings.

We are cooperating with certain government investigations as discussed in Note 15,*Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K. To our knowledge, no government agency in any such ongoing investigation has concluded that any wrongdoing occurred. However, we cannot predict the outcome or impact of any such ongoing matters, and there exists the possibility that we could be subject to liability, penalties and other restrictive sanctions and adverse consequences if the SEC, the U.S. Department of Justice or any other government agency were to pursue legal action in the future. Moreover, we expect to incur costs in responding to related requests for information and subpoenas, and if instituted, in defending against any governmental proceedings.

We may face regulatory challenges to or limitations on our ability to sell vehicles directly.

While we intend to continue to leverage our most effective sales strategies, including sales through our website, we may not be able to sell our vehicles through our own stores in certain states in the U.S. with laws that may be interpreted to impose limitations on this direct-to-consumer sales model. It has also been asserted that the laws in some states limit our ability to obtain dealer licenses from state motor vehicle regulators, and such assertions persist. In certain locations, decisions by regulators permitting us to sell vehicles have been, and may be, challenged by dealer associations and others as to whether such decisions comply with applicable state motor vehicle industry laws. We have prevailed in many of these lawsuits and such results have reinforced our continuing belief that state franchise laws were not intended to apply to a manufacturer that does not have franchise dealers anywhere in the world. In some states, there have also been regulatory and legislative efforts by dealer associations to propose laws that, if enacted, would prevent us from obtaining dealer licenses in their states given our current sales model. A few states have passed legislation that clarifies our ability to operate, but at the same time limits the number of dealer licenses we can obtain or stores that we can operate. The application of state laws applicable to our operations continues to be difficult to predict.

Internationally, there may be laws in jurisdictions we have not yet entered or laws we are unaware of in jurisdictions we have entered that may restrict our sales or other business practices. Even for those jurisdictions we have analyzed, the laws in this area can be complex, difficult to interpret and may change over time. Continued regulatory limitations and other obstacles interfering with our ability to sell vehicles directly to consumers may harm our financial condition and operating results.

Risks Related to the Ownership of Our Common Stock

The trading price of our common stock is likely to continue to be volatile.

The trading price of our common stock has been highly volatile and could continue to be subject to wide fluctuations in response to various factors, some of which are beyond our control. Our common stock has experienced over the last 52 weeks an intra-day trading high of $299.29 per share and a low of $152.37 per share. The stock market in general, and the market for technology companies in particular, has experienced extreme price and volume fluctuations that have often been unrelated or disproportionate to the operating performance of those companies. In particular, a large proportion of our common stock has been historically and may in the future be traded by short sellers which may put pressure on the supply and demand for our common stock, further influencing volatility in its market price. Public perception of our company or management and other factors outside of our control may additionally impact the stock price of companies like us that garner a disproportionate degree of public attention, regardless of actual operating performance. In addition, in the past, following periods of volatility in the overall market or the market price of our shares, securities class action litigation has been filed against us. While we defend such actions vigorously, any judgment against us or any future stockholder litigation could result in substantial costs and a diversion of our management's attention and resources.

Our financial results may vary significantly from period to period due to fluctuations in our operating costs and other factors.

We expect our period-to-period financial results to vary based on our operating costs, which we anticipate will fluctuate as the pace at which we continue to design, develop and manufacture new products and increase production capacity by expanding our current manufacturing facilities and adding future facilities, may not be consistent or linear between periods. Additionally, our revenues from period to period may fluctuate as we introduce existing products to new markets for the first time and as we develop and introduce new products. As a result of these factors, we believe that quarter-to-quarter comparisons of our financial results, especially in the short term, are not necessarily meaningful and that these comparisons cannot be relied upon as indicators of future performance. Moreover, our financial results may not meet expectations of equity research analysts, ratings agencies or investors, who may be focused only on short-term quarterly financial results. If any of this occurs, the trading price of our stock could fall substantially, either suddenly or over time.

We may fail to meet our publicly announced guidance or other expectations about our business, which could cause our stock price to decline.

We provide from time to time guidance regarding our expected financial and business performance. Correctly identifying key factors affecting business conditions and predicting future events is inherently an uncertain process, and our guidance may not ultimately be accurate and has in the past been inaccurate in certain respects, such as the timing of new product manufacturing ramps. Our guidance is based on certain assumptions such as those relating to anticipated production and sales volumes (which generally are not linear throughout a given period), average sales prices, supplier and commodity costs and planned cost reductions. If our guidance varies from actual results, such as due to our assumptions not being met or the impact on our financial performance that could occur as a result of various risks and uncertainties, the market value of our common stock could decline significantly.

If Elon Musk were forced to sell shares of our common stock, either that he has pledged to secure certain personal loan obligations, or in satisfaction of other obligations, such sales could cause our stock price to decline.

Certain banking institutions have made extensions of credit to Elon Musk, our Chief Executive Officer, a portion of which was used to purchase shares of common stock in certain of our public offerings and private placements at the same prices offered to third-party participants in such offerings and placements. We are not a party to these loans, which are partially secured by pledges of a portion of the Tesla common stock currently owned by Mr. Musk. If the price of our common stock were to decline substantially, Mr. Musk may be forced by one or more of the banking institutions to sell shares of Tesla common stock to satisfy his loan obligations if he could not do so through other means. Any such sales could cause the price of our common stock to decline further. Further, Mr. Musk from time to time may commit to investing in significant business or other ventures, and as a result, be required to sell shares of our common stock in satisfaction of such commitments.

Anti-takeover provisions contained in our governing documents, applicable laws and our convertible senior notes could impair a takeover attempt.

Our certificate of incorporation and bylaws afford certain rights and powers to our board of directors that may facilitate the delay or prevention of an acquisition that it deems undesirable. We are also subject to Section 203 of the Delaware General Corporation Law and other provisions of Delaware law that limit the ability of stockholders in certain situations to effect certain business combinations. In addition, the terms of our convertible senior notes may require us to repurchase such notes in the event of a fundamental change, including a takeover of our company. Any of the foregoing provisions and terms that has the effect of delaying or deterring a change in control could limit the opportunity for our stockholders to receive a premium for their shares of our common stock, and could also affect the price that some investors are willing to pay for our common stock.

ITEM 1B. UNRESOLVED STAFF COMMENTS

None.

ITEM 1C. CYBERSECURITY

Cybersecurity Risk Management and Strategy

We recognize the importance of assessing, identifying, and managing material risks associated with cybersecurity threats, as such term is defined in Item 106(a) of Regulation S-K. These risks include, among other things: operational risks, intellectual property theft, fraud, extortion, harm to employees or customers and violation of data privacy or security laws.

Identifying and assessing cybersecurity risk is integrated into our overall risk management systems and processes. Cybersecurity risks related to our business, technical operations, privacy and compliance issues are identified and addressed through a multi-faceted approach including third party assessments, internal IT Audit, IT security, governance, risk and compliance reviews. To defend, detect and respond to cybersecurity incidents, we, among other things: conduct proactive privacy and cybersecurity reviews of systems and applications, audit applicable data policies, perform penetration testing using external third-party tools and techniques to test security controls, operate a bug bounty program to encourage proactive vulnerability reporting, conduct employee training, monitor emerging laws and regulations related to data protection and information security (including our consumer products) and implement appropriate changes.

We have implemented incident response and breach management processes which have four overarching and interconnected stages: 1) preparation for a cybersecurity incident, 2) detection and analysis of a security incident, 3) containment, eradication and recovery, and 4) post-incident analysis. Such incident responses are overseen by leaders from our Information Security, Product Security, Compliance and Legal teams regarding matters of cybersecurity.

Security events and data incidents are evaluated, ranked by severity and prioritized for response and remediation. Incidents are evaluated to determine materiality as well as operational and business impact, and reviewed for privacy impact.

We also conduct tabletop exercises to simulate responses to cybersecurity incidents. Our team of cybersecurity professionals then collaborate with technical and business stakeholders across our business units to further analyze the risk to the company, and form detection, mitigation and remediation strategies.

As part of the above processes, we regularly engage external auditors and consultants to assess our internal cybersecurity programs and compliance with applicable practices and standards. As of 2023, our Information Security Management System has been certified to conform to the requirements of ISO/IEC 27001:2013.

Our risk management program also assesses third party risks, and we perform third-party risk management to identify and mitigate risks from third parties such as vendors, suppliers, and other business partners associated with our use of third-party service providers. Cybersecurity risks are evaluated when determining the selection and oversight of applicable third-party service providers and potential fourth-party risks when handling and/or processing our employee, business or customer data. In addition to new vendor onboarding, we perform risk management during third-party cybersecurity compromise incidents to identify and mitigate risks to us from third-party incidents.

We describe whether and how risks from identified cybersecurity threats, including as a result of any previous cybersecurity incidents, have materially affected or are reasonably likely to materially affect us, including our business strategy, results of operations, or financial condition, under the heading “Our information technology systems or data, or those of our service providers or customers or users could be subject to cyber-attacks or other security incidents, which could result in data breaches, intellectual property theft, claims, litigation, regulatory investigations, significant liability, reputational damage and other adverse consequences” included as part of our risk factor disclosures at Item 1A of this Annual Report on Form 10-K.

Cybersecurity Governance

Cybersecurity is an important part of our risk management processes and an area of focus for our Board and management. Our Audit Committee is responsible for the oversight of risks from cybersecurity threats. Members of the Audit Committee receive updates on a quarterly basis from senior management, including leaders from our Information Security, Product Security, Compliance and Legal teams regarding matters of cybersecurity. This includes existing and new cybersecurity risks, status on how management is addressing and/or mitigating those risks, cybersecurity and data privacy incidents (if any) and status on key information security initiatives. Our Board members also engage in ad hoc conversations with management on cybersecurity-related news events and discuss any updates to our cybersecurity risk management and strategy programs.

Our cybersecurity risk management and strategy processes are overseen by leaders from our Information Security, Product Security, Compliance and Legal teams. Such individuals have an average of over 15 years of prior work experience in various roles involving information technology, including security, auditing, compliance, systems and programming. These individuals are informed about, and monitor the prevention, mitigation, detection and remediation of cybersecurity incidents through their management of, and participation in, the cybersecurity risk management and strategy processes described above, including the operation of our incident response plan, and report to the Audit Committee on any appropriate items.

ITEM 2. PROPERTIES

We are headquartered in Austin, Texas. Our principal facilities include a large number of properties in North America, Europe and Asia utilized for manufacturing and assembly, warehousing, engineering, retail and service locations, Supercharger sites and administrative and sales offices. Our facilities are used to support both of our reporting segments, and are suitable and adequate for the conduct of our business. We generally lease such facilities with the primary exception of some manufacturing facilities. The following table sets forth the location of our primary owned and leased manufacturing facilities.

Primary Manufacturing Facilities	Location	Owned or Leased
Gigafactory Texas	Austin, Texas	Owned
Fremont Factory	Fremont, California	Owned
Gigafactory Nevada	Sparks, Nevada	Owned
Gigafactory Berlin-Brandenburg	Grunheide, Germany	Owned
Gigafactory Shanghai	Shanghai, China	*
Gigafactory New York	Buffalo, New York	Leased
Megafactory	Lathrop, California	Leased

* We own the building and the land use rights with an initial term of 50 years. The land use rights are treated as operating lease right-of-use assets.

ITEM 3. LEGAL PROCEEDINGS

For a description of our material pending legal proceedings, please see Note 15,*Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

In addition, each of the matters below is being disclosed pursuant to Item 103 of Regulation S-K because it relates to environmental regulations and aggregate civil penalties that we currently believe could potentially exceed $1 million. We believe that any proceeding that is material to our business or financial condition is likely to have potential penalties far in excess of such amount.

District attorneys in certain California counties conducted an investigation into Tesla's waste segregation practices pursuant to Cal. Health & Saf. Code § 25100 et seq. and Cal. Civil Code § 1798.80. Tesla has implemented various remedial measures, including conducting training and audits, and enhancements to its site waste management programs, and settlement discussions are ongoing. While the outcome of this matter cannot be determined at this time, it is not currently expected to have a material adverse impact on our business.

ITEM 4. MINE SAFETY DISCLOSURES

Not applicable.

PART II

ITEM 5. MARKET FOR REGISTRANT'S COMMON EQUITY, RELATED STOCKHOLDER MATTERS AND ISSUER PURCHASES OF EQUITY SECURITIES

Market Information

Our common stock has traded on The NASDAQ Global Select Market under the symbol "TSLA" since it began trading on June 29, 2010. Our initial public offering was priced at approximately $1.13 per share on June 28, 2010 as adjusted to give effect to the three-for-one stock split effected in the form of a stock dividend in August 2022 (the "2022 Stock Split") and the five-for-one stock split effected in the form of a stock dividend in August 2020 (the "2020 Stock Split").

Holders

As of January 22, 2024, there were 9,300 holders of record of our common stock. A substantially greater number of holders of our common stock are "street name" or beneficial holders, whose shares are held by banks, brokers and other financial institutions.

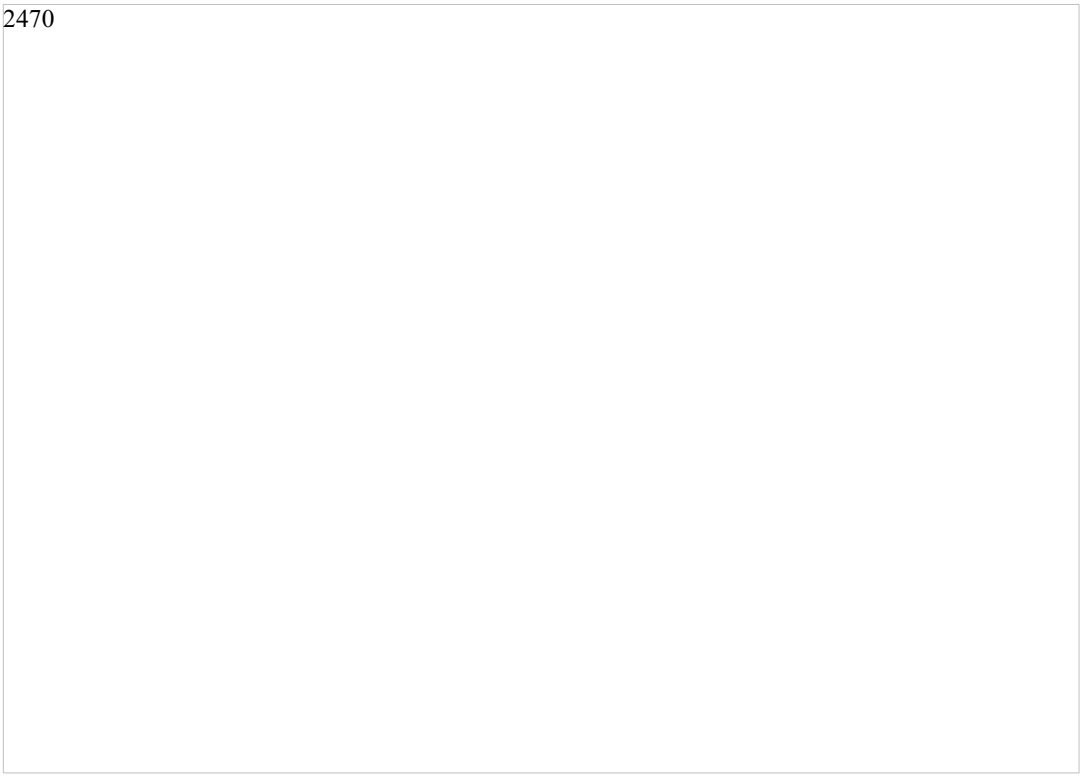
Dividend Policy

We have never declared or paid cash dividends on our common stock. We currently do not anticipate paying any cash dividends in the foreseeable future. Any future determination to declare cash dividends will be made at the discretion of our board of directors, subject to applicable laws, and will depend on our financial condition, results of operations, capital requirements, general business conditions and other factors that our board of directors may deem relevant.

Stock Performance Graph

This performance graph shall not be deemed "filed" for purposes of Section 18 of the Securities Exchange Act of 1934, as amended (the "Exchange Act"), or incorporated by reference into any filing of Tesla, Inc. under the Securities Act of 1933, as amended (the "Securities Act"), or the Exchange Act, except as shall be expressly set forth by specific reference in such filing.

The following graph shows a comparison, from January 1, 2019 through December 31, 2023, of the cumulative total return on our common stock, The NASDAQ Composite Index and a group of all public companies sharing the same SIC code as us, which is SIC code 3711, "Motor Vehicles and Passenger Car Bodies" (Motor Vehicles and Passenger Car Bodies Public Company Group). Such returns are based on historical results and are not intended to suggest future performance. Data for The NASDAQ Composite Index and the Motor Vehicles and Passenger Car Bodies Public Company Group assumes an investment of $100 on January 1, 2019 and reinvestment of dividends. We have never declared or paid cash dividends on our common stock nor do we anticipate paying any such cash dividends in the foreseeable future.



Unregistered Sales of Equity Securities and Use of Proceeds

None.

Purchases of Equity Securities by the Issuer and Affiliated Purchasers

None.

ITEM 6. [RESERVED]

ITEM 7. MANAGEMENT'S DISCUSSION AND ANALYSIS OF FINANCIAL CONDITION AND RESULTS OF OPERATIONS

The following discussion and analysis should be read in conjunction with the consolidated financial statements and the related notes included elsewhere in this Annual Report on Form 10-K. For further discussion of our products and services, technology and competitive strengths, refer to Item 1- Business. For discussion related to changes in financial condition and the results of operations for fiscal year 2022-related items, refer to Part II, Item 7. Management's Discussion and Analysis of Financial Condition and Results of Operations in our Annual Report on Form 10-K for fiscal year 2022, which was filed with the Securities and Exchange Commission on January 31, 2023.

Overview and 2023 Highlights

Our mission is to accelerate the world's transition to sustainable energy. We design, develop, manufacture, lease and sell high-performance fully electric vehicles, solar energy generation systems and energy storage products. We also offer maintenance, installation, operation, charging, insurance, financial and other services related to our products. Additionally, we are increasingly focused on products and services based on artificial intelligence, robotics and automation.

In 2023, we produced 1,845,985 consumer vehicles and delivered 1,808,581 consumer vehicles. We are currently focused on increasing vehicle production, capacity and delivery capabilities, reducing costs, improving and developing our vehicles and battery technologies, vertically integrating and localizing our supply chain, improving and further deploying our FSD capabilities, increasing the affordability and efficiency of our vehicles, bringing new products to market and expanding our global infrastructure, including our service and charging infrastructure.

In 2023, we deployed 14.72 GWh of energy storage products and 223 megawatts of solar energy systems. We are currently focused on ramping production of energy storage products, improving our Solar Roof installation capability and efficiency, and increasing market share of retrofit solar energy systems.

In 2023, we recognized total revenues of $96.77 billion, representing an increase of $15.31 billion, compared to the prior year. We continue to ramp production, build new manufacturing capacity and expand our operations to enable increased deliveries and deployments of our products, and invest in research and development to accelerate our AI, software and fleet-based profits for further revenue growth.

In 2023, our net income attributable to common stockholders was $15.00 billion, representing a favorable change of $2.44 billion, compared to the prior year. This included a one-time non-cash tax benefit of $5.93 billion for the release of valuation allowance on certain deferred tax assets. We continue to focus on further cost reductions and operational efficiencies while maximizing delivery volumes.

We ended 2023 with $29.09 billion in cash and cash equivalents and investments, representing an increase of $6.91 billion from the end of 2022. Our cash flows provided by operating activities in 2023 and 2022 were $13.26 billion and $14.72 billion, respectively, representing a decrease of $1.47 billion. Capital expenditures amounted to $8.90 billion in 2023, compared to $7.16 billion in 2022, representing an increase of $1.74 billion. Sustained growth has allowed our business to generally fund itself, and we will continue investing in a number of capital-intensive projects and research and development in upcoming periods.

Management Opportunities, Challenges and Uncertainties and 2024 Outlook

Automotive—Production

The following is a summary of the status of production of each of our announced vehicle models in production and under development, as of the date of this Annual Report on Form 10-K:

Production Location	Vehicle Model(s)	Production Status
Fremont Factory	Model S / Model X	Active
	Model 3 / Model Y	Active
Gigafactory Shanghai	Model 3 / Model Y	Active
Gigafactory Berlin-Brandenburg	Model Y	Active
Gigafactory Texas	Model Y	Active
	Cybertruck	Active
Gigafactory Nevada	Tesla Semi	Pilot production
Various	Next Generation Platform	In development
TBD	Tesla Roadster	In development

We are focused on growing our manufacturing capacity, which includes capacity for manufacturing new vehicle models such as our Cybertruck and next generation platform, and ramping all of our production vehicles to their installed production capacities as well as increasing production rate and efficiency at our current factories. The next phase of production growth will depend on the continued ramp at our factories and the introduction of our next generation platform, as well as our ability to add to our available sources of battery cell supply by manufacturing our own cells that we are developing to have high-volume output, lower capital and production costs and longer range. Our goals are to improve vehicle performance, decrease production costs and increase affordability and customer awareness.

These plans are subject to uncertainties inherent in establishing and ramping manufacturing operations, which may be exacerbated by new product and manufacturing technologies we introduce, the number of concurrent international projects, any industry-wide component constraints, labor shortages and any future impact from events outside of our control. For example, during the third quarter of 2023, we experienced a sequential decline in production volumes due to pre-planned shutdowns for upgrades at various factories. Moreover, we have set ambitious technological targets with our plans for battery cells as well as for iterative manufacturing and design improvements for our vehicles with each new factory.

Automotive—Demand, Sales, Deliveries and Infrastructure

Our cost reduction efforts, cost innovation strategies, and additional localized procurement and manufacturing are key to our vehicles' affordability and have allowed us to competitively price our vehicles. We will also continue to generate demand and brand awareness by improving our vehicles' performance and functionality, including through products based on artificial intelligence such as Autopilot, FSD Capability, and other software features and delivering new vehicles, such as our Cybertruck. Moreover, we expect to continue to benefit from ongoing electrification of the automotive sector and increasing environmental regulations and initiatives.

However, we operate in a cyclical industry that is sensitive to political and regulatory uncertainty, including with respect to trade and the environment, all of which can be compounded by inflationary pressures, rising energy prices, interest rate fluctuations and the liquidity of enterprise customers. For example, inflationary pressures have increased across the markets in which we operate. In an effort to curb this trend, central banks in developed countries raised interest rates rapidly and substantially, impacting the affordability of vehicle lease and finance arrangements. Further, sales of vehicles in the automotive industry also tend to be cyclical in many markets, which may expose us to increased volatility as we expand and adjust our operations. Moreover, as additional competitors enter the marketplace and help bring the world closer to sustainable transportation, we will have to adjust and continue to execute well to maintain our momentum. Additionally, our suppliers' liquidity and allocation plans may be affected by current challenges in the North American automotive industry, which could reduce our access to components or result in unfavorable changes to cost. These macroeconomic and industry trends have had, and will likely continue to have, an impact on the pricing of, and order rate for our vehicles, and in turn our operating margin. Changes in government and economic incentives in relation to electric vehicles may also impact our sales. We will continue to adjust accordingly to such developments, and we believe our ongoing cost reduction, including improved production innovation and efficiency at our newest factories and lower logistics costs, and focus on operating leverage will continue to benefit us in relation to our competitors, while our new products will help enable future growth.

As our production increases, we must work constantly to similarly increase vehicle delivery capability so that it does not become a bottleneck on our total deliveries. We are also committed to reducing the percentage of vehicles delivered in the third month of each quarter, which will help to reduce the cost per vehicle. As we expand our manufacturing operations globally, we will also have to continue to increase and staff our delivery, servicing and charging infrastructure accordingly, maintain our vehicle reliability and optimize our Supercharger locations to ensure cost effectiveness and customer satisfaction. In particular, as other automotive manufacturers have announced their adoption of the North American Charging Standard ("NACS") and agreements with us to utilize our Superchargers, we must correspondingly expand our network in order to ensure adequate availability to meet customer demands. We also remain focused on continued enhancements of the capability and efficiency of our servicing operations.

Energy Generation and Storage Demand, Production and Deployment

The long-term success of this business is dependent upon increasing margins through greater volumes. We continue to increase the production of our energy storage products to meet high levels of demand, including the construction of a new Megafactory in Shanghai and the ongoing ramp at our Megafactory in Lathrop, California. For Megapack, energy storage deployments can vary meaningfully quarter to quarter depending on the timing of specific project milestones. We remain committed to growing our retrofit solar energy business by offering a low-cost and simplified online ordering experience. In addition, we continue to seek to improve our installation capabilities and price efficiencies for Solar Roof. As these product lines grow, we will have to maintain adequate battery cell supply for our energy storage products and ensure the availability of qualified personnel, particularly skilled electricians, to support the ramp of Solar Roof.

Cash Flow and Capital Expenditure Trends

Our capital expenditures are typically difficult to project beyond the short-term given the number and breadth of our core projects at any given time, and may further be impacted by uncertainties in future global market conditions. We are simultaneously ramping new products, building or ramping manufacturing facilities on three continents, piloting the development and manufacture of new battery cell technologies, expanding our Supercharger network and investing in autonomy and other artificial intelligence enabled training and products, and the pace of our capital spend may vary depending on overall priority among projects, the pace at which we meet milestones, production adjustments to and among our various products, increased capital efficiencies and the addition of new projects. Owing and subject to the foregoing as well as the pipeline of announced projects under development, all other continuing infrastructure growth and varying levels of inflation, we currently expect our capital expenditures to exceed $10.00 billion in 2024 and be between $8.00 to $10.00 billion in each of the following two fiscal years.

Our business has been consistently generating cash flow from operations in excess of our level of capital spend, and with better working capital management resulting in shorter days sales outstanding than days payable outstanding, our sales growth is also generally facilitating positive cash generation. We have and will continue to utilize such cash flows, among other things, to do more vertical integration, expand our product roadmap and provide financing options to our customers. At the same time, we are likely to see heightened levels of capital expenditures during certain periods depending on the specific pace of our capital-intensive projects and other potential variables such as rising material prices and increases in supply chain and labor expenses resulting from changes in global trade conditions and labor availability. Overall, we expect our ability to be self-funding to continue as long as macroeconomic factors support current trends in our sales.

Critical Accounting Policies and Estimates

The consolidated financial statements are prepared in accordance with accounting principles generally accepted in the U.S. ("GAAP"). The preparation of the consolidated financial statements requires us to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues, costs and expenses and related disclosures. We base our estimates on historical experience, as appropriate, and on various other assumptions that we believe to be reasonable under the circumstances. Changes in the accounting estimates are reasonably likely to occur from period to period. Accordingly, actual results could differ significantly from the estimates made by our management. We evaluate our estimates and assumptions on an ongoing basis. To the extent that there are material differences between these estimates and actual results, our future financial statement presentation, financial condition, results of operations and cash flows may be affected.

The estimates used for, but not limited to, determining significant economic incentive for resale value guarantee arrangements, sales return reserves, the collectability of accounts and financing receivables, inventory valuation, warranties, fair value of long-lived assets, goodwill, fair value of financial instruments, fair value and residual value of operating lease vehicles and solar energy systems subject to leases could be impacted. We have assessed the impact and are not aware of any specific events or circumstances that required an update to our estimates and assumptions or materially affected the carrying value of our assets or liabilities as of the date of issuance of this Annual Report on Form 10-K. These estimates may change as new events occur and additional information is obtained. Actual results could differ materially from these estimates under different assumptions or conditions.

Revenue Recognition

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under Accounting Standards Codification ("ASC") 606, *Revenue from Contracts with Customers* ("ASC 606"), including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD Capability features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Inventory Valuation

Inventories are stated at the lower of cost or net realizable value. Cost is computed using standard cost for vehicles and energy products, which approximates actual cost on a first-in, first-out basis. We record inventory write-downs for excess or obsolete inventories based upon assumptions about current and future demand forecasts. If our inventory on-hand is in excess of our future demand forecast, the excess amounts are written-off.

We also review our inventory to determine whether its carrying value exceeds the net amount realizable upon the ultimate sale of the inventory. This requires us to determine the estimated selling price of our vehicles less the estimated cost to convert the inventory on-hand into a finished product. Once inventory is written-down, a new, lower cost basis for that inventory is established and subsequent changes in facts and circumstances do not result in the restoration or increase in that newly established cost basis.

Should our estimates of future selling prices or production costs change, additional and potentially material write-downs may be required. A small change in our estimates may result in a material charge to our reported financial results.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment.

Stock-Based Compensation

We use the fair value method of accounting for our stock options and restricted stock units ("RSUs") granted to employees and for our employee stock purchase plan (the "ESPP") to measure the cost of employee services received in exchange for the stock-based awards. The fair value of stock option awards with only service and/or performance conditions is estimated on the grant or offering date using the Black-Scholes option-pricing model. The Black-Scholes option-pricing model requires inputs such as the risk-free interest rate, expected term and expected volatility. These inputs are subjective and generally require significant judgment. The fair value of RSUs is measured on the grant date based on the closing fair market value of our common stock. The resulting cost is recognized over the period during which an employee is required to provide service in exchange for the awards, usually the vesting period, which is generally four years for stock options and RSUs and six months for the ESPP. Stock-based compensation expense is recognized on a straight-line basis, net of actual forfeitures in the period.

For performance-based awards, stock-based compensation expense is recognized over the expected performance achievement period of individual performance milestones when the achievement of each individual performance milestone becomes probable.

As we accumulate additional employee stock-based awards data over time and as we incorporate market data related to our common stock, we may calculate significantly different volatilities and expected lives, which could materially impact the valuation of our stock-based awards and the stock-based compensation expense that we will recognize in future periods. Stock-based compensation expense is recorded in Cost of revenues, Research and development expense and Selling, general and administrative expense in the consolidated statements of operations.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Significant judgment is required in determining our provision for income taxes, our deferred tax assets and liabilities and any valuation allowance recorded against our net deferred tax assets that are not more likely than not to be realized. We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. In completing our assessment of realizability of our deferred tax assets, we consider our history of income (loss) measured at pre-tax income (loss) adjusted for permanent book-tax differences on a jurisdictional basis, volatility in actual earnings, excess tax benefits related to stock-based compensation in recent prior years, and impacts of the timing of reversal of existing temporary differences. We also rely on our assessment of the Company's projected future results of business operations, including uncertainty in future operating results relative to historical results, volatility in the market price of our common stock and its performance over time, variable macroeconomic conditions impacting our ability to forecast future taxable income, and changes in business that may affect the existence and magnitude of future taxable income. Our valuation allowance assessment is based on our best estimate of future results considering all available information.

Furthermore, significant judgment is required in evaluating our tax positions. In the ordinary course of business, there are many transactions and calculations for which the ultimate tax settlement is uncertain. As a result, we recognize the effect of this uncertainty on our tax attributes or taxes payable based on our estimates of the eventual outcome. These effects are recognized when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that some of those positions may not be fully sustained upon review by tax authorities. We are required to file income tax returns in the U.S. and various foreign jurisdictions, which requires us to interpret the applicable tax laws and regulations in effect in such jurisdictions. Such returns are subject to audit by the various federal, state and foreign taxing authorities, who may disagree with respect to our tax positions. We believe that our consideration is adequate for all open audit years based on our assessment of many factors, including past experience and interpretations of tax law. We review and update our estimates in light of changing facts and circumstances, such as the closing of a tax audit, the lapse of a statute of limitations or a change in estimate. To the extent that the final tax outcome of these matters differs from our expectations, such differences may impact income tax expense in the period in which such determination is made.

Results of Operations

Revenues

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Automotive sales	$ 78,509	$ 67,210	$ 44,125	$ 11,299	17 %	$ 23,085	52 %
Automotive regulatory credits	1,790	1,776	1,465	14	1 %	311	21 %
Automotive leasing	2,120	2,476	1,642	(356)	(14)%	834	51 %
Total automotive revenues	82,419	71,462	47,232	10,957	15 %	24,230	51 %
Services and other	8,319	6,091	3,802	2,228	37 %	2,289	60 %
Total automotive & services and other segment revenue	90,738	77,553	51,034	13,185	17 %	26,519	52 %
Energy generation and storage segment revenue	6,035	3,909	2,789	2,126	54 %	1,120	40 %
Total revenues	$ 96,773	$ 81,462	$ 53,823	$ 15,311	19 %	$ 27,639	51 %

Automotive & Services and Other Segment

Automotive sales revenue includes revenues related to cash and financing deliveries of new Model S, Model X, Semi, Model 3, Model Y, and Cybertruck vehicles, including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. These deliveries are vehicles that are not subject to lease accounting.

Automotive regulatory credits includes sales of regulatory credits to other automotive manufacturers. Our revenue from automotive regulatory credits is directly related to our new vehicle production, sales and pricing negotiated with our customers. We monetize them proactively as new vehicles are sold based on standing arrangements with buyers of such credits, typically as close as possible to the production and delivery of the vehicle or changes in regulation impacting the credits.

Automotive leasing revenue includes the amortization of revenue for vehicles under direct operating lease agreements. Additionally, automotive leasing revenue includes direct sales-type leasing programs where we recognize all revenue associated with the sales-type lease upon delivery to the customer.

Services and other revenue consists of sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise.

2023 compared to 2022

Automotive sales revenue increased $11.30 billion, or 17%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to an increase of 473,382 combined Model 3 and Model Y cash deliveries from production ramping of Model Y globally. The increase was partially offset by a lower average selling price on our vehicles driven by overall price reductions year over year, sales mix, and a negative impact from the United States dollar strengthening against other foreign currencies in the year ended December 31, 2023 compared to the prior year.

Automotive regulatory credits revenue increased $14 million, or 1%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022.

Automotive leasing revenue decreased $356 million, or 14%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a decrease in direct sales-type leasing revenue driven by lower deliveries year over year, partially offset by an increase from our growing direct operating lease portfolio.

Services and other revenue increased $2.23 billion, or 37%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was primarily due to higher used vehicle revenue driven by increases in volume, body shop and part sales revenue, non-warranty maintenance services revenue, paid Supercharging revenue and insurance services revenue, all of which are primarily attributable to our growing fleet. The increases were partially offset by a decrease in the average selling price of used vehicles.

Energy Generation and Storage Segment

Energy generation and storage revenue includes sales and leasing of solar energy generation and energy storage products, financing of solar energy generation products, services related to such products and sales of solar energy systems incentives.

2023 compared to 2022

Energy generation and storage revenue increased $2.13 billion, or 54%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was primarily due to an increase in deployments of Megapack.

Cost of Revenues and Gross Margin

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Cost of revenues							
Automotive sales	$ 65,121	$ 49,599	$ 32,415	$ 15,522	31 %	$ 17,184	53 %
Automotive leasing	1,268	1,509	978	(241)	(16)%	531	54 %
Total automotive cost of revenues	66,389	51,108	33,393	15,281	30 %	17,715	53 %
Services and other	7,830	5,880	3,906	1,950	33 %	1,974	51 %
Total automotive & services and other segment cost of revenues	74,219	56,988	37,299	17,231	30 %	19,689	53 %
Energy generation and storage segment	4,894	3,621	2,918	1,273	35 %	703	24 %
Total cost of revenues	$ 79,113	$ 60,609	$ 40,217	$ 18,504	31 %	$ 20,392	51 %
Gross profit total automotive	$ 16,030	$ 20,354	$ 13,839				
Gross margin total automotive	19.4 %	28.5 %	29.3 %				
Gross profit total automotive & services and other segment	$ 16,519	$ 20,565	$ 13,735				
Gross margin total automotive & services and other segment	18.2 %	26.5 %	26.9 %				
Gross profit energy generation and storage segment	$ 1,141	$ 288	$ (129)				
Gross margin energy generation and storage segment	18.9 %	7.4 %	(4.6)%				
Total gross profit	$ 17,660	$ 20,853	$ 13,606				
Total gross margin	18.2 %	25.6 %	25.3 %				

Automotive & Services and Other Segment

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD ongoing maintenance costs, allocations of electricity and infrastructure costs related to our Supercharger network and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned.

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs for retail merchandise.

2023 compared to 2022

Cost of automotive sales revenue increased $15.52 billion, or 31%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. Cost of automotive sales revenue increased in line with the change in deliveries year over year, as discussed above. The increase was partially offset by a decrease in the average combined cost per unit of our vehicles primarily due to sales mix, lower inbound freight, a decrease in material costs and lower manufacturing costs from better fixed cost absorption. Our costs of revenue were also positively impacted by the United States dollar strengthening against our foreign currencies as compared to the prior periods and by the IRA manufacturing credits earned during the current year.

Cost of automotive leasing revenue decreased $241 million, or 16%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a decrease in direct sales-type leasing cost of revenue driven by lower deliveries year over year.

Cost of services and other revenue increased $1.95 billion, or 33%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was generally in line with the changes in services and other revenue as discussed above.

Gross margin for total automotive decreased from 28.5% to 19.4% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The decrease was primarily due to a lower average selling price on our vehicles partially offset by the favorable change in our average combined cost per unit of our vehicles and IRA manufacturing credits earned as discussed above.

Gross margin for total automotive & services and other segment decreased from 26.5% to 18.2% in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the automotive gross margin decrease discussed above.

Energy Generation and Storage Segment

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, warehouse rent, freight, warranty expense, other overhead costs and amortization of certain acquired intangible assets. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

2023 compared to 2022

Cost of energy generation and storage revenue increased $1.27 billion, or 35%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022, in line with the increase in Megapack deployments year over year, as discussed above. This increase was partially offset by an improvement in production ramping that drove down the average cost per MWh of Megapack as well as IRA manufacturing credits earned during the current year.

Gross margin for energy generation and storage increased from 7.4% to 18.9% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The increase was driven by an improvement in our Megapack gross margin from lower average cost per MWh and a higher proportion of Megapack, which operated at a higher gross margin, within the segment as compared to the prior year periods. Additionally, there was a margin benefit from IRA manufacturing credits earned.

Research and Development Expense

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Research and development	$ 3,969	$ 3,075	$ 2,593	$ 894	29 %	$ 482	19 %
As a percentage of revenues	4 %	4 %	5 %				

Research and development ("R&D") expenses consist primarily of personnel costs for our teams in engineering and research, manufacturing engineering and manufacturing test organizations, prototyping expense, contract and professional services and amortized equipment expense.

R&D expenses increased $894 million, or 29%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The overall increase was primarily driven by additional costs in the current year related to the pre-production phase for Cybertruck, AI and other programs.

R&D expenses as a percentage of revenue stayed consistent at 4% in the year ended December 31, 2023 as compared to the year ended December 31, 2022. Our R&D expenses have increased proportionately with total revenues as we continue to expand our product roadmap and technologies.

Selling, General and Administrative Expense

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Selling, general and administrative	$ 4,800	$ 3,946	$ 4,517	$ 854	22 %	$ (571)	(13)%
As a percentage of revenues	5 %	5 %	8 %				

Selling, general and administrative ("SG&A") expenses generally consist of personnel and facilities costs related to our stores, marketing, sales, executive, finance, human resources, information technology and legal organizations, as well as fees for professional and contract services and litigation settlements.

SG&A expenses increased $854 million, or 22%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. This was driven by a $447 million increase in employee and labor costs primarily from increased headcount, including professional services and a $363 million increase in facilities related expenses.

Restructuring and Other

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Restructuring and other	$ —	$ 176	$ (27)	$ (176)	(100)%	$ 203	Not meaningful

During the year ended December 31, 2022, we recorded an impairment loss of $204 million as well as realized gains of $64 million in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of $36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

Interest Income

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Interest income	$ 1,066	$ 297	$ 56	$ 769	259 %	$ 241	430 %

Interest income increased $769 million, or 259%, in the year ended December 31, 2023 as compared to the year ended December 31, 2022. This increase was primarily due to higher interest earned on our cash and cash equivalents and short-term investments in the year ended December 31, 2023 as compared to the prior year due to rising interest rates and our increasing portfolio balance.

Other Income (Expense), Net

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
Other income (expense), net	$ 172	$ (43)	$ 135	$ 215	Not meaningful	$ (178)	Not meaningful

Other income (expense), net, consists primarily of foreign exchange gains and losses related to our foreign currency-denominated monetary assets and liabilities. We expect our foreign exchange gains and losses will vary depending upon movements in the underlying exchange rates.

Other income, net, changed favorably by $215 million in the year ended December 31, 2023 as compared to the year ended December 31, 2022. The favorable change was primarily due to fluctuations in foreign currency exchange rates on our intercompany balances.

(Benefit from) Provision for Income Taxes

	Year Ended December 31,			2023 vs. 2022 Change		2022 vs. 2021 Change	
(Dollars in millions)	2023	2022	2021	$	%	$	%
(Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699	$ (6,133)	Not meaningful	$ 433	62 %
Effective tax rate	(50)%	8 %	11 %				

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. As of December 31, 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which is objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that our U.S. federal and certain state deferred tax assets are realizable. As such, we released $6.54 billion of our valuation allowance associated with the U.S. federal and state deferred tax assets, with the exception of our California deferred tax assets. Approximately $5.93 billion of the total valuation allowance release was related to deferred tax assets to be realized in the future years and the remainder benefited us during the year ended December 31, 2023. We continue to maintain a full valuation allowance against our California deferred tax assets as of December 31, 2023, because we concluded they are not more likely than not to be realized as we expect our California deferred tax assets generation in future years to exceed our ability to use these deferred tax assets.

Our (benefit from) provision for income taxes changed by $6.13 billion in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the release of $6.54 billion of our valuation allowance associated with the U.S. federal and certain state deferred tax assets.

Our effective tax rate changed from an expense of 8% to a benefit of 50% in the year ended December 31, 2023 as compared to the year ended December 31, 2022, primarily due to the release of the valuation allowance regarding our U.S. federal and certain state deferred tax assets.

See Note 14, *Income Taxes*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details.

Liquidity and Capital Resources

We expect to continue to generate net positive operating cash flow as we have done in the last five fiscal years. The cash we generate from our core operations enables us to fund ongoing operations and production, our research and development projects for new products and technologies including our proprietary battery cells, additional manufacturing ramps at existing manufacturing facilities, the construction of future factories, and the continued expansion of our retail and service locations, body shops, Mobile Service fleet, Supercharger, including to support NACS, energy product installation capabilities and autonomy and other artificial intelligence enabled products.

In addition, because a large portion of our future expenditures will be to fund our growth, we expect that if needed we will be able to adjust our capital and operating expenditures by operating segment. For example, if our near-term manufacturing operations decrease in scale or ramp more slowly than expected, including due to global economic or business conditions, we may choose to correspondingly slow the pace of our capital expenditures. Finally, we continually evaluate our cash needs and may decide it is best to raise additional capital or seek alternative financing sources to fund the rapid growth of our business, including through drawdowns on existing or new debt facilities or financing funds. Conversely, we may also from time to time determine that it is in our best interests to voluntarily repay certain indebtedness early.

Accordingly, we believe that our current sources of funds will provide us with adequate liquidity during the 12-month period following December 31, 2023, as well as in the long-term.

See the sections below for more details regarding the material requirements for cash in our business and our sources of liquidity to meet such needs.

Material Cash Requirements

From time to time in the ordinary course of business, we enter into agreements with vendors for the purchase of components and raw materials to be used in the manufacture of our products. However, due to contractual terms, variability in the precise growth curves of our development and production ramps, and opportunities to renegotiate pricing, we generally do not have binding and enforceable purchase orders under such contracts beyond the short-term, and the timing and magnitude of purchase orders beyond such period is difficult to accurately project.

As discussed in and subject to the considerations referenced in Part II, Item 7, Management's Discussion and Analysis of Financial Condition and Results of Operations—Management Opportunities, Challenges and Uncertainties and 2023 Outlook—Cash Flow and Capital Expenditure Trends in this Annual Report on Form 10-K, we currently expect our capital expenditures to support our projects globally to exceed $10.00 billion in 2024 and be between $8.00 to $10.00 billion in each of the following two fiscal years. In connection with our operations at Gigafactory New York, we have an agreement to spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York through December 31, 2029 (pursuant to a deferral of our required timelines to meet such obligations that was granted in April 2021, and which was memorialized in an amendment to our agreement with the SUNY Foundation in August 2021). For details regarding these obligations, refer to Note 15, *Commitments and Contingencies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

As of December 31, 2023, we and our subsidiaries had outstanding $4.68 billion in aggregate principal amount of indebtedness, of which $1.98 billion is scheduled to become due in the succeeding 12 months. As of December 31, 2023, our total minimum lease payments was $5.96 billion, of which $1.31 billion is due in the succeeding 12 months. For details regarding our indebtedness and lease obligations, refer to Note 11, *Debt*, and Note 12, *Leases*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

Sources and Conditions of Liquidity

Our sources to fund our material cash requirements are predominantly from our deliveries and servicing of new and used vehicles, sales and installations of our energy storage products and solar energy systems, proceeds from debt facilities and proceeds from equity offerings, when applicable.

As of December 31, 2023, we had $16.40 billion and $12.70 billion of cash and cash equivalents and short-term investments, respectively. Balances held in foreign currencies had a U.S. dollar equivalent of $4.43 billion and consisted primarily of Chinese yuan and euros. We had $5.03 billion of unused committed credit amounts as of December 31, 2023. For details regarding our indebtedness, refer to Note 11, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

We continue adapting our strategy to meet our liquidity and risk objectives, such as investing in U.S. government securities and other investments, to do more vertical integration, expand our product roadmap and provide financing options to our customers.

Summary of Cash Flows

(Dollars in millions)	Year Ended December 31, 2023	2022	2021
Net cash provided by operating activities	$ 13,256	$ 14,724	$ 11,497
Net cash used in investing activities	$ (15,584)	$ (11,973)	$ (7,868)
Net cash provided by (used in) financing activities	$ 2,589	$ (3,527)	$ (5,203)

Cash Flows from Operating Activities

Our cash flows from operating activities are significantly affected by our cash investments to support the growth of our business in areas such as research and development and selling, general and administrative and working capital. Our operating cash inflows include cash from vehicle sales and related servicing, customer lease and financing payments, customer deposits, cash from sales of regulatory credits and energy generation and storage products, and interest income on our cash and investments portfolio. These cash inflows are offset by our payments to suppliers for production materials and parts used in our manufacturing process, operating expenses, operating lease payments and interest payments on our financings.

Net cash provided by operating activities decreased by $1.47 billion to $13.26 billion during the year ended December 31, 2023 from $14.72 billion during the year ended December 31, 2022. This decrease was primarily due to the decrease in net income excluding non-cash expenses, gains and losses of $2.93 billion, partially offset by favorable changes in net operating assets and liabilities of $1.46 billion.

Cash Flows from Investing Activities

Cash flows from investing activities and their variability across each period related primarily to capital expenditures, which were $8.90 billion for the year ended December 31, 2023 and $7.16 billion for the year ended December 31, 2022, mainly for global factory expansion and machinery and equipment as we expand our product roadmap. We also purchased $6.62 billion and $5.81 billion of investments, net of proceeds from maturities and sales, for the year ended December 31, 2023 and 2022, respectively. Additionally, proceeds from sales of digital assets was $936 million in the year ended December 31, 2022.

Cash Flows from Financing Activities

Net cash from financing activities changed by $6.12 billion to $2.59 billion net cash provided by financing activities during the year ended December 31, 2023 from $3.53 billion net cash used in financing activities during the year ended December 31, 2022. The change was primarily due to a $3.93 billion increase in proceeds from issuances of debt and a $2.01 billion decrease in repayments of debt. See Note 11, *Debt*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K for further details regarding our debt obligations.

Recent Accounting Pronouncements

See Note 2, *Summary of Significant Accounting Policies*, to the consolidated financial statements included elsewhere in this Annual Report on Form 10-K.

ITEM 7A. QUANTITATIVE AND QUALITATIVE DISCLOSURES ABOUT MARKET RISK

Foreign Currency Risk

We transact business globally in multiple currencies and hence have foreign currency risks related to our revenue, costs of revenue and operating expenses denominated in currencies other than the U.S. dollar (primarily the Chinese yuan and euro in relation to our current year operations). In general, we are a net receiver of currencies other than the U.S. dollar for our foreign subsidiaries. Accordingly, changes in exchange rates affect our operating results as expressed in U.S. dollars as we do not typically hedge foreign currency risk.

We have also experienced, and will continue to experience, fluctuations in our net income as a result of gains (losses) on the settlement and the re-measurement of monetary assets and liabilities denominated in currencies that are not the local currency (primarily consisting of our intercompany and cash and cash equivalents balances).

We considered the historical trends in foreign currency exchange rates and determined that it is reasonably possible that adverse changes in foreign currency exchange rates of 10% for all currencies could be experienced in the near-term. These changes were applied to our total monetary assets and liabilities denominated in currencies other than our local currencies at the balance sheet date to compute the impact these changes would have had on our net income before income taxes. These changes would have resulted in a gain or loss of $1.01 billion at December 31, 2023 and $473 million at December 31, 2022, assuming no foreign currency hedging.

ITEM 8. FINANCIAL STATEMENTS AND SUPPLEMENTARY DATA

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Report of Independent Registered Public Accounting Firm

To the Board of Directors and Stockholders of Tesla, Inc.

Opinions on the Financial Statements and Internal Control over Financial Reporting

We have audited the accompanying consolidated balance sheets of Tesla, Inc. and its subsidiaries (the "Company") as of December 31, 2023 and 2022, and the related consolidated statements of operations, of comprehensive income, of redeemable noncontrolling interests and equity and of cash flows for each of the three years in the period ended December 31, 2023, including the related notes (collectively referred to as the "consolidated financial statements"). We also have audited the Company's internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission (COSO).

In our opinion, the consolidated financial statements referred to above present fairly, in all material respects, the financial position of the Company as of December 31, 2023 and 2022, and the results of its operations and its cash flows for each of the three years in the period ended December 31, 2023 in conformity with accounting principles generally accepted in the United States of America. Also in our opinion, the Company maintained, in all material respects, effective internal control over financial reporting as of December 31, 2023, based on criteria established in Internal Control - Integrated Framework (2013) issued by the COSO.

Changes in Accounting Principles

As discussed in Note 2 to the consolidated financial statements, the Company changed the manner in which it accounts for convertible debt in 2021.

Basis for Opinions

The Company's management is responsible for these consolidated financial statements, for maintaining effective internal control over financial reporting, and for its assessment of the effectiveness of internal control over financial reporting, included in Management's Report on Internal Control over Financial Reporting appearing under Item 9A. Our responsibility is to express opinions on the Company's consolidated financial statements and on the Company's internal control over financial reporting based on our audits. We are a public accounting firm registered with the Public Company Accounting Oversight Board (United States) (PCAOB) and are required to be independent with respect to the Company in accordance with the U.S. federal securities laws and the applicable rules and regulations of the Securities and Exchange Commission and the PCAOB.

We conducted our audits in accordance with the standards of the PCAOB. Those standards require that we plan and perform the audits to obtain reasonable assurance about whether the consolidated financial statements are free of material misstatement, whether due to error or fraud, and whether effective internal control over financial reporting was maintained in all material respects.

Our audits of the consolidated financial statements included performing procedures to assess the risks of material misstatement of the consolidated financial statements, whether due to error or fraud, and performing procedures that respond to those risks. Such procedures included examining, on a test basis, evidence regarding the amounts and disclosures in the consolidated financial statements. Our audits also included evaluating the accounting principles used and significant estimates made by management, as well as evaluating the overall presentation of the consolidated financial statements. Our audit of internal control over financial reporting included obtaining an understanding of internal control over financial reporting, assessing the risk that a material weakness exists, and testing and evaluating the design and operating effectiveness of internal control based on the assessed risk. Our audits also included performing such other procedures as we considered necessary in the circumstances. We believe that our audits provide a reasonable basis for our opinions.

Definition and Limitations of Internal Control over Financial Reporting

A company's internal control over financial reporting is a process designed to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles. A company's internal control over financial reporting includes those policies and procedures that (i) pertain to the maintenance of records that, in reasonable detail, accurately and fairly reflect the transactions and dispositions of the assets of the company; (ii) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that receipts and expenditures of the company are being made only in accordance with authorizations of management and directors of the company; and (iii) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use, or disposition of the company's assets that could have a material effect on the financial statements.

Because of its inherent limitations, internal control over financial reporting may not prevent or detect misstatements. Also, projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Critical Audit Matters

The critical audit matter communicated below is a matter arising from the current period audit of the consolidated financial statements that was communicated or required to be communicated to the audit committee and that (i) relates to accounts or disclosures that are material to the consolidated financial statements and (ii) involved our especially challenging, subjective, or complex judgments. The communication of critical audit matters does not alter in any way our opinion on the consolidated financial statements, taken as a whole, and we are not, by communicating the critical audit matter below, providing a separate opinion on the critical audit matter or on the accounts or disclosures to which it relates.

Automotive Warranty Reserve

As described in Note 2 to the consolidated financial statements, total accrued warranty, which primarily relates to the automotive segment, was $5,152 million as of December 31, 2023. The Company provides a manufacturer's warranty on all new and used Tesla vehicles. A warranty reserve is accrued for these products sold, which includes management's best estimate of the projected costs to repair or replace items under warranty and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims.

The principal considerations for our determination that performing procedures relating to the automotive warranty reserve is a critical audit matter are the significant judgment by management in determining the automotive warranty reserve for certain Tesla vehicle models; this in turn led to significant auditor judgment, subjectivity, and effort in performing procedures to evaluate management's significant assumptions related to the nature, frequency and costs of future claims for certain Tesla vehicle models, and the audit effort involved the use of professionals with specialized skill and knowledge.

Addressing the matter involved performing procedures and evaluating audit evidence in connection with forming our overall opinion on the consolidated financial statements. These procedures included testing the effectiveness of controls relating to management's estimate of the automotive warranty reserve for certain Tesla vehicle models, including controls over management's significant assumptions related to the nature, frequency and costs of future claims as well as the completeness and accuracy of actual claims incurred to date. These procedures also included, among others, performing one of the following: (i) testing management's process for determining the automotive warranty reserve for certain Tesla vehicle models or (ii) developing an independent estimate of the automotive warranty reserve for certain Tesla vehicle models and comparing the independent estimate to management's estimate to evaluate the reasonableness of the estimate. Testing management's process involved evaluating the reasonableness of significant assumptions related to the nature and frequency of future claims and the related costs to repair or replace items under warranty. Evaluating the assumptions related to the nature and frequency of future claims and the related costs to repair or replace items under warranty involved evaluating whether the assumptions used were reasonable by performing a lookback analysis comparing prior period forecasted claims to actual claims incurred. Developing the independent estimate involved testing the completeness and accuracy of historical vehicle claims processed and testing that such claims were appropriately used by management in the estimation of future claims. Professionals with specialized skill and knowledge were used to assist in developing an independent estimate of the automotive warranty reserve for certain Tesla vehicle models and in evaluating the appropriateness of certain aspects of management's significant assumptions related to the nature and frequency of future claims.

/s/ PricewaterhouseCoopers LLP

San Jose, California
January 26, 2024

We have served as the Company's auditor since 2005.

Tesla, Inc.

Consolidated Balance Sheets
(in millions, except per share data)

	December 31, 2023	December 31, 2022
Assets		
Current assets		
Cash and cash equivalents	$ 16,398	$ 16,253
Short-term investments	12,696	5,932
Accounts receivable, net	3,508	2,952
Inventory	13,626	12,839
Prepaid expenses and other current assets	3,388	2,941
Total current assets	49,616	40,917
Operating lease vehicles, net	5,989	5,035
Solar energy systems, net	5,229	5,489
Property, plant and equipment, net	29,725	23,548
Operating lease right-of-use assets	4,180	2,563
Digital assets, net	184	184
Intangible assets, net	178	215
Goodwill	253	194
Deferred tax assets	6,733	328
Other non-current assets	4,531	3,865
Total assets	$ 106,618	$ 82,338
Liabilities		
Current liabilities		
Accounts payable	$ 14,431	$ 15,255
Accrued liabilities and other	9,080	8,205
Deferred revenue	2,864	1,747
Current portion of debt and finance leases	2,373	1,502
Total current liabilities	28,748	26,709
Debt and finance leases, net of current portion	2,857	1,597
Deferred revenue, net of current portion	3,251	2,804
Other long-term liabilities	8,153	5,330
Total liabilities	43,009	36,440
Commitments and contingencies (Note 15)		
Redeemable noncontrolling interests in subsidiaries	242	409
Equity		
Stockholders' equity		
Preferred stock; $0.001 par value; 100 shares authorized; no shares issued and outstanding	—	—
Common stock; $0.001 par value; 6,000 shares authorized; 3,185 and 3,164 shares issued and outstanding as of December 31, 2023 and 2022, respectively	3	3
Additional paid-in capital	34,892	32,177
Accumulated other comprehensive loss	(143)	(361)
Retained earnings	27,882	12,885
Total stockholders' equity	62,634	44,704
Noncontrolling interests in subsidiaries	733	785
Total liabilities and equity	$ 106,618	$ 82,338

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Operations
(in millions, except per share data)

	Year Ended December 31,		
	2023	**2022**	**2021**
Revenues			
Automotive sales	$ 78,509	$ 67,210	$ 44,125
Automotive regulatory credits	1,790	1,776	1,465
Automotive leasing	2,120	2,476	1,642
Total automotive revenues	82,419	71,462	47,232
Energy generation and storage	6,035	3,909	2,789
Services and other	8,319	6,091	3,802
Total revenues	96,773	81,462	53,823
Cost of revenues			
Automotive sales	65,121	49,599	32,415
Automotive leasing	1,268	1,509	978
Total automotive cost of revenues	66,389	51,108	33,393
Energy generation and storage	4,894	3,621	2,918
Services and other	7,830	5,880	3,906
Total cost of revenues	79,113	60,609	40,217
Gross profit	17,660	20,853	13,606
Operating expenses			
Research and development	3,969	3,075	2,593
Selling, general and administrative	4,800	3,946	4,517
Restructuring and other	—	176	(27)
Total operating expenses	8,769	7,197	7,083
Income from operations	8,891	13,656	6,523
Interest income	1,066	297	56
Interest expense	(156)	(191)	(371)
Other income (expense), net	172	(43)	135
Income before income taxes	9,973	13,719	6,343
(Benefit from) provision for income taxes	(5,001)	1,132	699
Net income	14,974	12,587	5,644
Net (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	(23)	31	125
Net income attributable to common stockholders	$ 14,997	$ 12,556	$ 5,519
Net income per share of common stock attributable to common stockholders			
Basic	$ 4.73	$ 4.02	$ 1.87
Diluted	$ 4.30	$ 3.62	$ 1.63
Weighted average shares used in computing net income per share of common stock			
Basic	3,174	3,130	2,959
Diluted	3,485	3,475	3,386

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Comprehensive Income
(in millions)

	Year Ended December 31,		
	2023	**2022**	**2021**
Net income	$ 14,974	$ 12,587	$ 5,644
Other comprehensive income (loss):			
Foreign currency translation adjustment	198	(392)	(308)
Unrealized net gain (loss) on investments	16	(23)	(1)
Adjustment for net loss realized and included in net income	4	—	—
Comprehensive income	15,192	12,172	5,335
Less: Comprehensive (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries	(23)	31	125
Comprehensive income attributable to common stockholders	$ 15,215	$ 12,141	$ 5,210

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Redeemable Noncontrolling Interests and Equity
(in millions)

	Redeemable Noncontrolling Interests	Common Stock		Additional Paid-In Capital	Accumulated Other Comprehensive Income (Loss)	(Accumulated Deficit) Retained Earnings	Total Stockholders' Equity	Noncontrolling Interests in Subsidiaries	Total Equity
		Shares	Amount						
Balance as of December 31, 2020	$ 604	2,879	$ 3	$ 27,260	$ 363	$ (5,401)	$ 22,225	$ 850	$ 23,075
Adjustments for prior periods from adopting ASU 2020-06	—	—	—	(474)	—	211	(263)	—	(263)
Exercises of conversion feature of convertible senior notes	—	2	—	6	—	—	6	—	6
Settlements of warrants	—	112	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	107	—	707	—	—	707	—	707
Stock-based compensation	—	—	—	2,299	—	—	2,299	—	2,299
Contributions from noncontrolling interests	2	—	—	—	—	—	—	—	—
Distributions to noncontrolling interests	(66)	—	—	—	—	—	—	(106)	(106)
Buy-outs of noncontrolling interests	(15)	—	—	5	—	—	5	—	5
Net income	43	—	—	—	—	5,519	5,519	82	5,601
Other comprehensive loss	—	—	—	—	(309)	—	(309)	—	(309)
Balance as of December 31, 2021	$ 568	3,100	$ 3	$ 29,803	$ 54	$ 329	$ 30,189	$ 826	$ 31,015
Settlements of warrants	—	37	—	—	—	—	—	—	—
Issuance of common stock for equity incentive awards	—	27	—	541	—	—	541	—	541
Stock-based compensation	—	—	—	1,806	—	—	1,806	—	1,806
Distributions to noncontrolling interests	(46)	—	—	—	—	—	—	(113)	(113)
Buy-outs of noncontrolling interests	(11)	—	—	27	—	—	27	(61)	(34)
Net (loss) income	(102)	—	—	—	—	12,556	12,556	133	12,689
Other comprehensive loss	—	—	—	—	(415)	—	(415)	—	(415)
Balance as of December 31, 2022	$ 409	3,164	$ 3	$ 32,177	$ (361)	$ 12,885	$ 44,704	$ 785	$ 45,489
Issuance of common stock for equity incentive awards	—	21	—	700	—	—	700	—	700
Stock-based compensation	—	—	—	2,013	—	—	2,013	—	2,013
Distributions to noncontrolling interests	(32)	—	—	—	—	—	—	(108)	(108)
Buy-outs of noncontrolling interests	(39)	—	—	2	—	—	2	(17)	(15)
Net (loss) income	(96)	—	—	—	—	14,997	14,997	73	15,070
Other comprehensive income	—	—	—	—	218	—	218	—	218
Balance as of December 31, 2023	$ 242	3,185	$ 3	$ 34,892	$ (143)	$ 27,882	$ 62,634	$ 733	$ 63,367

The accompanying notes are an integral part of these consolidated financial statements.

Tesla, Inc.

Consolidated Statements of Cash Flows
(in millions)

	Year Ended December 31,		
	2023	2022	2021
Cash Flows from Operating Activities			
Net income	$ 14,974	$ 12,587	$ 5,644
Adjustments to reconcile net income to net cash provided by operating activities:			
Depreciation, amortization and impairment	4,667	3,747	2,911
Stock-based compensation	1,812	1,560	2,121
Inventory and purchase commitments write-downs	463	177	140
Foreign currency transaction net unrealized (gain) loss	(144)	81	(55)
Deferred income taxes	(6,349)	(196)	(149)
Non-cash interest and other operating activities	81	340	245
Digital assets loss (gain), net	—	140	(27)
Changes in operating assets and liabilities:			
Accounts receivable	(586)	(1,124)	(130)
Inventory	(1,195)	(6,465)	(1,709)
Operating lease vehicles	(1,952)	(1,570)	(2,114)
Prepaid expenses and other assets	(2,652)	(3,713)	(1,540)
Accounts payable, accrued and other liabilities	2,605	8,029	5,367
Deferred revenue	1,532	1,131	793
Net cash provided by operating activities	13,256	14,724	11,497
Cash Flows from Investing Activities			
Purchases of property and equipment excluding finance leases, net of sales	(8,898)	(7,158)	(6,482)
Purchases of solar energy systems, net of sales	(1)	(5)	(32)
Purchases of digital assets	—	—	(1,500)
Proceeds from sales of digital assets	—	936	272
Purchase of intangible assets	—	(9)	—
Purchases of investments	(19,112)	(5,835)	(132)
Proceeds from maturities of investments	12,353	22	—
Proceeds from sales of investments	138	—	—
Receipt of government grants	—	76	6
Business combinations, net of cash acquired	(64)	—	—
Net cash used in investing activities	(15,584)	(11,973)	(7,868)
Cash Flows from Financing Activities			
Proceeds from issuances of debt	3,931	—	8,883
Repayments of debt	(1,351)	(3,364)	(14,167)
Collateralized lease repayments	—	—	(9)
Proceeds from exercises of stock options and other stock issuances	700	541	707
Principal payments on finance leases	(464)	(502)	(439)
Debt issuance costs	(29)	—	(9)
Proceeds from investments by noncontrolling interests in subsidiaries	—	—	2
Distributions paid to noncontrolling interests in subsidiaries	(144)	(157)	(161)
Payments for buy-outs of noncontrolling interests in subsidiaries	(54)	(45)	(10)
Net cash provided by (used in) financing activities	2,589	(3,527)	(5,203)
Effect of exchange rate changes on cash and cash equivalents and restricted cash	4	(444)	(183)
Net increase (decrease) in cash and cash equivalents and restricted cash	265	(1,220)	(1,757)
Cash and cash equivalents and restricted cash, beginning of period	16,924	18,144	19,901
Cash and cash equivalents and restricted cash, end of period	$ 17,189	$ 16,924	$ 18,144
Supplemental Non-Cash Investing and Financing Activities			
Acquisitions of property and equipment included in liabilities	$ 2,272	$ 2,148	$ 2,251
Supplemental Disclosures			
Cash paid during the period for interest, net of amounts capitalized	$ 126	$ 152	$ 266
Cash paid during the period for income taxes, net of refunds	$ 1,119	$ 1,203	$ 561

The accompanying notes are an integral part of these consolidated financial statements.

Note 1 – Overview

Tesla, Inc. ("Tesla", the "Company", "we", "us" or "our") was incorporated in the State of Delaware on July 1, 2003. We design, develop, manufacture, sell and lease high-performance fully electric vehicles and energy generation and storage systems, and offer services related to our products. Our Chief Executive Officer, as the chief operating decision maker ("CODM"), organizes our company, manages resource allocations and measures performance among two operating and reportable segments: (i) automotive and (ii) energy generation and storage.

Note 2 – Summary of Significant Accounting Policies

Principles of Consolidation

The accompanying consolidated financial statements have been prepared in conformity with GAAP and reflect our accounts and operations and those of our subsidiaries in which we have a controlling financial interest. In accordance with the provisions of ASC 810, *Consolidation* ("ASC 810"), we consolidate any variable interest entity ("VIE") of which we are the primary beneficiary. We have formed VIEs with financing fund investors in the ordinary course of business in order to facilitate the funding and monetization of certain attributes associated with solar energy systems and leases under our direct vehicle leasing programs. The typical condition for a controlling financial interest ownership is holding a majority of the voting interests of an entity; however, a controlling financial interest may also exist in entities, such as VIEs, through arrangements that do not involve controlling voting interests. ASC 810 requires a variable interest holder to consolidate a VIE if that party has the power to direct the activities of the VIE that most significantly impact the VIE's economic performance and the obligation to absorb losses of the VIE that could potentially be significant to the VIE or the right to receive benefits from the VIE that could potentially be significant to the VIE. We do not consolidate a VIE in which we have a majority ownership interest when we are not considered the primary beneficiary. We have determined that we are the primary beneficiary of all the VIEs (see Note 16*, Variable Interest Entity Arrangements*). We evaluate our relationships with all the VIEs on an ongoing basis to ensure that we continue to be the primary beneficiary. All intercompany transactions and balances have been eliminated upon consolidation.

Use of Estimates

The preparation of financial statements in conformity with GAAP requires management to make estimates and assumptions that affect the reported amounts of assets, liabilities, revenues, costs and expenses and related disclosures in the accompanying notes. The estimates used for, but not limited to, determining significant economic incentive for resale value guarantee arrangements, sales return reserves, income taxes, the collectability of accounts and finance receivables, inventory valuation, warranties, fair value of long-lived assets, goodwill, fair value of financial instruments, fair value and residual value of operating lease vehicles and solar energy systems subject to leases could be impacted. We have assessed the impact and are not aware of any specific events or circumstances that required an update to our estimates and assumptions or materially affected the carrying value of our assets or liabilities as of the date of issuance of this Annual Report on Form 10-K. These estimates may change as new events occur and additional information is obtained. Actual results could differ materially from these estimates under different assumptions or conditions.

Reclassifications

Certain prior period balances have been reclassified to conform to the current period presentation in the consolidated financial statements and the accompanying notes.

Revenue Recognition

Revenue by source

The following table disaggregates our revenue by major source (in millions):

	Year Ended December 31,		
	2023	2022	2021
Automotive sales	$ 78,509	$ 67,210	$ 44,125
Automotive regulatory credits	1,790	1,776	1,465
Energy generation and storage sales	5,515	3,376	2,279
Services and other	8,319	6,091	3,802
Total revenues from sales and services	94,133	78,453	51,671
Automotive leasing	2,120	2,476	1,642
Energy generation and storage leasing	520	533	510
Total revenues	$ 96,773	$ 81,462	$ 53,823

Automotive Segment

Automotive Sales

Automotive sales revenue includes revenues related to cash and financing deliveries of new vehicles, and specific other features and services that meet the definition of a performance obligation under ASC 606, including access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates. We recognize revenue on automotive sales upon delivery to the customer, which is when the control of a vehicle transfers. Payments are typically received at the point control transfers or in accordance with payment terms customary to the business, except sales we finance for which payments are collected over the contractual loan term. We also recognize a sales return reserve based on historical experience plus consideration for expected future market values, when we offer resale value guarantees or similar buyback terms. Other features and services such as access to our internet connectivity, unlimited free Supercharging and over-the-air software updates are provisioned upon control transfer of a vehicle and recognized over time on a straight-line basis as we have a stand-ready obligation to deliver such services to the customer. Other limited free Supercharging incentives are recognized based on actual usage or expiration, whichever is earlier. We recognize revenue related to these other features and services over the performance period, which is generally the expected ownership life of the vehicle. Revenue related to FSD Capability features is recognized when functionality is delivered to the customer and their ongoing maintenance is recognized over time. For our obligations related to automotive sales, we estimate standalone selling price by considering costs used to develop and deliver the service, third-party pricing of similar options and other information that may be available.

Any fees that are paid or payable by us to a customer’s lender when we arrange the financing are recognized as an offset against automotive sales revenue. Costs to obtain a contract mainly relate to commissions paid to our sales personnel for the sale of vehicles. As our contract costs related to automotive sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred. Amounts billed to customers related to shipping and handling are classified as automotive sales revenue, and we have elected to recognize the cost for freight and shipping when control over vehicles, parts or accessories have transferred to the customer as an expense in cost of automotive sales revenue. Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

We offer resale value guarantees to our commercial banking partners in connection with certain vehicle leasing programs. Under these programs, we originate the lease with our end customer and immediately transfer the lease and the underlying vehicle to our commercial banking partner, with the transaction being accounted for as a sale under ASC 606. We receive upfront payment for the vehicle, do not bear casualty and credit risks during the lease term, and we provide a guarantee capped to a limit if they are unable to sell the vehicle at or above the vehicle's contract residual value at the end of the lease term. We estimate a guarantee liability in accordance with ASC 460, *Guarantees* and record it within other liabilities on our consolidated balance sheet. On a quarterly basis, we assess the estimated market value of vehicles sold under this program to determine whether there have been changes to the amount of expected resale value guarantee payments. As we accumulate more data related to the resale values of our vehicles or as market conditions change, there may be material changes to their estimated values. The total guarantee liability on vehicles sold under this program was immaterial as of December 31, 2023.

Deferred revenue related to the access to our FSD Capability features and their ongoing maintenance, internet connectivity, free Supercharging programs and over-the-air software updates primarily on automotive sales consisted of the following (in millions):

	Year Ended December 31,	
	2023	2022
Deferred revenue— beginning of period	$ 2,913	$ 2,382
Additions	1,201	1,178
Net changes in liability for pre-existing contracts	17	(67)
Revenue recognized	(595)	(580)
Deferred revenue— end of period	$ 3,536	$ 2,913

Deferred revenue is equivalent to the total transaction price allocated to the performance obligations that are unsatisfied, or partially unsatisfied, as of the balance sheet date. Revenue recognized from the deferred revenue balance as of December 31, 2022 was $469 million for the year ended December 31, 2023. We had recognized revenue of $472 million from the deferred revenue balance as of December 31, 2021, for the year ended December 31, 2022, primarily related to the general FSD Capability feature release in North America in the fourth quarter of 2022. Of the total deferred revenue balance as of December 31, 2023, we expect to recognize $926 million of revenue in the next 12 months. The remaining balance will be recognized at the time of transfer of control of the product or over the performance period as discussed above in Automotive Sales.

We have been providing loans for financing our automotive deliveries in volume since fiscal year 2022. As of December 31, 2023 and 2022, we have recorded net financing receivables on the consolidated balance sheets, of which $242 million and $128 million, respectively, is recorded within Accounts receivable, net, for the current portion and $1.04 billion and $665 million, respectively, is recorded within Other non-current assets for the long-term portion.

Automotive Regulatory Credits

We earn tradable credits in the operation of our automotive business under various regulations related to ZEVs, greenhouse gas, fuel economy and clean fuel. We sell these credits to other regulated entities who can use the credits to comply with emission standards and other regulatory requirements.

Payments for automotive regulatory credits are typically received at the point control transfers to the customer, or in accordance with payment terms customary to the business. We recognize revenue on the sale of automotive regulatory credits, which have negligible incremental costs associated with them, at the time control of the regulatory credits is transferred to the purchasing party. Deferred revenue related to sales of automotive regulatory credits was immaterial as of December 31, 2023 and 2022. Revenue recognized from the deferred revenue balance as of December 31, 2022 and 2021 was immaterial for the years ended December 31, 2023 and 2022. During the year ended December 31, 2022, we had also recognized $288 million in revenue due to changes in regulation which entitled us to additional consideration for credits sold previously.

Automotive Leasing Revenue

Direct Vehicle Operating Leasing Program

We have outstanding leases under our direct vehicle operating leasing programs in the U.S., Canada and in certain countries in Europe. Qualifying customers are permitted to lease a vehicle directly from Tesla for up to 48 months. At the end of the lease term, customers are generally required to return the vehicles to us. We account for these leasing transactions as operating leases. We record leasing revenues to automotive leasing revenue on a straight-line basis over the contractual term, and we record the depreciation of these vehicles to cost of automotive leasing revenue. For the years ended December 31, 2023, 2022 and 2021, we recognized $1.86 billion, $1.75 billion and $1.25 billion of direct vehicle leasing revenue, respectively. As of December 31, 2023 and 2022, we had deferred $458 million and $407 million, respectively, of lease-related upfront payments, which will be recognized on a straight-line basis over the contractual terms of the individual leases.

Our policy is to exclude taxes collected from a customer from the transaction price of automotive contracts.

Direct Sales-Type Leasing Program

We have outstanding direct leases and vehicles financed by us under loan arrangements accounted for as sales-type leases under ASC 842, *Leases* ("ASC 842"), in certain countries in Asia and Europe. Depending on the specific program, customers may or may not have a right to return the vehicle to us during or at the end of the lease term. If the customer does not have a right to return, the customer will take title to the vehicle at the end of the lease term after making all contractual payments. Under the programs for which there is a right to return, the purchase option is reasonably certain to be exercised by the lessee and we therefore expect the customer to take title to the vehicle at the end of the lease term after making all contractual payments. Our arrangements under these programs can have terms for up to 72 months. We recognize all revenue and costs associated with the sales-type lease as automotive leasing revenue and automotive leasing cost of revenue, respectively, upon delivery of the vehicle to the customer. Interest income based on the implicit rate in the lease is recorded to automotive leasing revenue over time as customers are invoiced on a monthly basis. For the years ended December 31, 2023, 2022 and 2021, we recognized $215 million, $683 million and $369 million, respectively, of sales-type leasing revenue and $164 million, $427 million and $234 million, respectively, of sales-type leasing cost of revenue.

Services and Other Revenue

Services and other revenue consists of sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise.

Revenues related to repair, maintenance and vehicle insurance services are recognized over time as services are provided and extended service plans are recognized over the performance period of the service contract as the obligation represents a stand-ready obligation to the customer. We sell used vehicles, services, service plans, vehicle components and merchandise separately and thus use standalone selling prices as the basis for revenue allocation to the extent that these items are sold in transactions with other performance obligations. Payment for used vehicles, services, vehicle components, and merchandise are typically received at the point when control transfers to the customer or in accordance with payment terms customary to the business. Payments received for prepaid plans are refundable upon customer cancellation of the related contracts and are included within Customer deposits on the consolidated balance sheets. We record in Deferred revenue any non-refundable prepayment amounts that are collected from customers and unearned insurance premiums, which is recognized as revenue ratably over the respective customer contract term. Deferred revenue excluding unearned insurance premiums was immaterial as of December 31, 2023 and 2022.

Energy Generation and Storage Segment

Energy Generation and Storage Sales

Energy generation and storage sales revenue consists of the sale of solar energy systems and energy storage systems to residential, small commercial, large commercial and utility grade customers. Sales of solar energy systems to residential and small scale commercial customers consist of the engineering, design and installation of the system. Residential and small scale commercial customers pay the full purchase price of the solar energy system upfront. Revenue for the design and installation obligation is recognized when control transfers, which is when we install a solar energy system and the system passes inspection by the utility or the authority having jurisdiction. Sales of energy storage systems to residential and small scale commercial customers consist of the installation of the energy storage system and revenue is recognized when control transfers, which is when the product has been delivered or, if we are performing installation, when installed and commissioned. Payment for such storage systems is made upon invoice or in accordance with payment terms customary to the business.

For large commercial and utility grade energy storage system sales which consist of the engineering, design and installation of the system, customers make milestone payments that are consistent with contract-specific phases of a project. Revenue from such contracts is recognized over time using the percentage of completion method based on cost incurred as a percentage of total estimated contract costs for energy storage system sales.

In instances where there are multiple performance obligations in a single contract, we allocate the consideration to the various obligations in the contract based on the relative standalone selling price method. Standalone selling prices are estimated based on estimated costs plus margin or by using market data for comparable products. Costs to obtain a contract relate mainly to commissions paid to our sales personnel related to the sale of energy storage systems. As our contract costs related to energy storage system sales are typically fulfilled within one year, the costs to obtain a contract are expensed as incurred.

As part of our energy storage system contracts, we may provide the customer with performance guarantees that warrant that the underlying system will meet or exceed the minimum energy performance requirements specified in the contract. If an energy storage system does not meet the performance guarantee requirements, we may be required to pay liquidated damages. Other forms of variable consideration related to our large commercial and utility grade energy storage system contracts include variable customer payments that will be made based on our energy market participation activities. Such guarantees and variable customer payments represent a form of variable consideration and are estimated at contract inception at their most likely amount and updated at the end of each reporting period as additional performance data becomes available. Such estimates are included in the transaction price only to the extent that it is probable a significant reversal of revenue will not occur.

We record as deferred revenue any non-refundable amounts that are collected from customers related to fees charged for prepayments, which is recognized as revenue ratably over the respective customer contract term. As of December 31, 2023 and 2022, deferred revenue related to such customer payments amounted to $1.60 billion and $863 million, respectively, mainly due to contractual payment terms. Revenue recognized from the deferred revenue balance as of December 31, 2022 and 2021 was $571 million and $171 million for the years ended December 31, 2023 and 2022, respectively. We have elected the practical expedient to omit disclosure of the amount of the transaction price allocated to remaining performance obligations for energy generation and storage sales with an original expected contract length of one year or less and the amount that we have the right to invoice when that amount corresponds directly with the value of the performance to date. As of December 31, 2023, total transaction price allocated to performance obligations that were unsatisfied or partially unsatisfied for contracts with an original expected length of more than one year was $3.43 billion. Of this amount, we expect to recognize $1.05 billion in the next 12 months and the rest over the remaining performance obligation period.

We have been providing loans for financing our energy generation products in volume since fiscal year 2022. As of December 31, 2023 and 2022, we have recorded net financing receivables on the consolidated balance sheets, of which $31 million and $24 million, respectively, is recorded within Accounts receivable, net, for the current portion and $578 million and $387 million, respectively, is recorded within Other non-current assets for the long-term portion.

Energy Generation and Storage Leasing

For revenue arrangements where we are the lessor under operating lease agreements for energy generation and storage products, we record lease revenue from minimum lease payments, including upfront rebates and incentives earned from such systems, on a straight-line basis over the life of the lease term, assuming all other revenue recognition criteria have been met. The difference between the payments received and the revenue recognized is recorded as deferred revenue or deferred asset on the consolidated balance sheet.

For solar energy systems where customers purchase electricity from us under PPAs prior to January 1, 2019, we have determined that these agreements should be accounted for as operating leases pursuant to ASC 840, *Leases*. Revenue is recognized based on the amount of electricity delivered at rates specified under the contracts, assuming all other revenue recognition criteria are met.

We record as deferred revenue any amounts that are collected from customers, including lease prepayments, in excess of revenue recognized, which is recognized as revenue ratably over the respective customer contract term. As of December 31, 2023 and 2022, deferred revenue related to such customer payments amounted to $181 million and $191 million, respectively. Deferred revenue also includes the portion of rebates and incentives received from utility companies and various local and state government agencies, which is recognized as revenue over the lease term. As of December 31, 2023 and 2022, deferred revenue from rebates and incentives was immaterial.

We capitalize initial direct costs from the execution of agreements for solar energy systems and PPAs, which include the referral fees and sales commissions, as an element of solar energy systems, net, and subsequently amortize these costs over the term of the related agreements.

Cost of Revenues

Automotive Segment

Automotive Sales

Cost of automotive sales revenue includes direct and indirect materials, labor costs, manufacturing overhead, including depreciation costs of tooling and machinery, shipping and logistic costs, vehicle connectivity costs, FSD Capability ongoing maintenance costs, allocations of electricity and infrastructure costs related to our Supercharger network and reserves for estimated warranty expenses. Cost of automotive sales revenues also includes adjustments to warranty expense and charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of automotive sales revenue benefits from manufacturing credits earned.

Automotive Leasing

Cost of automotive leasing revenue includes the depreciation of operating lease vehicles, cost of goods sold associated with direct sales-type leases and warranty expense related to leased vehicles.

Services and Other

Costs of services and other revenue includes cost of used vehicles including refurbishment costs, costs associated with providing non-warranty after-sales services, costs associated with our body shops and part sales, costs of paid Supercharging, costs to provide vehicle insurance and costs for retail merchandise.

Energy Generation and Storage Segment

Energy Generation and Storage

Cost of energy generation and storage revenue includes direct and indirect material and labor costs, overhead costs, freight, warranty expense, and amortization of certain acquired intangible assets. Cost of energy generation and storage revenue also includes charges to write down the carrying value of our inventory when it exceeds its estimated net realizable value and to provide for obsolete and on-hand inventory in excess of forecasted demand. Additionally, cost of energy generation and storage revenue benefits from manufacturing credits earned. In agreements for solar energy systems and PPAs where we are the lessor, the cost of revenue is primarily comprised of depreciation of the cost of leased solar energy systems, maintenance costs associated with those systems and amortization of any initial direct costs.

Research and Development Costs

Research and development costs are expensed as incurred.

Income Taxes

We are subject to income taxes in the U.S. and in many foreign jurisdictions. Income taxes are computed using the asset and liability method, under which deferred tax assets and liabilities are determined based on the difference between the financial statement and tax bases of assets and liabilities using enacted tax rates in effect for the year in which the differences are expected to affect taxable income. Valuation allowances are established when necessary to reduce deferred tax assets to the amount expected to be realized.

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. Significant judgment is required in determining our provision for income taxes, our deferred tax assets and liabilities and any valuation allowance recorded against our net deferred tax assets that are not more likely than not to be realized. In completing our assessment of realizability of our deferred tax assets, we consider our history of income (loss) measured at pre-tax income (loss) adjusted for permanent book-tax differences on a jurisdictional basis, volatility in actual earnings, excess tax benefits related to stock-based compensation in recent prior years, and impacts of the timing of reversal of existing temporary differences. We also rely on our assessment of the Company's projected future results of business operations, including uncertainty in future operating results relative to historical results, volatility in the market price of our common stock and its performance over time, variable macroeconomic conditions impacting our ability to forecast future taxable income, and changes in business that may affect the existence and magnitude of future taxable income. Our valuation allowance assessment is based on our best estimate of future results considering all available information.

We record liabilities related to uncertain tax positions when, despite our belief that our tax return positions are supportable, we believe that it is more likely than not that those positions may not be fully sustained upon review by tax authorities. Accrued interest and penalties related to unrecognized tax benefits are classified as income tax expense.

The Tax Cuts and Jobs Act subjects a U.S. shareholder to tax on global intangible low-taxed income ("GILTI") earned by certain foreign subsidiaries. Under GAAP, we can make an accounting policy election to either treat taxes due on the GILTI inclusion as a current period expense or factor such amounts into our measurement of deferred taxes. We elected the deferred method, under which we recorded the corresponding deferred tax assets and liabilities in our consolidated balance sheets.

Comprehensive Income

Comprehensive income is comprised of net income and other comprehensive income (loss). Other comprehensive income (loss) consists of foreign currency translation adjustments and unrealized net gains and losses on investments that have been excluded from the determination of net income.

Stock-Based Compensation

We use the fair value method of accounting for our stock options and RSUs granted to employees and for our ESPP to measure the cost of employee services received in exchange for the stock-based awards. The fair value of stock option awards with only service and/or performance conditions is estimated on the grant or offering date using the Black-Scholes option-pricing model. The Black-Scholes option-pricing model requires inputs such as the risk-free interest rate, expected term and expected volatility. These inputs are subjective and generally require significant judgment. The fair value of RSUs is measured on the grant date based on the closing fair market value of our common stock. The resulting cost is recognized over the period during which an employee is required to provide service in exchange for the awards, usually the vesting period, which is generally four years for stock options and RSUs andsix months for the ESPP. Stock-based compensation expense is recognized on a straight-line basis, net of actual forfeitures in the period.

For performance-based awards, stock-based compensation expense is recognized over the expected performance achievement period of individual performance milestones when the achievement of each individual performance milestone becomes probable.

As we accumulate additional employee stock-based awards data over time and as we incorporate market data related to our common stock, we may calculate significantly different volatilities and expected lives, which could materially impact the valuation of our stock-based awards and the stock-based compensation expense that we will recognize in future periods. Stock-based compensation expense is recorded in Cost of revenues, Research and development expense and Selling, general and administrative expense in the consolidated statements of operations.

Noncontrolling Interests and Redeemable Noncontrolling Interests

Noncontrolling interests and redeemable noncontrolling interests represent third-party interests in the net assets under certain funding arrangements, or funds, that we have entered into to finance the costs of solar energy systems and vehicles under operating leases. We have determined that the contractual provisions of the funds represent substantive profit-sharing arrangements. We have further determined that the methodology for calculating the noncontrolling interest and redeemable noncontrolling interest balances that reflects the substantive profit-sharing arrangements is a balance sheet approach using the hypothetical liquidation at book value ("HLBV") method. We, therefore, determine the amount of the noncontrolling interests and redeemable noncontrolling interests in the net assets of the funds at each balance sheet date using the HLBV method, which is presented on the consolidated balance sheet as noncontrolling interests in subsidiaries and redeemable noncontrolling interests in subsidiaries. Under the HLBV method, the amounts reported as noncontrolling interests and redeemable noncontrolling interests in the consolidated balance sheet represent the amounts the third parties would hypothetically receive at each balance sheet date under the liquidation provisions of the funds, assuming the net assets of the funds were liquidated at their recorded amounts determined in accordance with GAAP and with tax laws effective at the balance sheet date and distributed to the third parties. The third parties' interests in the results of operations of the funds are determined as the difference in the noncontrolling interest and redeemable noncontrolling interest balances in the consolidated balance sheets between the start and end of each reporting period, after taking into account any capital transactions between the funds and the third parties. However, the redeemable noncontrolling interest balance is at least equal to the redemption amount. The redeemable noncontrolling interest balance is presented as temporary equity in the mezzanine section of the consolidated balance sheet since these third parties have the right to redeem their interests in the funds for cash or other assets. For certain funds, there have been significant fluctuations in net (loss) income attributable to noncontrolling interests and redeemable noncontrolling interests in subsidiaries due to changes in the liquidation provisions as time-based milestones have been reached.

Net Income per Share of Common Stock Attributable to Common Stockholders

Basic net income per share of common stock attributable to common stockholders is calculated by dividing net income attributable to common stockholders by the weighted-average shares of common stock outstanding for the period. Potentially dilutive shares, which are based on the weighted-average shares of common stock underlying outstanding stock-based awards, warrants and convertible senior notes using the treasury stock method or the if-converted method, as applicable, are included when calculating diluted net income per share of common stock attributable to common stockholders when their effect is dilutive.

Furthermore, in connection with the offerings of our convertible senior notes, we entered into convertible note hedges and warrants (see Note 11*Debt*). However, our convertible note hedges are not included when calculating potentially dilutive shares since their effect is always anti-dilutive. The strike price on the warrants were below our average share price during the period and were included in the tables below. Warrants are included in the weighted-average shares used in computing basic net income per share of common stock in the period(s) they are settled.

The following table presents the reconciliation of net income attributable to common stockholders to net income used in computing basic and diluted net income per share of common stock (in millions):

	Year Ended December 31,		
	2023	2022	2021
Net income attributable to common stockholders	$ 14,997	$ 12,556	$ 5,519
Less: Buy-out of noncontrolling interest	(2)	(27)	(5)
Net income used in computing basic net income per share of common stock	14,999	12,583	5,524
Less: Dilutive convertible debt	—	(1)	(9)
Net income used in computing diluted net income per share of common stock	$ 14,999	$ 12,584	$ 5,533

The following table presents the reconciliation of basic to diluted weighted average shares used in computing net income per share of common stock attributable to common stockholders (in millions):

	Year Ended December 31,		
	2023	2022	2021
Weighted average shares used in computing net income per share of common stock, basic	3,174	3,130	2,959
Add:			
Stock-based awards	298	310	292
Convertible senior notes	2	3	29
Warrants	11	32	106
Weighted average shares used in computing net income per share of common stock, diluted	3,485	3,475	3,386

The following table presents the potentially dilutive shares that were excluded from the computation of diluted net income per share of common stock attributable to common stockholders, because their effect was anti-dilutive (in millions):

	Year Ended December 31,		
	2023	2022	2021
Stock-based awards	12	4	1

Business Combinations

We account for business acquisitions under ASC 805, *Business Combinations.* The total purchase consideration for an acquisition is measured as the fair value of the assets given, equity instruments issued and liabilities assumed at the acquisition date. Costs that are directly attributable to the acquisition are expensed as incurred. Identifiable assets (including intangible assets), liabilities assumed (including contingent liabilities) and noncontrolling interests in an acquisition are measured initially at their fair values at the acquisition date. We recognize goodwill if the fair value of the total purchase consideration and any noncontrolling interests is in excess of the net fair value of the identifiable assets acquired and the liabilities assumed. We recognize a bargain purchase gain within Other income (expense), net, in the consolidated statement of operations if the net fair value of the identifiable assets acquired and the liabilities assumed is in excess of the fair value of the total purchase consideration and any noncontrolling interests. We include the results of operations of the acquired business in the consolidated financial statements beginning on the acquisition date.

Cash and Cash Equivalents

All highly liquid investments with an original maturity of three months or less at the date of purchase are considered cash equivalents. Our cash equivalents are primarily comprised of U.S. government securities, money market funds and commercial paper.

Restricted Cash

We maintain certain cash balances restricted as to withdrawal or use. Our restricted cash is comprised primarily of cash held to service certain payments under various secured debt facilities. In addition, restricted cash includes cash held as collateral for sales to lease partners with a resale value guarantee, letters of credit, real estate leases and deposits held for our insurance services. We record restricted cash as other assets in the consolidated balance sheets and determine current or non-current classification based on the expected duration of the restriction.

Our total cash and cash equivalents and restricted cash, as presented in the consolidated statements of cash flows, was as follows (in millions):

	December 31, 2023	December 31, 2022	December 31, 2021
Cash and cash equivalents	$ 16,398	$ 16,253	$ 17,576
Restricted cash included in prepaid expenses and other current assets	543	294	345
Restricted cash included in other non-current assets	248	377	223
Total as presented in the consolidated statements of cash flows	$ 17,189	$ 16,924	$ 18,144

Investments

Investments may be comprised of a combination of marketable securities, includingU.S. government securities, corporate debt securities, commercial paper, time deposits, and certain certificates of deposit, which are all designated as available-for-sale and reported at estimated fair value, with unrealized gains and losses recorded in accumulated other comprehensive income which is included within stockholders' equity. Available-for-sale marketable securities with maturities greater than three months at the date of purchase are included in short-term investments in our consolidated balance sheets. Interest, dividends, amortization and accretion of purchase premiums and discounts on these investments are included within Interest income in our consolidated statements of operations.

The cost of available-for-sale investments sold is based on the specific identification method. Realized gains and losses on the sale of available-for-sale investments are recorded in Other income (expense), net.

We regularly review all of our investments for declines in fair value. The review includes but is not limited to (i) the consideration of the cause of the decline, (ii) any currently recorded expected credit losses and (iii) the creditworthiness of the respective security issuers. The amortized cost basis of our investments approximates its fair value.

Accounts Receivable and Allowance for Doubtful Accounts

Accounts receivable primarily include amounts related to receivables from financial institutions and leasing companies offering various financing products to our customers, sales of energy generation and storage products, sales of regulatory credits to other automotive manufacturers and government rebates already passed through to customers. We provide an allowance against accounts receivable for the amount we expect to be uncollectible. We write-off accounts receivable against the allowance when they are deemed uncollectible.

Depending on the day of the week on which the end of a fiscal quarter falls, our accounts receivable balance may fluctuate as we are waiting for certain customer payments to clear through our banking institutions and receipts of payments from our financing partners, which can take up to approximately two weeks based on the contractual payment terms with such partners. Our accounts receivable balances associated with our sales of regulatory credits are dependent on contractual payment terms. Additionally, government rebates can take up to a year or more to be collected depending on the customary processing timelines of the specific jurisdictions issuing them. These various factors may have a significant impact on our accounts receivable balance from period to period. As of December 31, 2023 and 2022, we had $207 million and $753 million, respectively, of long-term government rebates receivable in Other non-current assets in our consolidated balance sheets.

Financing Receivables

We provide financing options to our customers for our automotive and energy products. Financing receivables are carried at amortized cost, net of allowance for loan losses. Provisions for loan losses are charged to operations in amounts sufficient to maintain the allowance for loan losses at levels considered adequate to cover expected credit losses on the financing receivables. In determining expected credit losses, we consider our historical level of credit losses, current economic trends, and reasonable and supportable forecasts that affect the collectability of the future cash flows.

When originating consumer receivables, we review the credit application, the proposed contract terms, credit bureau information (e.g., FICO score) and other information. Our evaluation emphasizes the applicant's ability to pay and creditworthiness focusing on payment, affordability, and applicant credit history as key considerations. Generally, all customers in this portfolio have strong creditworthiness at loan origination.

After origination, we review the credit quality of retail financing based on customer payment activity and aging analysis. For all financing receivables, we define "past due" as any payment, including principal and interest, which is at least 31 days past the contractual due date. As of December 31, 2023 and 2022, the vast majority of our financing receivables were at current status with only an immaterial balance being past due. As of December 31, 2023, the majority of our financing receivables, excluding MyPower notes receivable, were originated in 2023 and 2022, and as of December 31, 2022, the majority of our financing receivables, excluding MyPower notes receivable, were originated in 2022.

We have customer notes receivable under the legacy MyPower loan program, which provided residential customers with the option to finance the purchase of a solar energy system through a 30-year loan and were all originated prior to year 2018. The outstanding balances, net of any allowance for expected credit losses, are presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. As of December 31, 2023 and 2022, the total outstanding balance of MyPower customer notes receivable, net of allowance for expected credit losses, was $266 million and $280 million, respectively, of which $5 million and $7 million were due in the next 12 months as of December 31, 2023 and 2022, respectively. As of December 31, 2023 and 2022, the allowance for expected credit losses was $36 million and $37 million, respectively.

Concentration of Risk

Credit Risk

Financial instruments that potentially subject us to a concentration of credit risk consist of cash, cash equivalents, investments, restricted cash, accounts receivable and other finance receivables. Our cash and investments balances are primarily on deposit at high credit quality financial institutions or invested in money market funds. These deposits are typically in excess of insured limits. As of December 31, 2023 and 2022, no entity represented 10% or more of our total receivables balance.

Supply Risk

We are dependent on our suppliers, including single source suppliers, and the inability of these suppliers to deliver necessary components of our products in a timely manner at prices, quality levels and volumes acceptable to us, or our inability to efficiently manage these components from these suppliers, could have a material adverse effect on our business, prospects, financial condition and operating results.

Inventory Valuation

Inventories are stated at the lower of cost or net realizable value. Cost is computed using standard cost for vehicles and energy products, which approximates actual cost on a first-in, first-out basis. We record inventory write-downs for excess or obsolete inventories based upon assumptions about current and future demand forecasts. If our inventory on-hand is in excess of our future demand forecast, the excess amounts are written-off.

We also review our inventory to determine whether its carrying value exceeds the net amount realizable upon the ultimate sale of the inventory. This requires us to determine the estimated selling price of our vehicles less the estimated cost to convert the inventory on-hand into a finished product. Once inventory is written-down, a new, lower cost basis for that inventory is established and subsequent changes in facts and circumstances do not result in the restoration or increase in that newly established cost basis.

Should our estimates of future selling prices or production costs change, additional and potentially material write-downs may be required. A small change in our estimates may result in a material charge to our reported financial results.

Operating Lease Vehicles

Vehicles that are leased as part of our direct vehicle leasing program are classified as operating lease vehicles at cost less accumulated depreciation. We generally depreciate their cost, less residual value, using the straight-line-method to cost of automotive leasing revenue over the contractual period. The gross cost of operating lease vehicles as of December 31, 2023 and 2022 was $7.36 billion and $6.08 billion, respectively. Operating lease vehicles on the consolidated balance sheets are presented net of accumulated depreciation of $1.38 billion and $1.04 billion as of December 31, 2023 and 2022, respectively.

Digital Assets, Net

We currently account for all digital assets held as indefinite-lived intangible assets in accordance with ASC 350 *Intangibles—Goodwill and Other*. We have ownership of and control over our digital assets and we may use third-party custodial services to secure it. The digital assets are initially recorded at cost and are subsequently remeasured on the consolidated balance sheet at cost, net of any impairment losses incurred since acquisition.

We determine the fair value of our digital assets on a nonrecurring basis in accordance with ASC 820 *Fair Value Measurement* ("ASC 820"), based on quoted prices on the active exchange(s) that we have determined is the principal market for such assets (Level I inputs). We perform an analysis each quarter to identify whether events or changes in circumstances, principally decreases in the quoted prices on active exchanges, indicate that it is more likely than not that our digital assets are impaired. In determining if an impairment has occurred, we consider the lowest market price of one unit of digital asset quoted on the active exchange since acquiring the digital asset. When the then current carrying value of a digital asset exceeds the fair value determined each quarter, an impairment loss has occurred with respect to those digital assets in the amount equal to the difference between their carrying values and the prices determined.

Impairment losses are recognized within Restructuring and other in the consolidated statements of operations in the period in which the impairment is identified. Gains are not recorded until realized upon sale(s), at which point they are presented net of any impairment losses for the same digital assets held within Restructuring and other. In determining the gain to be recognized upon sale, we calculate the difference between the sales price and carrying value of the digital assets sold immediately prior to sale.

See Note 3, *Digital Assets, Net*, for further information regarding digital assets.

Solar Energy Systems, Net

We are the lessor of solar energy systems. Solar energy systems are stated at cost less accumulated depreciation.

Depreciation and amortization is calculated using the straight-line method over the estimated useful lives of the respective assets, as follows:

Solar energy systems in service	30 to 35 years
Initial direct costs related to customer solar energy system lease acquisition costs	Lease term (up to 25 years)

Solar energy systems pending interconnection will be depreciated as solar energy systems in service when they have been interconnected and placed in-service. Solar energy systems under construction represents systems that are under installation, which will be depreciated as solar energy systems in service when they are completed, interconnected and placed in service. Initial direct costs related to customer solar energy system agreement acquisition costs are capitalized and amortized over the term of the related customer agreements.

Property, Plant and Equipment, Net

Property, plant and equipment, net, including leasehold improvements, are recognized at cost less accumulated depreciation. Depreciation is generally computed using the straight-line method over the estimated useful lives of the respective assets, as follows:

Machinery, equipment, vehicles and office furniture	3 to 15 years
Tooling	4 to 7 years
Building and building improvements	15 to 30 years
Computer equipment and software	3 to 10 years

Leasehold improvements are depreciated on a straight-line basis over the shorter of their estimated useful lives or the terms of the related leases.

Upon the retirement or sale of our property, plant and equipment, the cost and associated accumulated depreciation are removed from the consolidated balance sheet, and the resulting gain or loss is reflected on the consolidated statement of operations. Maintenance and repair expenditures are expensed as incurred while major improvements that increase the functionality, output or expected life of an asset are capitalized and depreciated ratably over the identified useful life.

Interest expense on outstanding debt is capitalized during the period of significant capital asset construction. Capitalized interest on construction in progress is included within Property, plant and equipment, net and is amortized over the life of the related assets.

Long-Lived Assets Including Acquired Intangible Assets

We review our property, plant and equipment, solar energy systems, long-term prepayments and intangible assets for impairment whenever events or changes in circumstances indicate that the carrying amount of an asset (or asset group) may not be recoverable. We measure recoverability by comparing the carrying amount to the future undiscounted cash flows that the asset is expected to generate. If the asset is not recoverable, its carrying amount would be adjusted down to its fair value. For the years ended December 31, 2023, 2022 and 2021, we have recognized no material impairments of our long-lived assets.

Intangible assets with definite lives are amortized on a straight-line basis over their estimated useful lives, which range fromseven to thirty years.

Goodwill

We assess goodwill for impairment annually in the fourth quarter, or more frequently if events or changes in circumstances indicate that it might be impaired, by comparing its carrying value to the reporting unit's fair value. For the years ended December 31, 2023, 2022, and 2021, we didnot recognize any impairment of goodwill.

Capitalization of Software Costs

We capitalize costs incurred in the development of internal use software, during the application development stage to Property, plant and equipment, net on the consolidated balance sheets. Costs related to preliminary project activities and post-implementation activities are expensed as incurred. Such costs are amortized on a straight-line basis over their estimated useful life of three to five years.

Software development costs incurred in development of software to be sold, leased, or otherwise marketed, incurred subsequent to the establishment of technological feasibility and prior to the general availability of the software are capitalized when they are expected to become significant. Such costs are amortized over the estimated useful life of the applicable software once it is made generally available to our customers.

We evaluate the useful lives of these assets on an annual basis, and we test for impairment whenever events or changes in circumstances occur that could impact the recoverability of these assets. For the years ended December 31, 2023, 2022, and 2021, we have recognized no impairments of capitalized software costs.

Foreign Currency

We determine the functional and reporting currency of each of our international subsidiaries and their operating divisions based on the primary currency in which they operate. In cases where the functional currency is not the U.S. dollar, we recognize a cumulative translation adjustment created by the different rates we apply to current period income or loss and the balance sheet. For each subsidiary, we apply the monthly average functional exchange rate to its monthly income or loss and the month-end functional currency rate to translate the balance sheet.

Foreign currency transaction gains and losses are a result of the effect of exchange rate changes on transactions denominated in currencies other than the functional currency of the respective subsidiary. Transaction gains and losses are recognized in Other income (expense), net, in the consolidated statements of operations. For the years ended December 31, 2023, 2022 and 2021, we recorded a net foreign currency transaction gain of $122 million, loss of $89 million and gain of $97 million, respectively.

Warranties

We provide a manufacturer's warranty on all new and used vehicles and a warranty on the installation and components of the energy generation and storage systems we sell for periods typically between 10 to 25 years. We accrue a warranty reserve for the products sold by us, which includes our best estimate of the projected costs to repair or replace items under warranties and recalls if identified. These estimates are based on actual claims incurred to date and an estimate of the nature, frequency and costs of future claims. These estimates are inherently uncertain and changes to our historical or projected warranty experience may cause material changes to the warranty reserve in the future. The warranty reserve does not include projected warranty costs associated with our vehicles subject to operating lease accounting and our solar energy systems under lease contracts or PPAs, as the costs to repair these warranty claims are expensed as incurred. The portion of the warranty reserve expected to be incurred within the next 12 months is included within Accrued liabilities and other, while the remaining balance is included within Other long-term liabilities on the consolidated balance sheets. For liabilities that we are entitled to receive indemnification from our suppliers, we record receivables for the contractually obligated amounts on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Warranty expense is recorded as a component of Cost of revenues in the consolidated statements of operations. Due to the magnitude of our automotive business, our accrued warranty balance is primarily related to our automotive segment. Accrued warranty activity consisted of the following (in millions):

	Year Ended December 31,		
	2023	2022	2021
Accrued warranty—beginning of period	$ 3,505	$ 2,101	$ 1,468
Warranty costs incurred	(1,225)	(803)	(525)
Net changes in liability for pre-existing warranties, including expirations and foreign exchange impact	539	522	102
Provision for warranty	2,333	1,685	1,056
Accrued warranty—end of period	$ 5,152	$ 3,505	$ 2,101

Customer Deposits

Customer deposits primarily consist of refundable cash payments from customers at the time they place an order or reservation for a vehicle or an energy product and any additional payments up to the point of delivery or the completion of installation. Customer deposits also include prepayments on contracts that can be cancelled without significant penalties, such as vehicle maintenance plans. Customer deposits are included in Accrued liabilities and other on the consolidated balance sheets until refunded, forfeited or applied towards the customer's purchase balance.

Government Assistance Programs and Incentives

Globally, the operation of our business is impacted by various government programs, incentives, and other arrangements. Government incentives are recorded in our consolidated financial statements in accordance with their purpose as a reduction of expense, or an offset to the related capital asset. The benefit is generally recorded when all conditions attached to the incentive have been met or are expected to be met and there is reasonable assurance of their receipt.

The IRA Incentives

On August 16, 2022, the IRA was enacted into law and is effective for taxable years beginning after December 31, 2022. The IRA includes multiple incentives to promote clean energy, electric vehicles, battery and energy storage manufacture or purchase, in addition to a new corporate alternative minimum tax of 15% on adjusted financial statement income of corporations with profits greater than $1 billion. Some of these measures are expected to materially affect our consolidated financial statements. For the year ended December 31, 2023, the impact from our IRA incentive was primarily a reduction of our material costs in our consolidated statement of operations. We will continue to evaluate the effects of the IRA as more guidance is issued and the relevant implications to our consolidated financial statements.

Gigafactory New York—New York State Investment and Lease

We have a lease through the Research Foundation for the SUNY Foundation with respect to Gigafactory New York. Under the lease and a related research and development agreement, we are continuing to designate further buildouts at the facility. We are required to comply with certain covenants, including hiring and cumulative investment targets. Under the terms of the arrangement, the SUNY Foundation paid for a majority of the construction costs related to the manufacturing facility and the acquisition and commissioning of certain manufacturing equipment; and we are responsible for any construction or equipment costs in excess of such amount (refer to Note 15, *Commitments and Contingencies*). This incentive reduces the related lease costs of the facility within the Energy generation and storage cost of revenues and operating expense line items in our consolidated statements of operations.

Gigafactory Shanghai—Land Use Rights and Economic Benefits

We have an agreement with the local government of Shanghai for land use rights at Gigafactory Shanghai. Under the terms of the arrangement, we are required to meet a cumulative capital expenditure target and an annual tax revenue target starting at the end of 2023. In addition, the Shanghai government has granted to our Gigafactory Shanghai subsidiary certain incentives to be used in connection with eligible capital investments at Gigafactory Shanghai (refer to Note 15, *Commitments and Contingencies*). For the year ended December 31, 2022, we received grant funding of $76 million. These incentives offset the related costs of our facilities and are recorded as a reduction of the cost of the capital investment within the Property, plant and equipment, net line item in our consolidated balance sheets. The incentive therefore reduces the depreciation expense over the useful lives of the related equipment.

Nevada Tax Incentives

In connection with the construction of Gigafactory Nevada, we entered into agreements with the State of Nevada and Storey County in Nevada that provide abatements for specified taxes, discounts to the base tariff energy rates and transferable tax credits of up to $195 million in consideration of capital investment and hiring targets that were met at Gigafactory Nevada.

Gigafactory Texas Tax Incentives

In connection with the construction of Gigafactory Texas, we entered into a20-year agreement with Travis County in Texas pursuant to which we would receive grant funding equal to 70-80% of property taxes paid by us to Travis County and a separate 10-year agreement with the Del Valle Independent School District in Texas pursuant to which a portion of the taxable value of our property would be capped at a specified amount, in each case subject to our meeting certain minimum economic development metrics through our construction and operations at Gigafactory Texas. This incentive is recorded as a reduction of the related expenses within the Cost of automotive revenues and operating expense line items of our consolidated statements of operations. As of December 31, 2023, the grant funding related to property taxes paid were immaterial.

Defined Contribution Plan

We have a 401(k) savings plan in the U.S. that is intended to qualify as a deferred salary arrangement under Section 401(k) of the Internal Revenue Code and a number of savings plans internationally. Under the 401(k) savings plan, participating employees may elect to contribute up to 90% of their eligible compensation, subject to certain limitations. Beginning in January 2022, we began to match 50% of each employee's contributions up to a maximum of 6% (capped at $3,000) of the employee's eligible compensation, vested upon one year of service. During the years ended December 31, 2023 and 2022, we recognized $99 million and $91 million, respectively, of expenses related to employer contributions for the 401(k) savings plan.

Recent Accounting Pronouncements

Recently issued accounting pronouncements not yet adopted

In November 2023, the Financial Accounting Standards Board ("FASB") issued ASU No. 2023-07, Improvements to Reportable Segment Disclosures (Topic 280). This ASU updates reportable segment disclosure requirements by requiring disclosures of significant reportable segment expenses that are regularly provided to the Chief Operating Decision Maker ("CODM") and included within each reported measure of a segment's profit or loss. This ASU also requires disclosure of the title and position of the individual identified as the CODM and an explanation of how the CODM uses the reported measures of a segment's profit or loss in assessing segment performance and deciding how to allocate resources. The ASU is effective for annual periods beginning after December 15, 2023, and interim periods within fiscal years beginning after December 15, 2024. Adoption of the ASU should be applied retrospectively to all prior periods presented in the financial statements. Early adoption is also permitted. This ASU will likely result in us including the additional required disclosures when adopted. We are currently evaluating the provisions of this ASU and expect to adopt them for the year ending December 31, 2024.

In December 2023, the FASB issued ASU No. 2023-08, Accounting for and Disclosure of Crypto Assets (Subtopic 350-60). This ASU requires certain crypto assets to be measured at fair value separately in the balance sheet and income statement each reporting period. This ASU also enhances the other intangible asset disclosure requirements by requiring the name, cost basis, fair value, and number of units for each significant crypto holding. The ASU is effective for annual periods beginning after December 15, 2024, including interim periods within those fiscal years. Adoption of the ASU requires a cumulative-effect adjustment to the opening balance of retained earnings as of the beginning of the annual reporting period in which an entity adopts the amendments. Early adoption is also permitted, including adoption in an interim period. However, if the ASU is early adopted in an interim period, an entity must adopt the ASU as of the beginning of the fiscal year that includes the interim period. This ASU will result in gains and losses recorded in the consolidated financial statements of operations and additional disclosures when adopted. We are currently evaluating the adoption of this ASU and it will affect the carrying value of our crypto assets held and the gains and losses relating thereto, once adopted.

In December 2023, the FASB issued ASU No. 2023-09, Improvements to Income Tax Disclosures (Topic 740). The ASU requires disaggregated information about a reporting entity's effective tax rate reconciliation as well as additional information on income taxes paid. The ASU is effective on a prospective basis for annual periods beginning after December 15, 2024. Early adoption is also permitted for annual financial statements that have not yet been issued or made available for issuance. This ASU will result in the required additional disclosures being included in our consolidated financial statements, once adopted.

Recently adopted accounting pronouncements

In October 2021, the FASB issued ASU No. 2021-08, Accounting for Contract Assets and Contract Liabilities from Contracts with Customers (Topic 805). This ASU requires an acquirer in a business combination to recognize and measure contract assets and contract liabilities (deferred revenue) from acquired contracts using the revenue recognition guidance in Topic 606. At the acquisition date, the acquirer applies the revenue model as if it had originated the acquired contracts. The ASU is effective for annual periods beginning after December 15, 2022, including interim periods within those fiscal years. We adopted this ASU prospectively on January 1, 2023. This ASU has not and is currently not expected to have a material impact on our consolidated financial statements.

In March 2022, the FASB issued ASU 2022-02, Troubled Debt Restructurings and Vintage Disclosures. This ASU eliminates the accounting guidance for troubled debt restructurings by creditors that have adopted ASU 2016-13, Measurement of Credit Losses on Financial Instruments, which we adopted on January 1, 2020. This ASU also enhances the disclosure requirements for certain loan refinancing and restructurings by creditors when a borrower is experiencing financial difficulty. In addition, the ASU amends the guidance on vintage disclosures to require entities to disclose current period gross write-offs by year of origination for financing receivables and net investments in leases within the scope of ASC 326-20. The ASU is effective for annual periods beginning after December 15, 2022, including interim periods within those fiscal years. We adopted the ASU prospectively on January 1, 2023. This ASU has not and is currently not expected to have a material impact on our consolidated financial statements.

ASU 2020-06

In August 2020, the FASB issued ASU 2020-06, Accounting for Convertible Instruments and Contracts in an Entity's Own Equity. The ASU simplifies the accounting for convertible instruments by removing certain separation models in ASC 470-20, Debt—Debt with Conversion and Other Options, for convertible instruments. The ASU updates the guidance on certain embedded conversion features that are not required to be accounted for as derivatives under Topic 815, Derivatives and Hedging, or that do not result in substantial premiums accounted for as paid-in capital, such that those features are no longer required to be separated from the host contract. The convertible debt instruments will be accounted for as a single liability measured at amortized cost. This will also result in the interest expense recognized for convertible debt instruments to be typically closer to the coupon interest rate when applying the guidance in Topic 835, Interest. Further, the ASU made amendments to the EPS guidance in Topic 260 for convertible debt instruments, the most significant impact of which is requiring the use of the if-converted method for diluted EPS calculation, and no longer allowing the net share settlement method. The ASU also made revisions to Topic 815-40, which provides guidance on how an entity must determine whether a contract qualifies for a scope exception from derivative accounting. The amendments to Topic 815-40 change the scope of contracts that are recognized as assets or liabilities.

On January 1, 2021, we adopted the ASU using the modified retrospective method. We recognized a favorable $211 million cumulative effect of initially applying the ASU as an adjustment to the January 1, 2021 opening balance of accumulated deficit. Due to the recombination of the equity conversion component of our convertible debt remaining outstanding, additional paid in capital was reduced by $474 million and convertible senior notes (mezzanine equity) was reduced by $51 million. The removal of the remaining debt discounts recorded for this previous separation had the effect of increasing our net debt balance by $269 million and we reduced property, plant and equipment by $45 million related to previously capitalized interest. The prior period consolidated financial statements have not been retrospectively adjusted and continue to be reported under the accounting standards in effect for those periods.

Note 3 – Digital Assets, Net

During the years ended December 31, 2023 and 2022, we purchased and/or received immaterial amounts of digital assets. During the year ended December 31, 2023, we recorded an immaterial amount of impairment losses on digital assets. During the year ended December 31, 2022, we recorded $204 million of impairment losses on digital assets and realized gains of $64 million in connection with converting our holdings of digital assets into fiat currency. The gains are presented net of impairment losses in Restructuring and other in the consolidated statements of operations. As of December 31, 2023 and 2022, the carrying value of our digital assets held reflects cumulative impairment of $204 million.

Note 4 – Goodwill and Intangible Assets

Goodwill increased $59 million within the automotive segment from $194 million as of December 31, 2022 to $253 million as of December 31, 2023 primarily from a business combination, net of the impact of a divestiture. There were no accumulated impairment losses as of December 31, 2023 and 2022.

The net carrying value of our intangible assets decreased from $215 million as of December 31, 2022 to $178 million as of December 31, 2023 mainly from amortization.

Note 5 – Fair Value of Financial Instruments

ASC 820, *Fair Value Measurements* ("ASC 820") states that fair value is an exit price, representing the amount that would be received to sell an asset or paid to transfer a liability in an orderly transaction between market participants. As such, fair value is a market-based measurement that should be determined based on assumptions that market participants would use in pricing an asset or a liability. The three-tiered fair value hierarchy, which prioritizes which inputs should be used in measuring fair value, is comprised of: (Level I) observable inputs such as quoted prices in active markets; (Level II) inputs other than quoted prices in active markets that are observable either directly or indirectly and (Level III) unobservable inputs for which there is little or no market data. The fair value hierarchy requires the use of observable market data when available in determining fair value. Our assets and liabilities that were measured at fair value on a recurring basis were as follows (in millions):

	December 31, 2023				December 31, 2022			
	Fair Value	Level I	Level II	Level III	Fair Value	Level I	Level II	Level III
Money market funds	$ 109	$ 109	$ —	$ —	$ 2,188	$ 2,188	$ —	$ —
U.S. government securities	5,136	—	5,136	—	894	—	894	—
Corporate debt securities	480	—	480	—	885	—	885	—
Certificates of deposit and time deposits	6,996	—	6,996	—	4,253	—	4,253	—
Commercial paper	470	—	470	—	—	—	—	—
Total	$ 13,191	$ 109	$ 13,082	$ —	$ 8,220	$ 2,188	$ 6,032	$ —

All of our money market funds were classified within Level I of the fair value hierarchy because they were valued using quoted prices in active markets. Our U.S. government securities, certificates of deposit, commercial paper, time deposits and corporate debt securities are classified within Level II of the fair value hierarchy and the market approach was used to determine fair value of these investments.

Our cash, cash equivalents and investments classified by security type as of December 31, 2023 and 2022 consisted of the following (in millions):

	December 31, 2023					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	$ 15,903	$ —	$ —	$ 15,903	$ 15,903	$ —
Money market funds	109	—	—	109	109	—
U.S. government securities	5,136	1	(1)	5,136	277	4,859
Corporate debt securities	485	1	(6)	480	—	480
Certificates of deposit and time deposits	6,995	1	—	6,996	—	6,996
Commercial paper	470	—	—	470	109	361
Total cash, cash equivalents and short-term investments	$ 29,098	$ 3	$ (7)	$ 29,094	$ 16,398	$ 12,696

	December 31, 2022					
	Adjusted Cost	Gross Unrealized Gains	Gross Unrealized Losses	Fair Value	Cash and Cash Equivalents	Short-Term Investments
Cash	$ 13,965	$ —	$ —	$ 13,965	$ 13,965	$ —
Money market funds	2,188	—	—	2,188	2,188	—
U.S. government securities	897	—	(3)	894	—	894
Corporate debt securities	907	—	(22)	885	—	885
Certificates of deposit and time deposits	4,252	1	—	4,253	100	4,153
Total cash, cash equivalents and short-term investments	$ 22,209	$ 1	$ (25)	$ 22,185	$ 16,253	$ 5,932

We record gross realized gains, losses and credit losses as a component of Other income (expense), net in the consolidated statements of operations. For the years ended December 31, 2023 and 2022, we did not recognize any material gross realized gains, losses or credit losses. The ending allowance balances for credit losses were immaterial as of December 31, 2023 and 2022. We have determined that the gross unrealized losses on our investments as of December 31, 2023 and 2022 were temporary in nature.

The following table summarizes the fair value of our investments by stated contractual maturities as of December 31, 2023 (in millions):

Due in 1 year or less	$ 12,374
Due in 1 year through 5 years	297
Due in 5 years through 10 years	25
Total	$ 12,696

Disclosure of Fair Values

Our financial instruments that are not re-measured at fair value include accounts receivable, financing receivables, other receivables, digital assets, accounts payable, accrued liabilities, customer deposits and debt. The carrying values of these financial instruments materially approximate their fair values, other than our 2.00% Convertible Senior Notes due in 2024 ("2024 Notes") and digital assets.

We estimate the fair value of the 2024 Notes using commonly accepted valuation methodologies and market-based risk measurements that are indirectly observable, such as credit risk (Level II). In addition, we estimate the fair values of our digital assets based on quoted prices in active markets (Level I). The following table presents the estimated fair values and the carrying values (in millions):

	December 31, 2023		December 31, 2022	
	Carrying Value	Fair Value	Carrying Value	Fair Value
2024 Notes	$ 37	$ 443	$ 37	$ 223
Digital assets, net	$ 184	$ 487	$ 184	$ 191

Note 6 – Inventory

Our inventory consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Raw materials	$ 5,390	$ 6,137
Work in process	2,016	2,385
Finished goods (1)	5,049	3,475
Service parts	1,171	842
Total	$ 13,626	$ 12,839

(1) Finished goods inventory includes products in transit to fulfill customer orders, new vehicles available for sale, used vehicles and energy products available for sale.

We write-down inventory for any excess or obsolete inventories or when we believe that the net realizable value of inventories is less than the carrying value. During the years ended December 31, 2023, 2022 and 2021 we recorded write-downs of $233 million, $144 million and $106 million, respectively, in Cost of revenues in the consolidated statements of operations.

Note 7 – Solar Energy Systems, Net

Our solar energy systems, net, consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Solar energy systems in service	$ 6,755	$ 6,785
Initial direct costs related to customer solar energy system lease acquisition costs	104	104
	6,859	6,889
Less: accumulated depreciation and amortization (1)	(1,643)	(1,418)
	5,216	5,471
Solar energy systems under construction	1	2
Solar energy systems pending interconnection	12	16
Solar energy systems, net (2)	$ 5,229	$ 5,489

(1) Depreciation and amortization expense during the years ended December 31, 2023, 2022 and 2021 was $235 million, $235 million and $236 million, respectively.

(2) As of December 31, 2023 and 2022, there were $740 million and $802 million, respectively, of gross solar energy systems under lease pass-through fund arrangements with accumulated depreciation of $157 million and $148 million, respectively.

Note 8 – Property, Plant and Equipment, Net

Our property, plant and equipment, net, consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Machinery, equipment, vehicles and office furniture	$ 16,372	$ 13,558
Tooling	3,147	2,579
Leasehold improvements	3,168	2,366
Land and buildings	9,505	7,751
Computer equipment, hardware and software	3,799	2,072
Construction in progress	5,791	4,263
	41,782	32,589
Less: Accumulated depreciation	(12,057)	(9,041)
Total	$ 29,725	$ 23,548

Construction in progress is primarily comprised of ongoing construction and expansion of our facilities, and equipment and tooling related to the manufacturing of our products. Completed assets are transferred to their respective asset classes and depreciation begins when an asset is ready for its intended use.

Depreciation expense during the years ended December 31, 2023, 2022 and 2021 was $3.33 billion, $2.42 billion and $1.91 billion, respectively.

Panasonic has partnered with us on Gigafactory Nevada with investments in the production equipment that it uses to manufacture and supply us with battery cells. Under our arrangement with Panasonic, we plan to purchase the full output from their production equipment at negotiated prices. As the terms of the arrangement convey a finance lease under ASC 842, we account for their production equipment as leased assets when production commences. We account for each lease and any non-lease components associated with that lease as a single lease component for all asset classes, except production equipment classes embedded in supply agreements. This results in us recording the cost of their production equipment within Property, plant and equipment, net, on the consolidated balance sheets with a corresponding liability recorded to debt and finance leases. Depreciation on Panasonic production equipment is computed using the units-of-production method whereby capitalized costs are amortized over the total estimated productive life of the respective assets. As of December 31, 2023 and 2022, we had cumulatively capitalized gross costs of $2.02 billion and $2.01 billion, respectively, on the consolidated balance sheets in relation to the production equipment under our Panasonic arrangement.

Note 9 – Accrued Liabilities and Other

Our accrued liabilities and other current liabilities consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Accrued purchases (1)	$ 2,721	$ 2,747
Accrued warranty reserve, current portion	1,546	1,025
Payroll and related costs	1,325	1,026
Taxes payable (2)	1,204	1,235
Customer deposits	876	1,063
Operating lease liabilities, current portion	672	485
Sales return reserve, current portion	219	270
Other current liabilities	517	354
Total	$ 9,080	$ 8,205

(1) Accrued purchases primarily reflects receipts of goods and services for which we had not yet been invoiced. As we are invoiced for these goods and services, this balance will reduce and accounts payable will increase.

(2) Taxes payable includes value added tax, income tax, sales tax, property tax and use tax payables.

Note 10 – Other Long-Term Liabilities

Our other long-term liabilities consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Operating lease liabilities	$ 3,671	$ 2,164
Accrued warranty reserve	3,606	2,480
Other non-current liabilities	876	686
Total other long-term liabilities	$ 8,153	$ 5,330

Note 11 – Debt

The following is a summary of our debt and finance leases as of December 31, 2023 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (1)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	$ 37	$ —	$ 37	$ —	2.00 %	May 2024
RCF Credit Agreement	—	—	—	5,000	Not applicable	January 2028
Solar Bonds	—	7	7	—	4.70-5.75%	March 2025 - January 2031
Other	—	—	—	28	Not applicable	December 2026
Total recourse debt	37	7	44	5,028		
Non-recourse debt:						
Automotive Asset-backed Notes	1,906	2,337	4,259	—	0.60-6.57%	July 2024-May 2031
Solar Asset-backed Notes	4	8	13	—	4.80 %	December 2026
Cash Equity Debt	28	330	367	—	5.25-5.81%	July 2033-January 2035
Total non-recourse debt	1,938	2,675	4,639	—		
Total debt	1,975	2,682	$ 4,683	$ 5,028		
Finance leases	398	175				
Total debt and finance leases	$ 2,373	$ 2,857				

The following is a summary of our debt and finance leases as of December 31, 2022 (in millions):

	Net Carrying Value		Unpaid Principal Balance	Unused Committed Amount (2)	Contractual Interest Rates	Contractual Maturity Date
	Current	Long-Term				
Recourse debt:						
2024 Notes	$ —	$ 37	$ 37	$ —	2.00 %	May 2024
Credit Agreement	—	—	—	2,266	Not applicable	July 2023
Solar Bonds	—	7	7	—	4.70-5.75%	March 2025 - January 2031
Total recourse debt	—	44	44	2,266		
Non-recourse debt:						
Automotive Asset-backed Notes	984	613	1,603	—	0.36-4.64%	December 2023-September 2025
Solar Asset-backed Notes	4	13	17	—	4.80 %	December 2026
Cash Equity Debt	28	359	397	—	5.25-5.81%	July 2033-January 2035
Automotive Lease-backed Credit Facilities	—	—	—	151	Not applicable	September 2024
Total non-recourse debt	1,016	985	2,017	151		
Total debt	1,016	1,029	$ 2,061	$ 2,417		
Finance leases	486	568				
Total debt and finance leases	$ 1,502	$ 1,597				

(1) There are no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our RCF Credit Agreement, except certain specified conditions prior to draw-down. Refer to the section below for the terms of the facility.

(2) There were no restrictions on draw-down or use for general corporate purposes with respect to any available committed funds under our credit facilities, except certain specified conditions prior to draw-down, including pledging to our lenders sufficient amounts of qualified receivables, inventories, leased vehicles and our interests in those leases or various other assets as described below.

Recourse debt refers to debt that is recourse to our general assets. Non-recourse debt refers to debt that is recourse to only assets of our subsidiaries. The differences between the unpaid principal balances and the net carrying values are due to debt discounts or deferred issuance costs. As of December 31, 2023, we were in material compliance with all financial debt covenants.

2024 Notes

The closing price of our common stock continued to exceed130% of the applicable conversion price of our 2024 Notes on at least20 of the last 30 consecutive trading days of each quarter in 2023, causing the 2024 Notes to be convertible by their holders in the subsequent quarter. As of December 31, 2023, the if-converted value of the notes exceeds the outstanding principal amount by $406 million. Upon conversion, the 2024 Notes will be settled in cash, shares of our common stock or a combination thereof, at our election.

Credit Agreement

In June 2015, we entered into a senior asset-based revolving credit agreement (as amended from time to time, the "Credit Agreement") with a syndicate of banks. Borrowed funds bear interest, at our option, at an annual rate of (a) 1% plus LIBOR or (b) the highest of (i) the federal funds rate plus0.50%, (ii) the lenders' "prime rate" or (iii) 1% plus LIBOR. The fee for undrawn amounts is0.25% per annum. The Credit Agreement is secured by certain of our accounts receivable, inventory and equipment. Availability under the Credit Agreement is based on the value of such assets, as reduced by certain reserves.

In January 2023, we entered into a5-year senior unsecured revolving credit facility (the "RCF Credit Agreement") with a syndicate of banks to replace the existing Credit Agreement, which was terminated. The RCF Credit Agreement contains two optional one-year extensions and has a total commitment of up to $5.00 billion, which could be increased up to $7.00 billion under certain circumstances. The underlying borrowings may be used for general corporate purposes. Borrowed funds accrue interest at a variable rate equal to: (i) for dollar-denominated loans, at our election, (a) Term SOFR (the forward-looking secured overnight financing rate) plus 0.10%, or (b) an alternate base rate; (ii) for loans denominated in pounds sterling, SONIA (the sterling overnight index average reference rate); or (iii) for loans denominated in euros, an adjusted EURIBOR (euro interbank offered rate); in each case, plus an applicable margin. The applicable margin will be based on the rating assigned to our senior, unsecured long-term indebtedness (the "Credit Rating") from time to time. The fee for undrawn amounts is variable based on the Credit Rating and is currently 0.125% per annum.

Automotive Asset-backed Notes

From time to time, we transfer receivables and/or beneficial interests related to certain vehicles (either leased or financed) into special purpose entities ("SPEs") and issue Automotive Asset-backed Notes, backed by these automotive assets to investors. The SPEs are consolidated in the financial statements. The cash flows generated by these automotive assets are used to service the principal and interest payments on the Automotive Asset-backed Notes and satisfy the SPEs' expenses, and any remaining cash is distributed to the owners of the SPEs. We recognize revenue earned from the associated customer lease or financing contracts in accordance with our revenue recognition policy. The SPEs' assets and cash flows are not available to our other creditors, and the creditors of the SPEs, including the Automotive Asset-backed Note holders, have no recourse to our other assets.

In 2023, we transferred beneficial interests related to certain leased vehicles and financing receivables into SPEs and issued $3.93 billion in aggregate principal amount of Automotive Asset-backed Notes, with terms similar to our other previously issued Automotive Asset-backed Notes. The proceeds from the issuance, net of debt issuance costs, were $3.92 billion.

Cash Equity Debt

In connection with the cash equity financing deals closed in 2016, our subsidiaries issued $502 million in aggregate principal amount of debt that bears interest at fixed rates. This debt is secured by, among other things, our interests in certain financing funds and is non-recourse to our other assets.

Automotive Lease-backed Credit Facilities

In the third quarter of 2023, we terminated our Automotive Lease-backed Credit Facilities and the previously committed funds are no longer available for future borrowings.

Pledged Assets

As of December 31, 2023 and 2022, we had pledged or restricted $4.64 billion and $2.02 billion of our assets (consisting principally of operating lease vehicles, financing receivables, restricted cash, and equity interests in certain SPEs) as collateral for our outstanding debt.

Schedule of Principal Maturities of Debt

The future scheduled principal maturities of debt as of December 31, 2023 were as follows (in millions):

	Recourse debt	Non-recourse debt	Total
2024	$ 37	$ 1,941	$ 1,978
2025	4	1,663	1,667
2026	—	494	494
2027	—	276	276
2028	—	44	44
Thereafter	3	221	224
Total	$ 44	$ 4,639	$ 4,683

Note 12 – Leases

We have entered into various operating and finance lease agreements for certain of our offices, manufacturing and warehouse facilities, retail and service locations, data centers, equipment, vehicles, and solar energy systems, worldwide. We determine if an arrangement is a lease, or contains a lease, at inception and record the leases in our financial statements upon lease commencement, which is the date when the underlying asset is made available for use by the lessor.

We have lease agreements with lease and non-lease components, and have elected to utilize the practical expedient to account for lease and non-lease components together as a single combined lease component, from both a lessee and lessor perspective with the exception of direct sales-type leases and production equipment classes embedded in supply agreements. From a lessor perspective, the timing and pattern of transfer are the same for the non-lease components and associated lease component and, the lease component, if accounted for separately, would be classified as an operating lease.

We have elected not to present short-term leases on the consolidated balance sheet as these leases have a lease term of 12 months or less at lease inception and do not contain purchase options or renewal terms that we are reasonably certain to exercise. All other lease assets and lease liabilities are recognized based on the present value of lease payments over the lease term at commencement date. Because most of our leases do not provide an implicit rate of return, we used our incremental borrowing rate based on the information available at lease commencement date in determining the present value of lease payments.

Our leases, where we are the lessee, often include options to extend the lease term for up to10 years. Some of our leases also include options to terminate the lease prior to the end of the agreed upon lease term. For purposes of calculating lease liabilities, lease terms include options to extend or terminate the lease when it is reasonably certain that we will exercise such options.

Lease expense for operating leases is recognized on a straight-line basis over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Certain operating leases provide for annual increases to lease payments based on an index or rate. We calculate the present value of future lease payments based on the index or rate at the lease commencement date for new leases. Differences between the calculated lease payment and actual payment are expensed as incurred. Amortization of finance lease assets is recognized over the lease term as cost of revenues or operating expenses depending on the nature of the leased asset. Interest expense on finance lease liabilities is recognized over the lease term within Interest expense in the consolidated statements of operations.

The balances for the operating and finance leases where we are the lessee are presented as follows (in millions) within our consolidated balance sheets:

	December 31, 2023	December 31, 2022
Operating leases:		
Operating lease right-of-use assets	$ 4,180	$ 2,563
Accrued liabilities and other	$ 672	$ 485
Other long-term liabilities	3,671	2,164
Total operating lease liabilities	$ 4,343	$ 2,649
Finance leases:		
Solar energy systems, net	$ 23	$ 25
Property, plant and equipment, net	601	1,094
Total finance lease assets	$ 624	$ 1,119
Current portion of long-term debt and finance leases	$ 398	$ 486
Long-term debt and finance leases, net of current portion	175	568
Total finance lease liabilities	$ 573	$ 1,054

The components of lease expense are as follows (in millions) within our consolidated statements of operations:

	Year Ended December 31,		
	2023	2022	2021
Operating lease expense:			
Operating lease expense (1)	$ 1,153	$ 798	$ 627
Finance lease expense:			
Amortization of leased assets	$ 506	$ 493	$ 415
Interest on lease liabilities	45	72	89
Total finance lease expense	$ 551	$ 565	$ 504
Total lease expense	$ 1,704	$ 1,363	$ 1,131

(1) Includes short-term leases and variable lease costs, which are immaterial.

Other information related to leases where we are the lessee is as follows:

	December 31, 2023	December 31, 2022
Weighted-average remaining lease term:		
Operating leases	7.4 years	6.4 years
Finance leases	2.3 years	3.1 years
Weighted-average discount rate:		
Operating leases	5.6 %	5.3 %
Finance leases	5.5 %	5.7 %

Supplemental cash flow information related to leases where we are the lessee is as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cash paid for amounts included in the measurement of lease liabilities:			
Operating cash outflows from operating leases	$ 1,084	$ 754	$ 616
Operating cash outflows from finance leases (interest payments)	$ 47	$ 75	$ 89
Leased assets obtained in exchange for finance lease liabilities	$ 10	$ 58	$ 486
Leased assets obtained in exchange for operating lease liabilities	$ 2,170	$ 1,059	$ 818

As of December 31, 2023, the maturities of our operating and finance lease liabilities (excluding short-term leases) are as follows (in millions):

	Operating Leases	Finance Leases
2024	$ 892	$ 418
2025	831	81
2026	706	57
2027	603	38
2028	508	2
Thereafter	1,820	4
Total minimum lease payments	5,360	600
Less: Interest	1,017	27
Present value of lease obligations	4,343	573
Less: Current portion	672	398
Long-term portion of lease obligations	$ 3,671	$ 175

As of December 31, 2023, we have excluded from the table above additional operating leases that have not yet commenced with aggregate rent payments of $1.53 billion. These operating leases will commence between fiscal year 2024 and 2025 with lease terms of 2 years to 20 years.

Operating Lease and Sales-type Lease Receivables

We are the lessor of certain vehicle and solar energy system arrangements as described in Note 2, *Summary of Significant Accounting Policies*. As of December 31, 2023, maturities of our operating lease and sales-type lease receivables from customers for each of the next five years and thereafter were as follows (in millions):

	Operating Leases	Sales-type Leases
2024	$ 1,405	$ 227
2025	960	214
2026	461	210
2027	227	102
2028	197	25
Thereafter	1,492	2
Gross lease receivables	$ 4,742	$ 780

The above table does not include vehicle sales to customers or leasing partners with a resale value guarantee as the cash payments were received upfront. For our solar PPA arrangements, customers are charged solely based on actual power produced by the installed solar energy system at a predefined rate per kilowatt-hour of power produced. The future payments from such arrangements are not included in the above table as they are a function of the power generated by the related solar energy systems in the future.

Net Investment in Sales-type Leases

Net investment in sales-type leases, which is the sum of the present value of the future contractual lease payments, is presented on the consolidated balance sheets as a component of Prepaid expenses and other current assets for the current portion and as Other non-current assets for the long-term portion. Lease receivables relating to sales-type leases are presented on the consolidated balance sheets as follows (in millions):

	December 31, 2023	December 31, 2022
Gross lease receivables	$ 780	$ 837
Unearned interest income	(78)	(95)
Allowance for expected credit losses	(6)	(4)
Net investment in sales-type leases	$ 696	$ 738
Reported as:		
Prepaid expenses and other current assets	$ 189	$ 164
Other non-current assets	507	574
Net investment in sales-type leases	$ 696	$ 738

Lease Pass-Through Financing Obligation

As of December 31, 2023, we have five transactions referred to as "lease pass-through fund arrangements." Under these arrangements, our wholly owned subsidiaries finance the cost of solar energy systems with investors through arrangements contractually structured as master leases for an initial term ranging between 10 and 25 years. These solar energy systems are subject to lease or PPAs with customers with an initial term not exceeding25 years.

Under a lease pass-through fund arrangement, the investor makes a large upfront payment to the lessor, which is one of our subsidiaries, and in some cases, subsequent periodic payments. As of December 31, 2023, the future minimum master lease payments to be received from investors, for each of the next five years and thereafter, were as follows (in millions):

2024	$ 18
2025	27
2026	28
2027	29
2028	29
Thereafter	337
Total	$ 468

Note 13 – Equity Incentive Plans

In June 2019, we adopted the 2019 Equity Incentive Plan (the "2019 Plan"). The 2019 Plan provides for the grant of stock options, restricted stock, RSUs, stock appreciation rights, performance units and performance shares to our employees, directors and consultants. Stock options granted under the 2019 Plan may be either incentive stock options or nonstatutory stock options. Incentive stock options may only be granted to our employees. Nonstatutory stock options may be granted to our employees, directors and consultants. Generally, our stock options and RSUs vest over four years and our stock options are exercisable over a maximum period of10 years from their grant dates. Vesting typically terminates when the employment or consulting relationship ends.

As of December 31, 2023, 131.1 million shares were reserved and available for issuance under the 2019 Plan.

The following table summarizes our stock option and RSU activity for the year ended December 31, 2023:

	Stock Options				RSUs	
	Number of Options (in thousands)	Weighted-Average Exercise Price	Weighted-Average Remaining Contractual Life (years)	Aggregate Intrinsic Value (in billions)	Number of RSUs (in thousands)	Weighted-Average Grant Date Fair Value
Beginning of period	343,564	$ 30.65			21,333	$ 162.32
Granted	9,521	$ 226.50			11,743	$ 228.33
Exercised or released	(7,626)	$ 43.07			(11,085)	$ 116.47
Cancelled	(1,438)	$ 194.23			(2,903)	$ 192.22
End of period	344,021	$ 35.11	4.31	$ 73.57	19,088	$ 225.01
Vested and expected to vest, December 31, 2023	340,884	$ 33.38	4.27	$ 73.45	18,446	$ 225.76
Exercisable and vested, December 31, 2023	329,124	$ 27.07	4.11	$ 72.90		

The weighted-average grant date fair value of RSUs granted in the years ended December 31, 2023, 2022 and 2021 was $228.33, $239.85 and $261.33, respectively. The aggregate release date fair value of RSUs in the years ended December 31, 2023, 2022 and 2021 was $2.50 billion, $4.32 billion and $5.70 billion, respectively.

The aggregate intrinsic value of options exercised in the years ended December 31, 2023, 2022, and 2021 was $1.33 billion, $1.90 billion and $26.88 billion, respectively. During the year ended December 31, 2021, our CEO exercised all of the remaining vested options from the 2012 CEO Performance Award, which amounted to an intrinsic value of $23.45 billion.

ESPP

Our employees are eligible to purchase our common stock through payroll deductions of up to 15% of their eligible compensation, subject to any plan limitations. The purchase price would be 85% of the lower of the fair market value on the first and last trading days of each six-month offering period. During the years ended December 31, 2023, 2022 and 2021, under the ESPP we issued 2.1 million, 1.4 million and 1.5 million shares, respectively. As of December 31, 2023, there were 97.8 million shares available for issuance under the ESPP.

Fair Value Assumptions

We use the fair value method in recognizing stock-based compensation expense. Under the fair value method, we estimate the fair value of each stock option award with service or service and performance conditions and the ESPP on the grant date generally using the Black-Scholes option pricing model. The weighted-average assumptions used in the Black-Scholes model for stock options are as follows:

	Year Ended December 31,		
	2023	2022	2021
Risk-free interest rate	3.90 %	3.11 %	0.66 %
Expected term (in years)	4.5	4.1	4.3
Expected volatility	63 %	63 %	59 %
Dividend yield	0.0 %	0.0 %	0.0 %
Grant date fair value per share	$ 121.62	$ 114.51	$ 128.02

The fair value of RSUs with service or service and performance conditions is measured on the grant date based on the closing fair market value of our common stock. The risk-free interest rate is based on the U.S. Treasury yield for zero-coupon U.S. Treasury notes with maturities approximating each grant's expected life. We use our historical data in estimating the expected term of our employee grants. The expected volatility is based on the average of the implied volatility of publicly traded options for our common stock and the historical volatility of our common stock.

2018 CEO Performance Award

In March 2018, our stockholders approved the Board of Directors' grant of304.0 million stock option awards, as adjusted to give effect to the 2020 Stock Split and the 2022 Stock Split, to our CEO (the "2018 CEO Performance Award"). The 2018 CEO Performance Award consisted of 12 vesting tranches with a vesting schedule based entirely on the attainment of both operational milestones (performance conditions) and market conditions, assuming continued employment either as the CEO or as both Executive Chairman and Chief Product Officer and service through each vesting date. Each of the 12 vesting tranches of the 2018 CEO Performance Award vested upon certification by the Board of Directors that both (i) the market capitalization milestone for such tranche, which began at $100.0 billion for the first tranche and increases by increments of $50.0 billion thereafter (based on both a six calendar month trailing average and a 30 calendar day trailing average, counting only trading days), had been achieved, and (ii) any one of the following eight operational milestones focused on total revenue or any one of theeight operational milestones focused on Adjusted EBITDA had been achieved for the four consecutive fiscal quarters on an annualized basis and subsequently reported by us in our consolidated financial statements filed with our Forms 10-Q and/or 10-K. Adjusted EBITDA was defined as net income (loss) attributable to common stockholders before interest expense, provision (benefit) for income taxes, depreciation and amortization and stock-based compensation. Upon vesting and exercise, including the payment of the exercise price of $23.34 per share as adjusted to give effect to the 2020 Stock Split and the 2022 Stock Split, our CEO must hold shares that he acquires fofive years post-exercise, other than a cashless exercise where shares are simultaneously sold to pay for the exercise price and any required tax withholding.

The achievement status of the operational milestones as of December 31, 2023 is provided below.

Total Annualized Revenue			Annualized Adjusted EBITDA		
Milestone (in billions)		Achievement Status	Milestone (in billions)		Achievement Status
$	20.0	Achieved	$	1.5	Achieved
$	35.0	Achieved	$	3.0	Achieved
$	55.0	Achieved	$	4.5	Achieved
$	75.0	Achieved	$	6.0	Achieved
$	100.0	-	$	8.0	Achieved
$	125.0	-	$	10.0	Achieved
$	150.0	-	$	12.0	Achieved
$	175.0	-	$	14.0	Achieved

Stock-based compensation under the 2018 CEO Performance Award represented a non-cash expense and was recorded as a Selling, general, and administrative operating expense in our consolidated statements of operations. In each quarter since the grant of the 2018 CEO Performance Award, we had recognized expense, generally on a pro-rated basis, for only the number of tranches (up to the maximum of 12 tranches) that corresponded to the number of operational milestones that had been achieved or had been determined probable of being achieved in the future, in accordance with the following principles.

On the grant date, a Monte Carlo simulation was used to determine for each tranche (i) a fixed amount of expense for such tranche and (ii) the future time when the market capitalization milestone for such tranche was expected to be achieved, or its "expected market capitalization milestone achievement time." Separately, based on a subjective assessment of our future financial performance each quarter, we determined whether it was probable that we would achieve each operational milestone that had not previously been achieved or deemed probable of achievement and if so, the future time when we expected to achieve that operational milestone, or its "expected operational milestone achievement time."

As of December 31, 2022, all remaining unrecognized stock-based compensation expense under the 2018 CEO Performance Award had been recognized. For the years ended December 31, 2022 and 2021, we recorded stock-based compensation expense of $66 million and $910 million, respectively, related to the 2018 CEO Performance Award.

Other Performance-Based Grants

From time to time, the Compensation Committee of our Board of Directors grants certain employees performance-based RSUs and stock options.

As of December 31, 2023, we had unrecognized stock-based compensation expense of $655 million under these grants to purchase or receive an aggregate 5.3 million shares of our common stock. For awards probable of achievement, we estimate the unrecognized stock-based compensation expense of $110 million will be recognized over a weighted-average period of 4.0 years.

For the years ended December 31, 2023 and 2022, we recorded $57 million and $159 million, respectively, of stock-based compensation expense related to these grants, net of forfeitures.

Summary Stock-Based Compensation Information

The following table summarizes our stock-based compensation expense by line item in the consolidated statements of operations (in millions):

	Year Ended December 31,		
	2023	2022	2021
Cost of revenues	$ 741	$ 594	$ 421
Research and development	689	536	448
Selling, general and administrative	382	430	1,252
Total	$ 1,812	$ 1,560	$ 2,121

Our income tax benefits recognized from stock-based compensation arrangements wereimmaterial while we were under full valuation allowances on our U.S. deferred tax assets during the years ended December 31, 2022 and 2021. With the release of the valuation allowance associated with our federal and certain state deferred tax assets in 2023, income tax benefits recognized from stock-based compensation expense were $326 million during the year ended December 31, 2023. During the years ended December 31, 2023, 2022 and 2021, stock-based compensation expense capitalized to our consolidated balance sheets was $199 million, $245 million and $182 million, respectively. As of December 31, 2023, we had $4.82 billion of total unrecognized stock-based compensation expense related to non-performance awards, which will be recognized over a weighted-average period of 2.8 years.

Note 14 – Income Taxes

Our income before (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 was as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Domestic	$ 3,196	$ 5,524	$ (130)
Noncontrolling interest and redeemable noncontrolling interest	(23)	31	125
Foreign	6,800	8,164	6,348
Income before income taxes	$ 9,973	$ 13,719	$ 6,343

A (benefit from) provision for income taxes of $(5.00) billion, $1.13 billion and $699 million has been recognized for the years ended December 31, 2023, 2022 and 2021, respectively. The components of the (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 consisted of the following (in millions):

	Year Ended December 31,		
	2023	2022	2021
Current:			
Federal	$ 48	$ —	$ —
State	57	62	9
Foreign	1,243	1,266	839
Total current	1,348	1,328	848
Deferred:			
Federal	(5,246)	26	—
State	(653)	1	—
Foreign	(450)	(223)	(149)
Total deferred	(6,349)	(196)	(149)
Total (Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699

The reconciliation of taxes at the federal statutory rate to our (benefit from) provision for income taxes for the years ended December 31, 2023, 2022 and 2021 was as follows (in millions):

	Year Ended December 31,		
	2023	2022	2021
Tax at statutory federal rate	$ 2,094	$ 2,881	$ 1,332
State tax, net of federal benefit	(372)	51	6
Nondeductible executive compensation	23	14	201
Excess tax benefits related to stock-based compensation	(288)	(745)	(7,123)
Nontaxable manufacturing credit	(101)	—	—
Foreign income rate differential	(816)	(923)	(668)
U.S. tax credits	(593)	(276)	(328)
GILTI inclusion	670	1,279	1,008
Unrecognized tax benefits	183	252	28
Change in valuation allowance	(5,962)	(1,532)	6,165
Other	161	131	78
(Benefit from) provision for income taxes	$ (5,001)	$ 1,132	$ 699

We monitor the realizability of our deferred tax assets taking into account all relevant factors at each reporting period. As of December 31, 2023, based on the relevant weight of positive and negative evidence, including the amount of our taxable income in recent years which is objective and verifiable, and consideration of our expected future taxable earnings, we concluded that it is more likely than not that our U.S. federal and certain state deferred tax assets are realizable. As such, we released $6.54 billion of our valuation allowance associated with the U.S. federal and state deferred tax assets, with the exception of our California deferred tax assets. We continue to maintain a full valuation allowance against our California deferred tax assets as of December 31, 2023, because we concluded they are not more likely than not to be realized as we expect our California deferred tax assets generation in future years to exceed our ability to use these deferred tax assets.

Deferred tax assets (liabilities) as of December 31, 2023 and 2022 consisted of the following (in millions):

	December 31, 2023	December 31, 2022
Deferred tax assets:		
Net operating loss carry-forwards	$ 2,826	$ 4,486
Research and development credits	1,358	1,184
Other tax credits and attributes	827	217
Deferred revenue	1,035	751
Inventory and warranty reserves	1,258	819
Stock-based compensation	230	185
Operating lease right-of-use liabilities	930	554
Capitalized research and development costs	1,344	693
Deferred GILTI tax assets	760	466
Accruals and others	206	178
Total deferred tax assets	10,774	9,533
Valuation allowance	(892)	(7,349)
Deferred tax assets, net of valuation allowance	9,882	2,184
Deferred tax liabilities:		
Depreciation and amortization	(2,122)	(1,178)
Investment in certain financing funds	(133)	(238)
Operating lease right-of-use assets	(859)	(506)
Other	(116)	(15)
Total deferred tax liabilities	(3,230)	(1,937)
Deferred tax assets (liabilities), net of valuation allowance	$ 6,652	$ 247

As of December 31, 2023, we maintained valuation allowances of $892 million for deferred tax assets that are not more likely than not to be realized, which primarily included deferred tax assets in the state of California and certain foreign operating losses. The valuation allowance on our net deferred tax assets decreased by $6.46 billion and $1.73 billion during the years ended December 31, 2023 and 2022, respectively, and increased by $6.14 billion during the year ended December 31, 2021. The valuation allowance decrease during the year ended December 31, 2023 was primarily due to the release of our valuation allowance with respect to our U.S. federal and certain state deferred tax assets. The changes in valuation allowances during the years ended December 31, 2022 and 2021 were primarily due to changes in our U.S. deferred tax assets and liabilities in the respective year. Among our deferred tax assets in foreign jurisdictions, we recorded a valuation allowance on certain foreign net operating losses that are not more likely than not to be realized. The remainder of our foreign deferred tax assets are more likely than not to be realized given the expectation of future earnings in these jurisdictions.

As of December 31, 2023, we had $10.31 billion of federal and $10.36 billion of state net operating loss carry-forwards available to offset future taxable income, some of which, if not utilized, will begin to expire in 2024 for federal and state purposes. Federal and state laws can impose substantial restrictions on the utilization of net operating loss and tax credit carry-forwards in the event of an “ownership change,” as defined in Section 382 of the Internal Revenue Code. We have determined that no significant limitation would be placed on the utilization of our net operating loss and tax credit carry-forwards due to prior ownership changes or expirations.

As of December 31, 2023, we had federal research and development tax credits of $1.10 billion, federal renewable energy tax credits of $605 million, and state research and development tax credits of $923 million. Most of our state research and development tax credits were in the state of California. If not utilized, some of the federal tax credits may expire in various amounts beginning in 2036. However, California research and development tax credits can be carried forward indefinitely.

The local government of Shanghai granted a beneficial corporate income tax rate of 15% to certain eligible enterprises, compared to the 25% statutory corporate income tax rate in China. Our Gigafactory Shanghai subsidiary was granted this beneficial income tax rate of 15% for 2019 through 2023. Starting in 2024, Gigafactory Shanghai is subject to 25% statutory corporate income tax rate in China.

As of December 31, 2023, we intend to indefinitely reinvest our foreign earnings and cash unless such repatriation results in no or minimal tax costs. We have recorded the taxes associated with the foreign earnings we intend to repatriate in the future. For the earnings we intend to indefinitely reinvest, no deferred tax liabilities for foreign withholding or other taxes have been recorded. The estimated amount of such unrecognized withholding tax liability associated with the indefinitely reinvested earnings is approximately $245 million.

Uncertain Tax Positions

The changes to our gross unrecognized tax benefits were as follows (in millions):

December 31, 2020	$	380
Increases in balances related to prior year tax positions		117
Decreases in balances related to prior year tax positions		(90)
Increases in balances related to current year tax positions		124
December 31, 2021		531
Increases in balances related to prior year tax positions		136
Decreases in balances related to prior year tax positions		(12)
Increases in balances related to current year tax positions		222
Decreases in balances related to expiration of the statute of limitations		(7)
December 31, 2022		870
Increases in balances related to prior year tax positions		59
Decreases related to settlement with tax authorities		(6)
Increases in balances related to current year tax positions		255
Decreases in balances related to expiration of the statute of limitations		(4)
December 31, 2023	$	1,174

We include interest and penalties related to unrecognized tax benefits in income tax expense. We recognized net interest and penalties related to unrecognized tax benefits in provision for income taxes line of our consolidated statements of operations of $17 million, $27 million and $4 million for the years ended December 31, 2023, 2022 and 2021, respectively. As of December 31, 2023, and 2022, we have accrued $47 million and $31 million, respectively, related to interest and penalties on our unrecognized tax benefits. Unrecognized tax benefits of $901 million, if recognized, would affect our effective tax rate.

We file income tax returns in the U.S. and various state and foreign jurisdictions. We are currently under examination by the Internal Revenue Service ("IRS") for the years 2015 to 2018. Additional tax years within the periods 2004 to 2014 and 2019 to 2022 remain subject to examination for federal income tax purposes. All net operating losses and tax credits generated to date are subject to adjustment for U.S. federal and state income tax purposes. Our returns for 2004 and subsequent tax years remain subject to examination in U.S. state and foreign jurisdictions.

Given the uncertainty in timing and outcome of our tax examinations, an estimate of the range of the reasonably possible change in gross unrecognized tax benefits within twelve months cannot be made at this time.

Note 15 – Commitments and Contingencies

Operating Lease Arrangement in Buffalo, New York

We have an operating lease arrangement through the Research Foundation for the SUNY Foundation with respect to Gigafactory New York. Under the lease and a related research and development agreement, we are continuing to further develop the facility.

Under this agreement, we are obligated to, among other things, meet employment targets as well as specified minimum numbers of personnel in the State of New York and in Buffalo, New York and spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York during the 10-year period beginning April 30, 2018. On an annual basis during the initial lease term, as measured on each anniversary of such date, if we fail to meet these specified investment and job creation requirements, then we would be obligated to pay a $41 million "program payment" to the SUNY Foundation for each year that we fail to meet these requirements. Furthermore, if the arrangement is terminated due to a material breach by us, then additional amounts may become payable by us.

In 2021, an amendment was executed to extend our overall agreement to spend or incur $5.00 billion in combined capital, operational expenses, costs of goods sold and other costs in the State of New York through December 31, 2029. On February 1, 2022, we reported to the State of New York that we had met and exceeded our annual requirements for jobs and investment in Buffalo and New York State. As of December 31, 2023, we have met and expect to meet the requirements under this arrangement based on our current and anticipated level of operations. However, if our expectations as to the costs and timelines of our investment and operations at Buffalo prove incorrect, we may incur additional expenses or be required to make substantial payments to the SUNY Foundation.

Operating Lease Arrangement in Shanghai, China

We have an operating lease arrangement for an initial term of 50 years with the local government of Shanghai for land use rights where we have been constructing Gigafactory Shanghai. Under the terms of the arrangement, we are required to spend RMB 14.08 billion in capital expenditures by the end of 2023, which has been achieved in 2023, and to generate RMB 2.23 billion of annual tax revenues starting at the end of 2023. As of December 31, 2023, we have met and expect to meet the tax revenue requirements based on our current level of spend and sales.

Legal Proceedings

Litigation Relating to 2018 CEO Performance Award

On June 4, 2018, a purported Tesla stockholder filed a putative class and derivative action in the Delaware Court of Chancery against Elon Musk and the members of Tesla's board of directors as then constituted, alleging corporate waste, unjust enrichment and that such board members breached their fiduciary duties by approving the stock-based compensation plan awarded to Elon Musk in 2018. Trial was held November 14-18, 2022. Post-trial briefing and argument are now complete.

Litigation Related to Directors' Compensation

On June 17, 2020, a purported Tesla stockholder filed a derivative action in the Delaware Court of Chancery, purportedly on behalf of Tesla, against certain of Tesla's current and former directors regarding compensation awards granted to Tesla's directors, other than Elon Musk, between 2017 and 2020. The suit asserts claims for breach of fiduciary duty and unjust enrichment and seeks declaratory and injunctive relief, unspecified damages and other relief. Defendants filed their answer on September 17, 2020.

On July 14, 2023, the parties filed a Stipulation and Agreement of Compromise and Settlement, which does not involve an admission of any wrongdoing by any party. If the settlement is approved by the Court, this action will be fully settled and dismissed with prejudice. Pursuant to the terms of the agreement, Tesla provided notice of the proposed settlement to stockholders of record as of July 14, 2023. The Court held a hearing regarding the settlement on October 13, 2023, after which it took the settlement and plaintiff counsels' fee request under advisement. The settlement is not expected to have an adverse impact on our results of operations, cash flows or financial position.

Litigation Relating to Potential Going Private Transaction

Between August 10, 2018 and September 6, 2018, nine purported stockholder class actions were filed against Tesla and Elon Musk in connection with Mr. Musk's August 7, 2018 Twitter post that he was considering taking Tesla private. On January 16, 2019, Plaintiffs filed their consolidated complaint in the United States District Court for the Northern District of California and added as defendants the members of Tesla's board of directors. The consolidated complaint asserts claims for violations of the federal securities laws and seeks unspecified damages and other relief. The parties stipulated to certification of a class of stockholders, which the court granted on November 25, 2020. Trial started on January 17, 2023, and on February 3, 2023, a jury rendered a verdict in favor of the defendants on all counts. After trial, plaintiffs filed a motion for judgment as a matter of law and a motion for new trial, which the Court denied and judgement was entered in favor of defendants on July 11, 2023. On July 14, 2023, plaintiffs filed a notice of appeal.

Between October 17, 2018 and March 8, 2021, seven derivative lawsuits were filed in the Delaware Court of Chancery, purportedly on behalf of Tesla, against Mr. Musk and the members of Tesla's board of directors, as constituted at relevant times, in relation to statements made and actions connected to a potential going private transaction, with certain of the lawsuits challenging additional Twitter posts by Mr. Musk, among other things. Five of those actions were consolidated, and all seven actions have been stayed pending resolution of the appeal in the above-referenced consolidated purported stockholder class action. In addition to these cases, two derivative lawsuits were filed on October 25, 2018 and February 11, 2019 in the U.S. District Court for the District of Delaware, purportedly on behalf of Tesla, against Mr. Musk and the members of the Tesla board of directors as then constituted. Those cases have also been consolidated and stayed pending resolution of the appeal in the above-referenced consolidated purported stockholder class action.

On October 21, 2022, a lawsuit was filed in the Delaware Court of Chancery by a purported shareholder of Tesla alleging, among other things, that board members breached their fiduciary duties in connection with their oversight of the Company's 2018 settlement with the SEC, as amended. Among other things, the plaintiff seeks reforms to the Company's corporate governance and internal procedures, unspecified damages, and attorneys' fees. The parties reached an agreement to stay the case until March 5, 2024.

On November 15, 2021, JPMorgan Chase Bank ("JP Morgan") filed a lawsuit against Tesla in the Southern District of New York alleging breach of a stock warrant agreement that was entered into as part of a convertible notes offering in 2014. In 2018, JP Morgan informed Tesla that it had adjusted the strike price based upon Mr. Musk's August 7, 2018 Twitter post that he was considering taking Tesla private. Tesla disputed JP Morgan's adjustment as a violation of the parties' agreement. In 2021, Tesla delivered shares to JP Morgan per the agreement, which they duly accepted. JP Morgan now alleges that it is owed approximately $162 million as the value of additional shares that it claims should have been delivered as a result of the adjustment to the strike price in 2018. On January 24, 2022, Tesla filed multiple counterclaims as part of its answer to the underlying lawsuit, asserting among other points that JP Morgan should have terminated the stock warrant agreement in 2018 rather than make an adjustment to the strike price that it should have known would lead to a commercially unreasonable result. Tesla believes that the adjustments made by JP Morgan were neither proper nor commercially reasonable, as required under the stock warrant agreements. JP Morgan filed a motion for judgment on the pleadings, which Tesla opposed, and that motion is currently pending before the Court.

Litigation and Investigations Relating to Alleged Discrimination and Harassment

On October 4, 2021, in a case captioned Diaz v. Tesla, a jury in the Northern District of California returned a verdict against Tesla on claims by a former contingent worker that he was subjected to race discrimination while assigned to work at Tesla's Fremont Factory from 2015-2016. A retrial was held starting on March 27, 2023, after which a jury returned a verdict of $3,175,000. As a result, the damages awarded against Tesla were reduced from an initial $136.9 million (October 4, 2021) down to $15 million (April 13, 2022), and then further down to $3.175 million (April 3, 2023). On November 2, 2023, the plaintiff filed a notice of appeal, and on November 16, 2023, Tesla filed a notice of cross appeal.

On February 9, 2022, shortly after the first Diaz jury verdict, the California Civil Rights Department ("CRD," formerly "DFEH") filed a civil complaint against Tesla in Alameda County, California Superior Court, alleging systemic race discrimination, hostile work environment and pay equity claims, among others. CRD's amended complaint seeks monetary damages and injunctive relief. On September 22, 2022, Tesla filed a cross complaint against CRD, alleging that it violated the Administrative Procedures Act by failing to follow statutory pre-requisites prior to filing suit and that cross complaint was subject to a sustained demurrer, which Tesla later amended and refiled. The case is currently in discovery.

Additionally, on June 1, 2022 the Equal Employment Opportunity Commission ("EEOC") issued a cause finding against Tesla that closely parallels the CRD's allegations. On September 28, 2023, the EEOC filed a civil complaint against Tesla in the United States District Court for the Northern District of California asserting claims for race harassment and retaliation and seeking, among other things, monetary and injunctive relief. On December 18, 2023, Tesla filed a motion to stay the case. Separately, on December 26, 2023, Tesla filed a motion to dismiss the case.

On June 16, 2022, two Tesla stockholders filed separate derivative actions in the U.S. District Court for the Western District of Texas, purportedly on behalf of Tesla, against certain of Tesla's current and former directors. Both suits assert claims for breach of fiduciary duty, unjust enrichment, and violation of the federal securities laws in connection with alleged race and gender discrimination and sexual harassment. Among other things, plaintiffs seek declaratory and injunctive relief, unspecified damages payable to Tesla, and attorneys' fees. On July 22, 2022, the Court consolidated the two cases and on September 6, 2022, plaintiffs filed a consolidated complaint. On November 7, 2022, the defendants filed a motion to dismiss the case and on September 15, 2023, the Court dismissed the action but granted plaintiffs leave to file an amended complaint. On November 2, 2023, plaintiff filed an amended complaint purportedly on behalf of Tesla, against Elon Musk. On December 19, 2023, the defendants moved to dismiss the amended complaint.

Other Litigation Related to Our Products and Services

We are also subject to various lawsuits that seek monetary and other injunctive relief. These lawsuits include proposed class actions and other consumer claims that allege, among other things, purported defects and misrepresentations related to our products and services. For example, on September 14, 2022, a proposed class action was filed against Tesla, Inc. and related entities in the U.S. District Court for the Northern District of California, alleging various claims about the Company's driver assistance technology systems under state and federal law. This case was later consolidated with several other proposed class actions, and a Consolidated Amended Complaint was filed on October 28, 2022, which seeks damages and other relief on behalf of all persons who purchased or leased from Tesla between January 1, 2016 to the present. On October 5, 2022 a proposed class action complaint was filed in the U.S. District Court for the Eastern District of New York asserting similar state and federal law claims against the same defendants. On September 30, 2023, the Court dismissed this action with leave to amend the complaint. On November 20, 2023, the plaintiff moved to amend the complaint, which Tesla opposed. On March 22, 2023, the plaintiffs in the Northern District of California consolidated action filed a motion for a preliminary injunction to order Tesla to (1) cease using the term "Full Self-Driving Capability" (FSD Capability), (2) cease the sale and activation of FSD Capability and deactivate FSD Capability on Tesla vehicles, and (3) provide certain notices to consumers about proposed court-findings about the accuracy of the use of the terms Autopilot and FSD Capability. Tesla opposed the motion. On September 30, 2023, the Court denied the request for a preliminary injunction, compelled four of five plaintiffs to arbitration, and dismissed the claims of the fifth plaintiff with leave to amend the complaint. On October 31, 2023, the remaining plaintiff in the Northern District of California action filed an amended complaint, which Tesla has moved to dismiss. On October 2, 2023, a similar proposed class action was filed in San Diego County Superior Court in California. Tesla subsequently removed the San Diego County case to federal court and on January 8, 2024, the federal court granted Tesla's motion to transfer the case to the U.S. District Court for the Northern District of California.

On February 27, 2023, a proposed class action was filed in the U.S. District Court for the Northern District of California against Tesla, Inc., Elon Musk and certain current and former Company executives. The complaint alleges that the defendants made material misrepresentations and omissions about the Company's Autopilot and FSD Capability technologies and seeks money damages and other relief on behalf of persons who purchased Tesla stock between February 19, 2019 and February 17, 2023. An amended complaint was filed on September 5, 2023, naming only Tesla, Inc. and Elon Musk as defendants. On November 6, 2023, Tesla moved to dismiss the amended complaint.

On March 14, 2023, a proposed class action was filed against Tesla, Inc. in the U.S. District Court for the Northern District of California. Several similar complaints have also been filed in the same court and these cases have now all been consolidated. These complaints allege that Tesla violates federal antitrust and warranty laws through its repair, service, and maintenance practices and seeks, among other relief, damages for persons who paid Tesla for repairs services or Tesla compatible replacement parts from March 2019 to March 2023. On July 17, 2023, these plaintiffs filed a consolidated amended complaint. On September 27, 2023, the court granted Tesla's motion to compel arbitration as to three of the plaintiffs, and on November 17, 2023, the court granted Tesla's motion to dismiss without prejudice. The plaintiffs filed a Consolidated Second Amended Complaint on December 12, 2023, which Tesla has moved to dismiss. Plaintiffs have also appealed the court's arbitration order. Trial is currently set for July 7, 2025.

The Company intends to vigorously defend itself in these matters; however, we cannot predict the outcome or impact. We are unable to reasonably estimate the possible loss or range of loss, if any, associated with these claims, unless noted.

Certain Investigations and Other Matters

We regularly receive requests for information, including subpoenas, from regulators and governmental authorities such as the National Highway Traffic Safety Administration, the National Transportation Safety Board, the Securities and Exchange Commission ("SEC"), the Department of Justice ("DOJ"), and various local, state, federal, and international agencies. The ongoing requests for information include topics such as operations, technology (e.g., vehicle functionality, Autopilot and FSD Capability), compliance, finance, data privacy, and other matters related to Tesla's business, its personnel, and related parties. We routinely cooperate with such formal and informal requests for information, investigations, and other inquiries. To our knowledge no government agency in any ongoing investigation has concluded that any wrongdoing occurred. We cannot predict the outcome or impact of any ongoing matters. Should the government decide to pursue an enforcement action, there exists the possibility of a material adverse impact on our business, results of operation, prospects, cash flows, financial position or brand.

We are also subject to various other legal proceedings, risks and claims that arise from the normal course of business activities. For example, during the second quarter of 2023, a foreign news outlet reported that it obtained certain misappropriated data including, purportedly non-public Tesla business and personal information. Tesla has made notifications to potentially affected individuals (current and former employees) and regulatory authorities and we are working with certain law enforcement and other authorities. On August 5, 2023, a putative class action was filed in the United States District Court for the Northern District of California, purportedly on behalf of all U.S. individuals impacted by the data incident, followed by several additional lawsuits, that each assert claims under various state laws and seeks monetary damages and other relief. If an unfavorable ruling or development were to occur in these or other possible legal proceedings, risks and claims, there exists the possibility of a material adverse impact on our business, results of operations, prospects, cash flows, financial position or brand.

Letters of Credit

As of December 31, 2023, we had $525 million of unused letters of credit outstanding.

Note 16 – Variable Interest Entity Arrangements

We have entered into various arrangements with investors to facilitate the funding and monetization of our solar energy systems and vehicles. In particular, our wholly owned subsidiaries and fund investors have formed and contributed cash and assets into various financing funds and entered into related agreements. We have determined that the funds are VIEs and we are the primary beneficiary of these VIEs by reference to the power and benefits criterion under ASC 810. We have considered the provisions within the agreements, which grant us the power to manage and make decisions that affect the operation of these VIEs, including determining the solar energy systems and the associated customer contracts to be sold or contributed to these VIEs, redeploying solar energy systems and managing customer receivables. We consider that the rights granted to the fund investors under the agreements are more protective in nature rather than participating.

As the primary beneficiary of these VIEs, we consolidate in the financial statements the financial position, results of operations and cash flows of these VIEs, and all intercompany balances and transactions between us and these VIEs are eliminated in the consolidated financial statements. Cash distributions of income and other receipts by a fund, net of agreed upon expenses, estimated expenses, tax benefits and detriments of income and loss and tax credits, are allocated to the fund investor and our subsidiary as specified in the agreements.

Generally, our subsidiary has the option to acquire the fund investor’s interest in the fund for an amount based on the market value of the fund or the formula specified in the agreements.

Upon the sale or liquidation of a fund, distributions would occur in the order and priority specified in the agreements.

Pursuant to management services, maintenance and warranty arrangements, we have been contracted to provide services to the funds, such as operations and maintenance support, accounting, lease servicing and performance reporting. In some instances, we have guaranteed payments to the fund investors as specified in the agreements. A fund’s creditors have no recourse to our general credit or to that of other funds. Certain assets of the funds have been pledged as collateral for their obligations.

The aggregate carrying values of the VIEs' assets and liabilities, after elimination of any intercompany transactions and balances, in the consolidated balance sheets were as follows (in millions):

	December 31, 2023	December 31, 2022
Assets		
Current assets		
Cash and cash equivalents	$ 66	$ 68
Accounts receivable, net	13	22
Prepaid expenses and other current assets	361	274
Total current assets	440	364
Solar energy systems, net	3,278	4,060
Other non-current assets	369	404
Total assets	$ 4,087	$ 4,828
Liabilities		
Current liabilities		
Accrued liabilities and other	$ 67	$ 69
Deferred revenue	6	10
Current portion of debt and finance leases	1,564	1,013
Total current liabilities	1,637	1,092
Deferred revenue, net of current portion	99	149
Debt and finance leases, net of current portion	2,041	971
Other long-term liabilities	—	3
Total liabilities	$ 3,777	$ 2,215

Note 17 – Related Party Transactions

In relation to our CEO's exercise of stock options and sale of common stock from the 2012 CEO Performance Award, Tesla withheld the appropriate amount of taxes. However, given the significant amounts involved, our CEO entered into an indemnification agreement with us in November 2021 for additional taxes owed, if any.

Tesla periodically does business with certain entities with which its CEO and directors are affiliated, such as SpaceX and X Corp., in accordance with our Related Person Transactions Policy. Such transactions have not had to date, and are not currently expected to have, a material impact on our consolidated financial statements.

Note 18 – Segment Reporting and Information about Geographic Areas

We have two operating and reportable segments: (i) automotive and (ii) energy generation and storage. The automotive segment includes the design, development, manufacturing, sales and leasing of electric vehicles as well as sales of automotive regulatory credits. Additionally, the automotive segment is also comprised of services and other, which includes sales of used vehicles, non-warranty after-sales vehicle services, body shop and parts, paid Supercharging, vehicle insurance revenue and retail merchandise. The energy generation and storage segment includes the design, manufacture, installation, sales and leasing of solar energy generation and energy storage products and related services and sales of solar energy systems incentives. Our CODM does not evaluate operating segments using asset or liability information. The following table presents revenues and gross profit by reportable segment (in millions):

	Year Ended December 31,		
	2023	2022	2021
Automotive segment			
Revenues	$ 90,738	$ 77,553	$ 51,034
Gross profit	$ 16,519	$ 20,565	$ 13,735
Energy generation and storage segment			
Revenues	$ 6,035	$ 3,909	$ 2,789
Gross profit	$ 1,141	$ 288	$ (129)

The following table presents revenues by geographic area based on the sales location of our products (in millions):

	Year Ended December 31,		
	2023	2022	2021
United States	$ 45,235	$ 40,553	$ 23,973
China	21,745	18,145	13,844
Other international	29,793	22,764	16,006
Total	$ 96,773	$ 81,462	$ 53,823

The following table presents long-lived assets by geographic area (in millions):

	December 31, 2023	December 31, 2022
United States	$ 26,629	$ 21,667
Germany	4,258	3,547
China	2,820	2,978
Other international	1,247	845
Total	$ 34,954	$ 29,037

The following table presents inventory by reportable segment (in millions):

	December 31, 2023	December 31, 2022
Automotive	$ 11,139	$ 10,996
Energy generation and storage	2,487	1,843
Total	$ 13,626	$ 12,839

Note 19 – Restructuring and Other

During the years ended December 31, 2022 and 2021, werecorded $204 million and $101 million, respectively, of impairment losses on digital assets. During the years ended December 31, 2022 and 2021 we also realized gains of $64 million and $128 million, respectively, in connection with converting our holdings of digital assets into fiat currency. We also recorded other expenses of $36 million during the second quarter of the year ended December 31, 2022, related to employee terminations.

ITEM 9. CHANGES IN AND DISAGREEMENTS WITH ACCOUNTANTS ON ACCOUNTING AND FINANCIAL DISCLOSURE

None.

ITEM 9A. CONTROLS AND PROCEDURES

Evaluation of Disclosure Controls and Procedures

Our management, with the participation of our Chief Executive Officer and our Chief Financial Officer, evaluated the effectiveness of our disclosure controls and procedures pursuant to Rule 13a-15 under the Securities Exchange Act of 1934, as amended (the "Exchange Act"). In designing and evaluating the disclosure controls and procedures, our management recognizes that any controls and procedures, no matter how well designed and operated, can provide only reasonable assurance of achieving the desired control objectives. In addition, the design of disclosure controls and procedures must reflect the fact that there are resource constraints and that our management is required to apply its judgment in evaluating the benefits of possible controls and procedures relative to their costs.

Based on this evaluation, our Chief Executive Officer and our Chief Financial Officer concluded that, as of December 31, 2023, our disclosure controls and procedures were designed at a reasonable assurance level and were effective to provide reasonable assurance that the information we are required to disclose in reports that we file or submit under the Exchange Act is recorded, processed, summarized and reported within the time periods specified in the SEC rules and forms, and that such information is accumulated and communicated to our management, including our Chief Executive Officer and our Chief Financial Officer, as appropriate, to allow timely decisions regarding required disclosures.

Management's Report on Internal Control over Financial Reporting

Our management is responsible for establishing and maintaining adequate internal control over financial reporting. Internal control over financial reporting is a process designed by, or under the supervision of, our Chief Executive Officer and Chief Financial Officer to provide reasonable assurance regarding the reliability of financial reporting and the preparation of financial statements for external purposes in accordance with generally accepted accounting principles and includes those policies and procedures that (1) pertain to the maintenance of records that in reasonable detail accurately and fairly reflect the transactions and dispositions of our assets; (2) provide reasonable assurance that transactions are recorded as necessary to permit preparation of financial statements in accordance with generally accepted accounting principles, and that our receipts and expenditures are being made only in accordance with authorizations of our management and directors and (3) provide reasonable assurance regarding prevention or timely detection of unauthorized acquisition, use or disposition of our assets that could have a material effect on the financial statements.

Under the supervision and with the participation of our management, including our Chief Executive Officer and Chief Financial Officer, we conducted an evaluation of the effectiveness of our internal control over financial reporting based on criteria established in Internal Control – Integrated Framework (2013) issued by the Committee of Sponsoring Organizations of the Treadway Commission ("COSO"). Our management concluded that our internal control over financial reporting was effective as of December 31, 2023.

Our independent registered public accounting firm, PricewaterhouseCoopers LLP, has audited the effectiveness of our internal control over financial reporting as of December 31, 2023, as stated in their report which is included herein.

Limitations on the Effectiveness of Controls

Because of inherent limitations, internal control over financial reporting may not prevent or detect misstatements and projections of any evaluation of effectiveness to future periods are subject to the risk that controls may become inadequate because of changes in conditions, or that the degree of compliance with the policies or procedures may deteriorate.

Changes in Internal Control over Financial Reporting

There was no change in our internal control over financial reporting that occurred during the quarter ended December 31, 2023, which has materially affected, or is reasonably likely to materially affect, our internal control over financial reporting.

ITEM 9B. OTHER INFORMATION

None of the Company's directors or officers adopted, modified or terminated a Rule 10b5-1 trading arrangement or a non-Rule 10b5-1 trading arrangement during the Company's fiscal quarter ended December 31, 2023, as such terms are defined under Item 408(a) of Regulation S-K, except as follows:

On October 23, 2023, Robyn Denholm, one of our directors, adopted a Rule 10b5-1 trading arrangement for the potential sale of up to281,116 shares of our common stock, subject to certain conditions. The trading arrangement covers stock options that expire in August 2024. The arrangement's expiration date is August 16, 2024.

On November 13, 2023, Andrew Baglino, Senior Vice President, Powertrain and Energy Engineering, adopted a Rule 10b5-1 trading arrangement for the potential sale of up to 115,500 shares of our common stock, subject to certain conditions. The arrangement's expiration date is December 31, 2024.

ITEM 9C. DISCLOSURE REGARDING FOREIGN JURISDICTIONS THAT PREVENT INSPECTIONS

Not applicable.

PART III

ITEM 10. DIRECTORS, EXECUTIVE OFFICERS AND CORPORATE GOVERNANCE

The information required by this Item 10 of Form 10-K will be included in our 2024 Proxy Statement to be filed with the Securities and Exchange Commission in connection with the solicitation of proxies for our 2024 Annual Meeting of Stockholders and is incorporated herein by reference. The 2024 Proxy Statement will be filed with the Securities and Exchange Commission within 120 days after the end of the fiscal year to which this report relates.

ITEM 11. EXECUTIVE COMPENSATION

The information required by this Item 11 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 12. SECURITY OWNERSHIP OF CERTAIN BENEFICIAL OWNERS AND MANAGEMENT AND RELATED STOCKHOLDER MATTERS

The information required by this Item 12 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 13. CERTAIN RELATIONSHIPS AND RELATED TRANSACTIONS AND DIRECTOR INDEPENDENCE

The information required by this Item 13 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

ITEM 14. PRINCIPAL ACCOUNTANT FEES AND SERVICES

The information required by this Item 14 of Form 10-K will be included in our 2024 Proxy Statement and is incorporated herein by reference.

PART IV

ITEM 15. EXHIBITS AND FINANCIAL STATEMENT SCHEDULES

1. Financial statements (see *Index to Consolidated Financial Statements* in Part II, Item 8 of this report)
2. All financial statement schedules have been omitted since the required information was not applicable or was not present in amounts sufficient to require submission of the schedules, or because the information required is included in the consolidated financial statements or the accompanying notes
3. The exhibits listed in the following *Index to Exhibits* are filed or incorporated by reference as part of this report

INDEX TO EXHIBITS

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
3.1	Amended and Restated Certificate of Incorporation of the Registrant.	10-K	001-34756	3.1	March 1, 2017	
3.2	Certificate of Amendment to the Amended and Restated Certificate of Incorporation of the Registrant.	10-K	001-34756	3.2	March 1, 2017	
3.3	Amended and Restated Bylaws of the Registrant.	8-K	001-34756	3.1	April 5, 2023	
4.1	Specimen common stock certificate of the Registrant.	10-K	001-34756	4.1	March 1, 2017	
4.2	Fifth Amended and Restated Investors' Rights Agreement, dated as of August 31, 2009, between Registrant and certain holders of the Registrant's capital stock named therein.	S-1	333-164593	4.2	January 29, 2010	
4.3	Amendment to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 20, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.	S-1/A	333-164593	4.2A	May 27, 2010	
4.4	Amendment to Fifth Amended and Restated Investors' Rights Agreement between Registrant, Toyota Motor Corporation and certain holders of the Registrant's capital stock named therein.	S-1/A	333-164593	4.2B	May 27, 2010	
4.5	Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of June 14, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.	S-1/A	333-164593	4.2C	June 15, 2010	
4.6	Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of November 2, 2010, between Registrant and certain holders of the Registrant's capital stock named therein.	8-K	001-34756	4.1	November 4, 2010	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.7	Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 22, 2011, between Registrant and certain holders of the Registrant's capital stock named therein.	S-1/A	333-174466	4.2E	June 2, 2011	
4.8	Amendment to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 30, 2011, between Registrant and certain holders of the Registrant's capital stock named therein.	8-K	001-34756	4.1	June 1, 2011	
4.9	Sixth Amendment to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 15, 2013 among the Registrant, the Elon Musk Revocable Trust dated July 22, 2003 and certain other holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 20, 2013	
4.10	Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of May 14, 2013, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.2	May 20, 2013	
4.11	Waiver to Fifth Amended and Restated Investor's Rights Agreement, dated as of August 13, 2015, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	August 19, 2015	
4.12	Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 18, 2016, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 24, 2016	
4.13	Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of March 15, 2017, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	March 17, 2017	
4.14	Waiver to Fifth Amended and Restated Investors' Rights Agreement, dated as of May 1, 2019, between the Registrant and certain holders of the capital stock of the Registrant named therein.	8-K	001-34756	4.1	May 3, 2019	
4.15	Indenture, dated as of May 22, 2013, by and between the Registrant and U.S. Bank National Association.	8-K	001-34756	4.1	May 22, 2013	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.16	Fifth Supplemental Indenture, dated as of May 7, 2019, by and between Registrant and U.S. Bank National Association, related to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	4.2	May 8, 2019	
4.17	Form of 2.00% Convertible Senior Notes due May 15, 2024 (included in Exhibit 4.16).	8-K	001-34756	4.2	May 8, 2019	
4.18	Indenture, dated as of October 15, 2014, between SolarCity and U.S. Bank National Association, as trustee.	S-3ASR(1)	333-199321	4.1	October 15, 2014	
4.19	Tenth Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/6-10.	8-K(1)	001-35758	4.3	March 9, 2015	
4.20	Eleventh Supplemental Indenture, dated as of March 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/7-15.	8-K(1)	001-35758	4.4	March 9, 2015	
4.21	Fifteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C4-10.	8-K(1)	001-35758	4.5	March 19, 2015	
4.22	Sixteenth Supplemental Indenture, dated as of March 19, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C5-15.	8-K(1)	001-35758	4.6	March 19, 2015	
4.23	Twentieth Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C9-10.	8-K(1)	001-35758	4.5	March 26, 2015	
4.24	Twenty-First Supplemental Indenture, dated as of March 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C10-15.	8-K(1)	001-35758	4.6	March 26, 2015	
4.25	Twenty-Sixth Supplemental Indenture, dated as of April 2, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C14-10.	8-K(1)	001-35758	4.5	April 2, 2015	
4.26	Thirtieth Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C19-10.	8-K(1)	001-35758	4.5	April 9, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.27	Thirty-First Supplemental Indenture, dated as of April 9, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C20-15.	8-K(1)	001-35758	4.6	April 9, 2015	
4.28	Thirty-Fifth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C24-10.	8-K(1)	001-35758	4.5	April 14, 2015	
4.29	Thirty-Sixth Supplemental Indenture, dated as of April 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C25-15.	8-K(1)	001-35758	4.6	April 14, 2015	
4.30	Thirty-Eighth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C27-10.	8-K(1)	001-35758	4.3	April 21, 2015	
4.31	Thirty-Ninth Supplemental Indenture, dated as of April 21, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C28-15.	8-K(1)	001-35758	4.4	April 21, 2015	
4.32	Forty-Third Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C32-10.	8-K(1)	001-35758	4.5	April 27, 2015	
4.33	Forty-Fourth Supplemental Indenture, dated as of April 27, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C33-15.	8-K(1)	001-35758	4.6	April 27, 2015	
4.34	Forty-Eighth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/12-10.	8-K(1)	001-35758	4.5	May 1, 2015	
4.35	Forty-Ninth Supplemental Indenture, dated as of May 1, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/13-15.	8-K(1)	001-35758	4.6	May 1, 2015	
4.36	Fifty-Second Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C36-10.	8-K(1)	001-35758	4.4	May 11, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.37	Fifty-Third Supplemental Indenture, dated as of May 11, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C37-15.	8-K(1)	001-35758	4.5	May 11, 2015	
4.38	Fifty-Seventh Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C40-10.	8-K(1)	001-35758	4.4	May 18, 2015	
4.39	Fifty-Eighth Supplemental Indenture, dated as of May 18, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C41-15.	8-K(1)	001-35758	4.5	May 18, 2015	
4.40	Sixty-First Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C44-10.	8-K(1)	001-35758	4.4	May 26, 2015	
4.41	Sixty-Second Supplemental Indenture, dated as of May 26, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C45-15.	8-K(1)	001-35758	4.5	May 26, 2015	
4.42	Seventieth Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C52-10.	8-K(1)	001-35758	4.4	June 16, 2015	
4.43	Seventy-First Supplemental Indenture, dated as of June 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C53-15.	8-K(1)	001-35758	4.5	June 16, 2015	
4.44	Seventy-Fourth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C56-10.	8-K(1)	001-35758	4.4	June 23, 2015	
4.45	Seventy-Fifth Supplemental Indenture, dated as of June 22, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C57-15.	8-K(1)	001-35758	4.5	June 23, 2015	
4.46	Eightieth Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C61-10.	8-K(1)	001-35758	4.5	June 29, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.47	Eighty-First Supplemental Indenture, dated as of June 29, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C62-15.	8-K(1)	001-35758	4.6	June 29, 2015	
4.48	Ninetieth Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C71-10.	8-K(1)	001-35758	4.5	July 21, 2015	
4.49	Ninety-First Supplemental Indenture, dated as of July 20, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C72-15.	8-K(1)	001-35758	4.6	July 21, 2015	
4.50	Ninety-Fifth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/20-10.	8-K(1)	001-35758	4.5	July 31, 2015	
4.51	Ninety-Sixth Supplemental Indenture, dated as of July 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/21-15.	8-K(1)	001-35758	4.6	July 31, 2015	
4.52	One Hundred-and-Fifth Supplemental Indenture, dated as of August 10, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C81-10.	8-K(1)	001-35758	4.5	August 10, 2015	
4.53	One Hundred-and-Eleventh Supplemental Indenture, dated as of August 17, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C87-15.	8-K(1)	001-35758	4.6	August 17, 2015	
4.54	One Hundred-and-Sixteenth Supplemental Indenture, dated as of August 24, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C92-15.	8-K(1)	001-35758	4.6	August 24, 2015	
4.55	One Hundred-and-Twenty-First Supplemental Indenture, dated as of August 31, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C97-15.	8-K(1)	001-35758	4.6	August 31, 2015	
4.56	One Hundred-and-Twenty-Eighth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C101-10.	8-K(1)	001-35758	4.5	September 15, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.57	One Hundred-and-Twenty-Ninth Supplemental Indenture, dated as of September 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C102-15.	8-K(1)	001-35758	4.6	September 15, 2015	
4.58	One Hundred-and-Thirty-Third Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C106-10.	8-K(1)	001-35758	4.5	September 29, 2015	
4.59	One Hundred-and-Thirty-Fourth Supplemental Indenture, dated as of September 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C107-15.	8-K(1)	001-35758	4.6	September 29, 2015	
4.60	One Hundred-and-Thirty-Eighth Supplemental Indenture, dated as of October 13, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C111-10.	8-K(1)	001-35758	4.5	October 13, 2015	
4.61	One Hundred-and-Forty-Third Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2015/25-10.	8-K(1)	001-35758	4.5	October 30, 2015	
4.62	One Hundred-and-Forty-Fourth Supplemental Indenture, dated as of October 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2015/26-15.	8-K(1)	001-35758	4.6	October 30, 2015	
4.63	One Hundred-and-Forty-Eighth Supplemental Indenture, dated as of November 4, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C116-10.	8-K(1)	001-35758	4.5	November 4, 2015	
4.64	One Hundred-and-Fifty-Third Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C121-10.	8-K(1)	001-35758	4.5	November 17, 2015	
4.65	One Hundred-and-Fifty-Fourth Supplemental Indenture, dated as of November 16, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C122-15.	8-K(1)	001-35758	4.6	November 17, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
4.66	One Hundred-and-Fifty-Eighth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C126-10.	8-K(1)	001-35758	4.5	November 30, 2015	
4.67	One Hundred-and-Fifty-Ninth Supplemental Indenture, dated as of November 30, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C127-15.	8-K(1)	001-35758	4.6	November 30, 2015	
4.68	One Hundred-and-Sixty-Third Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C131-10.	8-K(1)	001-35758	4.5	December 14, 2015	
4.69	One Hundred-and-Sixty-Fourth Supplemental Indenture, dated as of December 14, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C132-15.	8-K(1)	001-35758	4.6	December 14, 2015	
4.70	One Hundred-and-Sixty-Eighth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 4.70% Solar Bonds, Series 2015/C136-10.	8-K(1)	001-35758	4.5	December 28, 2015	
4.71	One Hundred-and-Sixty-Ninth Supplemental Indenture, dated as of December 28, 2015, by and between SolarCity and the Trustee, related to SolarCity's 5.45% Solar Bonds, Series 2015/C137-15.	8-K(1)	001-35758	4.6	December 28, 2015	
4.72	One Hundred-and-Seventy-Third Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.00% Solar Bonds, Series 2016/4-10.	8-K(1)	001-35758	4.5	January 29, 2016	
4.73	One Hundred-and-Seventy-Fourth Supplemental Indenture, dated as of January 29, 2016, by and between SolarCity and the Trustee, related to SolarCity's 5.75% Solar Bonds, Series 2016/5-15.	8-K(1)	001-35758	4.6	January 29, 2016	
4.74	Description of Registrant's Securities	10-K	001-34756	4.119	February 13, 2020	
10.1**	Form of Indemnification Agreement between the Registrant and its directors and officers.	S-1/A	333-164593	10.1	June 15, 2010	
10.2**	2003 Equity Incentive Plan.	S-1/A	333-164593	10.2	May 27, 2010	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.3**	Form of Stock Option Agreement under 2003 Equity Incentive Plan.	S-1	333-164593	10.3	January 29, 2010	
10.4**	Amended and Restated 2010 Equity Incentive Plan.	10-K	001-34756	10.4	February 23, 2018	
10.5**	Form of Stock Option Agreement under 2010 Equity Incentive Plan.	10-K	001-34756	10.6	March 1, 2017	
10.6**	Form of Restricted Stock Unit Award Agreement under 2010 Equity Incentive Plan.	10-K	001-34756	10.7	March 1, 2017	
10.7**	Amended and Restated 2010 Employee Stock Purchase Plan, effective as of February 1, 2017.	10-K	001-34756	10.8	March 1, 2017	
10.8**	2019 Equity Incentive Plan.	S-8	333-232079	4.2	June 12, 2019	
10.9**	Form of Stock Option Agreement under 2019 Equity Incentive Plan.	S-8	333-232079	4.3	June 12, 2019	
10.10**	Form of Restricted Stock Unit Award Agreement under 2019 Equity Incentive Plan.	S-8	333-232079	4.4	June 12, 2019	
10.11**	Employee Stock Purchase Plan, effective as of June 12, 2019.	S-8	333-232079	4.5	June 12, 2019	
10.12**	2007 SolarCity Stock Plan and form of agreements used thereunder.	S-1(1)	333-184317	10.2	October 5, 2012	
10.13**	2012 SolarCity Equity Incentive Plan and form of agreements used thereunder.	S-1(1)	333-184317	10.3	October 5, 2012	
10.14**	2010 Zep Solar, Inc. Equity Incentive Plan and form of agreements used thereunder.	S-8(1)	333-192996	4.5	December 20, 2013	
10.15**	Offer Letter between the Registrant and Elon Musk dated October 13, 2008.	S-1	333-164593	10.9	January 29, 2010	
10.16**	Performance Stock Option Agreement between the Registrant and Elon Musk dated January 21, 2018.	DEF 14A	001-34756	Appendix A	February 8, 2018	
10.17**	Maxwell Technologies, Inc. 2005 Omnibus Equity Incentive Plan, as amended through May 6, 2010	8-K(2)	001-15477	10.1	May 10, 2010	
10.18**	Maxwell Technologies, Inc. 2013 Omnibus Equity Incentive Plan	DEF 14A(2)	001-15477	Appendix A	June 2, 2017	
10.19	Indemnification Agreement, effective as of June 23, 2020, between Registrant and Elon R. Musk.	10-Q	001-34756	10.4	July 28, 2020	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.20	Indemnification Agreement, dated as of February 27, 2014, by and between the Registrant and J.P. Morgan Securities LLC.	8-K	001-34756	10.1	March 5, 2014	
10.21	Form of Call Option Confirmation relating to 1.25% Convertible Senior Notes Due March 1, 2021.	8-K	001-34756	10.3	March 5, 2014	
10.22	Form of Warrant Confirmation relating to 1.25% Convertible Senior Notes Due March 1, 2021.	8-K	001-34756	10.5	March 5, 2014	
10.23	Form of Call Option Confirmation relating to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	10.1	May 3, 2019	
10.24	Form of Warrant Confirmation relating to 2.00% Convertible Senior Notes due May 15, 2024.	8-K	001-34756	10.2	May 3, 2019	
10.25†	Supply Agreement between Panasonic Corporation and the Registrant dated October 5, 2011.	10-K	001-34756	10.50	February 27, 2012	
10.26†	Amendment No. 1 to Supply Agreement between Panasonic Corporation and the Registrant dated October 29, 2013.	10-K	001-34756	10.35A	February 26, 2014	
10.27	Agreement between Panasonic Corporation and the Registrant dated July 31, 2014.	10-Q	001-34756	10.1	November 7, 2014	
10.28†	General Terms and Conditions between Panasonic Corporation and the Registrant dated October 1, 2014.	8-K	001-34756	10.2	October 11, 2016	
10.29	Letter Agreement, dated as of February 24, 2015, regarding addition of co-party to General Terms and Conditions, Production Pricing Agreement and Investment Letter Agreement between Panasonic Corporation and the Registrant.	10-K	001-34756	10.25A	February 24, 2016	
10.30†	Amendment to Gigafactory General Terms, dated March 1, 2016, by and among the Registrant, Panasonic Corporation and Panasonic Energy Corporation of North America.	8-K	001-34756	10.1	October 11, 2016	
10.31††	Amended and Restated General Terms and Conditions for Gigafactory, entered into on June 10, 2020, by and among Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and Panasonic Corporation of North America.	10-Q	001-34756	10.2	July 28, 2020	
10.32†	Production Pricing Agreement between Panasonic Corporation and the Registrant dated October 1, 2014.	10-Q	001-34756	10.3	November 7, 2014	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.33†	Investment Letter Agreement between Panasonic Corporation and the Registrant dated October 1, 2014.	10-Q	001-34756	10.4	November 7, 2014	
10.34	Amendment to Gigafactory Documents, dated April 5, 2016, by and among the Registrant, Panasonic Corporation, Panasonic Corporation of North America and Panasonic Energy Corporation of North America.	10-Q	001-34756	10.2	May 10, 2016	
10.35††	2019 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed September 20, 2019, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and SANYO Electric Co., Ltd.	10-Q	001-34756	10.6	October 29, 2019	
10.36††	2020 Pricing Agreement (Gigafactory 2170 Cells), entered into on June 9, 2020, by and among Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation and Panasonic Corporation of North America.	10-Q	001-34756	10.3	July 28, 2020	
10.37††	2021 Pricing Agreement (Japan Cells) with respect to 2011 Supply Agreement, executed December 29, 2020, by and among the Registrant, Tesla Motors Netherlands B.V., Panasonic Corporation of North America and SANYO Electric Co., Ltd.	10-K	001-34756	10.39	February 8, 2021	
10.38††	Amended and Restated Factory Lease, executed as of March 26, 2019, by and between the Registrant and Panasonic Energy North America, a division of Panasonic Corporation of North America, as tenant.	10-Q	001-34756	10.3	July 29, 2019	
10.39††	Lease Amendment, executed September 20, 2019, by and among the Registrant, Panasonic Corporation of North America, on behalf of its division Panasonic Energy of North America, with respect to the Amended and Restated Factory Lease, executed as of March 26, 2019.	10-Q	001-34756	10.7	October 29, 2019	
10.40††	Second Lease Amendment, entered into on June 9, 2020, by and between the Registrant and Panasonic Energy of North America, a division of Panasonic Corporation of North America, with respect to the Amended and Restated Factory Lease dated January 1, 2017.	10-Q	001-34756	10.1	July 28, 2020	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.41	Amendment and Restatement in respect of ABL Credit Agreement, dated as of March 6, 2019, by and among certain of the Registrant's and Tesla Motors Netherlands B.V.'s direct or indirect subsidiaries from time to time party thereto, as borrowers, Wells Fargo Bank, National Association, as documentation agent, JPMorgan Chase Bank, N.A., Goldman Sachs Bank USA, Morgan Stanley Senior Funding Inc. and Bank of America, N.A., as syndication agents, the lenders from time to time party thereto, and Deutsche Bank AG New York Branch, as administrative agent and collateral agent.	S-4/A	333-229749	10.68	April 3, 2019	
10.42	First Amendment to Amended and Restated ABL Credit Agreement, dated as of December 23, 2020, in respect of the Amended and Restated ABL Credit Agreement, dated as of March 6, 2019, by and among certain of the Registrant's and Tesla Motors Netherlands B.V.'s direct or indirect subsidiaries from time to time party thereto, as borrowers, Wells Fargo Bank, National Association, as documentation agent, JPMorgan Chase Bank, N.A., Goldman Sachs Bank USA, Morgan Stanley Senior Funding Inc. and Bank of America, N.A., as syndication agents, the lenders from time to time party thereto, and Deutsche Bank AG New York Branch, as administrative agent and collateral agent.	10-K	001-34756	10.44	February 8, 2021	
10.43†	Agreement for Tax Abatement and Incentives, dated as of May 7, 2015, by and between Tesla Motors, Inc. and the State of Nevada, acting by and through the Nevada Governor's Office of Economic Development.	10-Q	001-34756	10.1	August 7, 2015	
10.44	Purchase Agreement, dated as of August 11, 2017, by and among the Registrant, SolarCity and Goldman Sachs & Co. LLC and Morgan Stanley & Co. LLC as representatives of the several initial purchasers named therein.	8-K	001-34756	10.1	August 23, 2017	
10.45	Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 2, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16	November 6, 2014	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.46	First Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 31, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-K(1)	001-35758	10.16a	February 24, 2015	
10.47	Second Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 15, 2014, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-K(1)	001-35758	10.16b	February 24, 2015	
10.48	Third Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of February 12, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16c	May 6, 2015	
10.49	Fourth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, Inc.	10-Q(1)	001-35758	10.16d	May 6, 2015	
10.50	Fifth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of June 30, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16e	July 30, 2015	
10.51	Sixth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of September 1, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16f	October 30, 2015	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.52	Seventh Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16g	October 30, 2015	
10.53	Eighth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of October 26, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q(1)	001-35758	10.16h	October 30, 2015	
10.54	Ninth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of December 9, 2015, by and between The Research Foundation For The State University of New York, on behalf of the College of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-K(1)	001-35758	10.16i	February 10, 2016	
10.55	Tenth Amendment to Amended and Restated Agreement For Research & Development Alliance on Triex Module Technology, effective as of March 31, 2017, by and between The Research Foundation For The State University of New York, on behalf of the Colleges of Nanoscale Science and Engineering of the State University of New York, and Silevo, LLC.	10-Q	001-34756	10.8	May 10, 2017	
10.56	Eleventh Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of July 22, 2020, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.	10-Q	001-34756	10.6	July 28, 2020	
10.57	Twelfth Amendment to Amended and Restated Agreement for Research & Development Alliance on Triex Module Technology, effective as of May 1, 2021, among the Research Foundation for the State University of New York, Silevo, LLC and Tesla Energy Operations, Inc.	10-Q	001-34756	10.1	October 25, 2021	

Exhibit Number	Exhibit Description	Incorporated by Reference				Filed Herewith
		Form	File No.	Exhibit	Filing Date	
10.58††	Grant Contract for State-Owned Construction Land Use Right, dated as of October 17, 2018, by and between Shanghai Planning and Land Resource Administration Bureau, as grantor, and Tesla (Shanghai) Co., Ltd., as grantee (English translation).	10-Q	001-34756	10.2	July 29, 2019	
10.59	Credit Agreement, dated as of January 20, 2023, among Tesla, Inc., the Lenders and Issuing Banks from time to time party thereto, Citibank, N.A., as Administrative Agent and Deutsche Bank Securities, Inc., as Syndication Agent	10-K	001-34756	10.59	January 31, 2023	
21.1	List of Subsidiaries of the Registrant	—	—	—	—	X
23.1	Consent of PricewaterhouseCoopers LLP, Independent Registered Public Accounting Firm	—	—	—	—	X
31.1	Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Executive Officer	—	—	—	—	X
31.2	Rule 13a-14(a) / 15(d)-14(a) Certification of Principal Financial Officer	—	—	—	—	X
32.1*	Section 1350 Certifications	—	—	—	—	X
97	Tesla, Inc. Clawback Policy	—	—	—	—	X
101.INS	Inline XBRL Instance Document	—	—	—	—	X
101.SCH	Inline XBRL Taxonomy Extension Schema Document	—	—	—	—	X
101.CAL	Inline XBRL Taxonomy Extension Calculation Linkbase Document.	—	—	—	—	X
101.DEF	Inline XBRL Taxonomy Extension Definition Linkbase Document	—	—	—	—	X
101.LAB	Inline XBRL Taxonomy Extension Label Linkbase Document	—	—	—	—	X
101.PRE	Inline XBRL Taxonomy Extension Presentation Linkbase Document	—	—	—	—	X
104	Cover Page Interactive Data File (formatted as inline XBRL with applicable taxonomy extension information contained in Exhibits 101)					

* Furnished herewith
** Indicates a management contract or compensatory plan or arrangement
† Confidential treatment has been requested for portions of this exhibit
†† Portions of this exhibit have been redacted in compliance with Regulation S-K Item 601(b)(10).

(1) Indicates a filing of SolarCity
(2) Indicates a filing of Maxwell Technologies, Inc.

ITEM 16. SUMMARY

None.

SIGNATURES

Pursuant to the requirements of Section 13 or 15(d) the Securities Exchange Act of 1934, the registrant has duly caused this report to be signed on its behalf by the undersigned, thereunto duly authorized.

Tesla, Inc.

Date: January 26, 2024

/s/ Elon Musk
Elon Musk
Chief Executive Officer
(Principal Executive Officer)

Pursuant to the requirements of the Securities Exchange Act of 1934, this report has been signed below by the following persons on behalf of the registrant and in the capacities and on the dates indicated.

Signature	Title	Date
/s/ Elon Musk Elon Musk	Chief Executive Officer and Director (Principal Executive Officer)	January 26, 2024
/s/ Vaibhav Taneja Vaibhav Taneja	Chief Financial Officer (Principal Financial Officer and Principal Accounting Officer)	January 26, 2024
/s/ Robyn Denholm Robyn Denholm	Director	January 26, 2024
/s/ Ira Ehrenpreis Ira Ehrenpreis	Director	January 26, 2024
/s/ Joseph Gebbia Joseph Gebbia	Director	January 26, 2024
/s/ James Murdoch James Murdoch	Director	January 26, 2024
/s/ Kimbal Musk Kimbal Musk	Director	January 26, 2024
/s/ JB Straubel JB Straubel	Director	January 26, 2024
/s/ Kathleen Wilson-Thompson Kathleen Wilson-Thompson	Director	January 26, 2024